Robotics & State Estimation (RiSE) Tutorials

Lecture Notes

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Robotics & State Estimation Tutorials
https://github.com/yangyulin/rise-tutorial

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Contents

1 Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems

1.1 Probability

1.1.1 Definition

For an event A, the probability $P(A)$ is

$$P(A) = \frac{N_A}{N}, \tag{1}$$

where N_A is the number of occurrences of A and N is the number of all possible outcomes.

1.1.2 Conditional probability

The probability of event A given that event B is true is

$$P(A \mid B) = \frac{P(A, B)}{P(B)}, \qquad P(B) \neq 0, \tag{2}$$

which yields the product rule

$$\begin{aligned} P(A, B) &= P(A \mid B)\, P(B) &(3)\\ &= P(B \mid A)\, P(A). &(4) \end{aligned}$$

1.1.3 Total probability theorem

If $\{A_1, \ldots, A_n\}$ is a partition (mutually exclusive, $P(A_i, A_j) = 0$ for $i \neq j$) and B is an arbitrary event, then

$$\begin{aligned} P(B) &= P(B \mid A_1)P(A_1) + \cdots + P(B \mid A_n)P(A_n) &(5)\\ &= \sum_i P(B \mid A_i)\, P(A_i). &(6) \end{aligned}$$

1.1.4 Bayes' theorem

$$\begin{aligned} P(A_i \mid B) &= \frac{P(B \mid A_i)\, P(A_i)}{P(B)} &(7)\\ &= \frac{P(B \mid A_i)\, P(A_i)}{\sum_i P(B \mid A_i)\, P(A_i)}. &(8) \end{aligned}$$

1.1.5 Independence

With the inclusion–exclusion relation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, two events A and B are independent when

$$\begin{aligned} P(A, B) &= P(A \mid B)\, P(B) &(9)\\ &= P(B \mid A)\, P(A) &(10)\\ &= P(A)\, P(B) &(11)\\ &= P(B)\, P(A). &(12) \end{aligned}$$

1.2 Random Variables

A random variable defines a function that quantifies an experiment. For a sample space $S = \{s_1, \dots, s_n\}$, a random variable is a map $\mathrm{x} : S \to \mathbb{R}$ with $\mathrm{x}(s_i) = x_i$.

1.2.1 Distribution

The (cumulative) distribution of a random variable x is

$$F_{\mathrm{x}}(x) = P(\mathrm{x} \leq x), \tag{13}$$

i.e. the probability that x takes a value lower than or equal to x.

1.2.2 Probability density function (pdf)

$$\begin{aligned} p(x) &= \lim_{\varepsilon \to 0} \frac{F_{\mathrm{x}}(x) - F_{\mathrm{x}}(x - \varepsilon)}{\varepsilon} & (14) \\ &= \frac{\mathrm{d}F_{\mathrm{x}}(x)}{\mathrm{d}x} & (15) \\ &= \frac{\mathrm{d}P(\mathrm{x} \leq x)}{\mathrm{d}x}, & (16) \end{aligned}$$

hence

$$\begin{aligned} P(\mathrm{x} \leq x) &= \int_{-\infty}^{x} p(\xi)\,\mathrm{d}\xi, & (17) \\ P(\mathrm{x} \leq \infty) &= \int_{-\infty}^{+\infty} p(\xi)\,\mathrm{d}\xi = 1, & (18) \end{aligned}$$

and for an interval

$$P(x_1 \leq \mathrm{x} < x_2) = \int_{x_1}^{x_2} p(\xi)\,\mathrm{d}\xi. \tag{19}$$

1.2.3 Example distributions

- Gaussian: $\mathrm{x} \sim \mathcal{N}(m, \sigma) \iff p(x) = \dfrac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-m)^2}{2\sigma^2}}$.
- Uniform: $p(x) = \begin{cases} 0 & x < x_1 \\ \dfrac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$

1.2.4 Conditional distribution and density

$$\begin{aligned} P(\mathrm{x} \leq x \mid A) &= \frac{P(\mathrm{x} \leq x,\ A)}{P(A)}, & (20) \\ p(x \mid A) &= \frac{\mathrm{d}P(\mathrm{x} \leq x \mid A)}{\mathrm{d}x}. & (21) \end{aligned}$$

1.2.5 Total probability theorem for pdfs

For a mutually exclusive partition $\{A_1, \ldots, A_n\}$,

$$p(x) = p(x \mid A_1)P(A_1) + \cdots + p(x \mid A_n)P(A_n). \tag{22}$$

1.2.6 Continuous Bayes' theorem

Using the infinitesimal relations $P(\mathrm{x} = x) = p(x)\,\delta x$ and $P(\mathrm{x} = x \mid A) = p(x \mid A)\,\delta x$,

$$\begin{aligned}
P(A \mid \mathrm{x} = x) &= \frac{P(A, \mathrm{x} = x)}{P(\mathrm{x} = x)} &(23)\\
&= \frac{P(\mathrm{x} = x \mid A)\,P(A)}{P(\mathrm{x} = x)} &(24)\\
&= \frac{p(x \mid A)}{p(x)}\,P(A), &(25)
\end{aligned}$$

so that $P(A \mid \mathrm{x} = x)\,p(x) = p(x \mid A)\,P(A)$, which gives the continuous total-probability and Bayes' relations

$$\begin{aligned}
P(A) &= \int_{-\infty}^{+\infty} P(A \mid \mathrm{x} = x)\,p(x)\,\mathrm{d}x, &(26)\\
p(x \mid A) &= \frac{P(A \mid \mathrm{x} = x)}{P(A)}\,p(x) &(27)\\
&= \frac{P(A \mid \mathrm{x} = x)\,p(x)}{\int_{-\infty}^{+\infty} P(A \mid \mathrm{x} = x)\,p(x)\,\mathrm{d}x}. &(28)
\end{aligned}$$

1.3 Expectation, Variance, and Random Vectors

1.3.1 Mean

The expected value (mean) of a continuous random variable, and its conditional version, are

$$\begin{aligned}
m_x &= \mathbb{E}(\mathrm{x}) &(29)\\
&= \int_{-\infty}^{+\infty} x\,p(x)\,\mathrm{d}x, &(30)\\
m_{x|A} &= \mathbb{E}(\mathrm{x} \mid A) &(31)\\
&= \int_{-\infty}^{+\infty} x\,p(x \mid A)\,\mathrm{d}x. &(32)
\end{aligned}$$

1.3.2 Variance

$$\begin{aligned}
\sigma_x^2 &= \mathbb{E}\big((\mathrm{x} - m_x)^2\big) &(33)\\
&= \int_{-\infty}^{+\infty} (x - m_x)^2\,p(x)\,\mathrm{d}x &(34)\\
&= \mathbb{E}(\mathrm{x}^2) - \big(\mathbb{E}(\mathrm{x})\big)^2, &(35)
\end{aligned}$$

with the conditional variance

$$\sigma_{x|A}^2 = \mathbb{E}\big\{(\mathrm{x} - m_{x|A})^2 \mid A\big\} \tag{36}$$
$$= \int_{-\infty}^{+\infty} (x - m_{x|A})^2\, p(x \mid A)\,\mathrm{d}x. \tag{37}$$

1.3.3 A set of random variables

The joint pdf of $x_1, \ldots, x_n$ satisfies

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p(x_1, \ldots, x_n)\,\mathrm{d}x_1 \cdots \mathrm{d}x_n = 1. \tag{38}$$

If x_i, x_j are independent, $p(x_i, x_j) = p(x_i)\,p(x_j)$. Expectation is linear: for $y = \alpha x_i + \beta x_j$,

$$\mathbb{E}(y) = \alpha\,\mathbb{E}(x_i) + \beta\,\mathbb{E}(x_j). \tag{39}$$

1.3.4 Covariance

$$\sigma_{ij}^2 = \mathbb{E}\big((x_i - m_i)(x_j - m_j)\big) \tag{40}$$
$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\,\mathbb{E}(x_j). \tag{41}$$

When $\sigma_{ij} = 0$ we have $\mathbb{E}(x_i x_j) = \mathbb{E}(x_i)\,\mathbb{E}(x_j)$ and x_i, x_j are *uncorrelated.*

Discussion questions.

1. If x_i, x_j are independent, then $p(x_i, x_j) = p(x_i)p(x_j) \Rightarrow \mathbb{E}(x_i x_j) = \mathbb{E}(x_i)\mathbb{E}(x_j)$ — so independence implies uncorrelated.
2. If x_i, x_j are uncorrelated, are they necessarily independent?
3. If x_i, x_j are jointly Gaussian, is independence equivalent to being uncorrelated?

1.3.5 Random vector

Let $\mathbf{x} = [x_1, \ldots, x_n]^\top$, where all elements are random variables, with mean $\mathbb{E}(\mathbf{x}) \triangleq \mathbf{m}_x$ and covariance

$$\mathbf{P} = \mathbb{E}\big((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top\big) \tag{42}$$
$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\,\mathbb{E}(\mathbf{x}^\top), \tag{43}$$

with entries

$$P_{ii} = \mathbb{E}\big((x_i - m_i)^2\big) \tag{44}$$
$$= \int_{-\infty}^{+\infty} (x_i - m_i)^2 p(x_i)\,\mathrm{d}x_i, \tag{45}$$
$$P_{ij} = \mathbb{E}\big((x_i - m_i)(x_j - m_j)\big). \tag{46}$$

Since $P_{ij} = P_{ji}$, the covariance $\mathbf{P}$ is a symmetric matrix; it is also positive (semi-)definite.

1.3.6 Gaussian vector pdf

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{|\mathbf{P}|}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\mathbf{m}_x)^\top \mathbf{P}^{-1}(\mathbf{x}-\mathbf{m}_x)\Big), \tag{47}$$

where $|\mathbf{P}|$ is the determinant of $\mathbf{P}$, $\mathbf{m}_x = \mathbb{E}(\mathbf{x})$, and $\mathbf{P} = \mathbb{E}\big((\mathbf{x}-\mathbf{m}_x)(\mathbf{x}-\mathbf{m}_x)^\top\big)$.

Homework. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$. Define

$$\begin{aligned} q &= (\mathbf{x}-\mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x}-\mathbf{m}_x), \\ \mathbf{y} &= \mathbf{P}_{xx}^{-1/2}(\mathbf{x}-\mathbf{m}_x), \\ q &= \mathbf{y}^\top\mathbf{y} = \sum_i y_i^2. \end{aligned}$$

What are the mean and covariance of q?

1.4 Maximum Likelihood and Maximum a Posteriori Estimation

For a measurement model with additive noise $z_i = h_i(\mathbf{x}) + n_i$, stacked as $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, two estimators are

- **Maximum likelihood estimation (MLE):** $\max_{\mathbf{x}} p(\mathbf{z} \mid \mathbf{x})$.
- **Maximum a posteriori (MAP):** $\max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{z})$, with $p(\mathbf{x} \mid \mathbf{z}) = \dfrac{p(\mathbf{z} \mid \mathbf{x})\, p(\mathbf{x})}{p(\mathbf{z})}$.

For zero-mean Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{P})$,

$$\max_{\mathbf{x}} p(\mathbf{z} \mid \mathbf{x}) = \max_{\mathbf{x}} \frac{1}{(2\pi)^{n/2}\sqrt{|\mathbf{P}|}} \exp\Big(-(\mathbf{z}-\mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z}-\mathbf{h}(\mathbf{x}))\Big), \tag{48}$$

which is equivalent to the weighted least-squares problem

$$\min_{\mathbf{x}}\ C = (\mathbf{z}-\mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z}-\mathbf{h}(\mathbf{x})). \tag{49}$$

Linearizing about $\hat{\mathbf{x}}$ with $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}(\mathbf{x}-\hat{\mathbf{x}}) + \mathbf{n}$, and defining the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$ and error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$,

$$\min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}})^\top \mathbf{P}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} \big(\tilde{\mathbf{z}}^\top\mathbf{P}^{-1}\tilde{\mathbf{z}} - 2\,\tilde{\mathbf{z}}^\top\mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top\mathbf{H}^\top\mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}}\big). \tag{50}$$

Setting the gradient to zero gives the normal equations $\mathbf{H}^\top\mathbf{P}^{-1}\mathbf{H}\,\tilde{\mathbf{x}} = \mathbf{H}^\top\mathbf{P}^{-1}\tilde{\mathbf{z}}$, hence

$$\boxed{\begin{aligned} \mathbf{x}^\oplus &= \hat{\mathbf{x}} + \tilde{\mathbf{x}} && (51)\\ &= \hat{\mathbf{x}} + (\mathbf{H}^\top\mathbf{P}^{-1}\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{P}^{-1}\tilde{\mathbf{z}}. && (52) \end{aligned}}$$

This is the classical weighted least-squares solution.

Questions.

1. What is the covariance of the new estimate $\mathbf{x}^\oplus$?
2. For a prior $\mathbf{x}_p = \mathbf{x} + \mathbf{n}_p$ together with $z_i = h_i(\mathbf{x}) + n_i$, $i = 1, \dots, n$, what is the MAP solution $\max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{z})$?

1.4.1 Optimization

Nonlinear least-squares problems are solved iteratively, e.g. Gauss–Newton (GN) and Levenberg–Marquardt (LM).

1.5 Linear Systems

A linear (continuous-time) state-space system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \tag{53}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}, \tag{54}$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ the input, and $\mathbf{y}$ the output. Its discrete-time counterpart is

$$\mathbf{x}_{k+1} = \mathbf{A}'\mathbf{x}_k + \mathbf{B}'\mathbf{u}_k, \tag{55}$$
$$\mathbf{y}_{k+1} = \mathbf{C}'\mathbf{x}_{k+1} + \mathbf{D}'\mathbf{u}_{k+1}. \tag{56}$$

If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are constant, the system is *linear time-invariant* (LTI); if they depend on time, it is *linear time-variant* (LTV).

1.5.1 Homogeneous solution

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\mathbf{y} = \mathbf{C}\mathbf{x}$. From $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{0}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(e^{-\mathbf{A}t}\mathbf{x}\big) = e^{-\mathbf{A}t}\dot{\mathbf{x}} + e^{-\mathbf{A}t}(-\mathbf{A})\mathbf{x} \tag{57}$$
$$= e^{-\mathbf{A}t}\big(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}\big) \tag{58}$$
$$= \mathbf{0}. \tag{59}$$

Integrating from 0 to t gives $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(t_0) = \mathbf{0}$, so

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 \tag{60}$$
$$= \mathbf{\Phi}(t,0)\,\mathbf{x}_0, \tag{61}$$
$$\mathbf{\Phi}(t,0) = e^{\mathbf{A}t}, \tag{62}$$

where $\mathbf{\Phi}(t,0)$ is the *state transition matrix*, and $\mathbf{y} = \mathbf{C}\,\mathbf{\Phi}(t,0)\,\mathbf{x}_0$.

1.5.2 Properties of the state transition matrix

$$\mathbf{\Phi}(t_2,t_0) = \mathbf{\Phi}(t_2,t_1)\,\mathbf{\Phi}(t_1,t_0), \tag{63}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{\Phi}(t,0) = \mathbf{A}\,\mathbf{\Phi}(t,0). \tag{64}$$

The second follows from $\dot{\mathbf{x}} = \dot{\mathbf{\Phi}}(t,0)\mathbf{x}_0 = \mathbf{A}\mathbf{x} = \mathbf{A}\,\mathbf{\Phi}(t,0)\mathbf{x}_0$.

1.5.3 Output sequence and observability

Sampling the homogeneous response at $t_0, t_1, \ldots, t_n$ with $\mathbf{x}_k = \mathbf{\Phi}(k,0)\mathbf{x}_0$ and $\mathbf{y}_k = \mathbf{C}\mathbf{x}_k$ stacks into

$$\begin{bmatrix}\mathbf{y}_1\\ \vdots\\ \mathbf{y}_n\end{bmatrix} = \begin{bmatrix}\mathbf{C}\,\mathbf{\Phi}(1,0)\\ \vdots\\ \mathbf{C}\,\mathbf{\Phi}(n,0)\end{bmatrix}\mathbf{x}_0, \tag{65}$$

which relates the measured outputs to the initial state (an observability-type relation).

1.5.4 Full LTV solution

For the complete system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}(t)$, write $\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x} = \mathbf{B}(t)\mathbf{u}(t)$ and use the integrating factor $e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\mathbf{x}\Big) = e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\big(\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x}\big) \tag{66}$$

$$= e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\mathbf{B}(t)\mathbf{u}(t). \tag{67}$$

Integrating,

$$e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\int_0^\tau \mathbf{A}(s)\mathrm{d}s}\mathbf{B}(\tau)\mathbf{u}(\tau)\,\mathrm{d}\tau, \tag{68}$$

so that, with $\boldsymbol{\Phi}(t,0) = e^{\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}$ and $\boldsymbol{\Phi}(t,\tau) = e^{\int_\tau^t \mathbf{A}(s)\mathrm{d}s}$,

$$\boxed{\begin{aligned}&\mathbf{x}(t) = \boldsymbol{\Phi}(t,0)\,\mathbf{x}_0 + \int_0^t \boldsymbol{\Phi}(t,\tau)\,\mathbf{B}(\tau)\,\mathbf{u}(\tau)\,\mathrm{d}\tau, && (69)\\ &\mathbf{y}(t) = \mathbf{C}(t)\Big(\boldsymbol{\Phi}(t,0)\mathbf{x}_0 + \int_0^t \boldsymbol{\Phi}(t,\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)\,\mathrm{d}\tau\Big) + \mathbf{D}(t)\mathbf{u}(t). && (70)\end{aligned}}$$

This is one of the most important properties of linear systems.

1.6 Homework Problems and Solutions

1.6.1 Variance identity

Problem. Show that the variance of a scalar random variable can be written as $\sigma_x^2 = \mathbb{E}(\mathrm{x}^2) - \big(\mathbb{E}(\mathrm{x})\big)^2$, and discuss when it vanishes.

Solution. Expanding the definition with $m_x = \mathbb{E}(\mathrm{x})$,

$$\sigma_x^2 = \mathbb{E}\big((\mathrm{x} - m_x)^2\big) \tag{71}$$

$$= \mathbb{E}\big(\mathrm{x}^2 - 2m_x\mathrm{x} + m_x^2\big) \tag{72}$$

$$= \mathbb{E}(\mathrm{x}^2) - 2m_x\mathbb{E}(\mathrm{x}) + m_x^2 \tag{73}$$

$$= \mathbb{E}(\mathrm{x}^2) - m_x^2. \tag{74}$$

Since $\sigma_x^2 \geq 0$, we have $\mathbb{E}(\mathrm{x}^2) \geq \big(\mathbb{E}(\mathrm{x})\big)^2$, with equality if and only if $\sigma_x^2 = 0$, i.e. $\mathrm{x} = m_x$ almost surely (a deterministic variable) — a degenerate special case.

1.6.2 Covariance matrix identity

Problem. For a random vector $\mathbf{x}$ with mean $\mathbf{m}_x$, show that $\mathbf{P} = \mathbb{E}\big((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top\big) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top)$.

Solution. Expanding the outer product and using linearity of expectation, together with $\mathbb{E}(\mathbf{x}) = \mathbf{m}_x$,

$$\mathbf{P} = \mathbb{E}\big(\mathbf{x}\mathbf{x}^\top - \mathbf{x}\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{x}^\top + \mathbf{m}_x\mathbf{m}_x^\top\big) \tag{75}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbf{m}_x^\top - \mathbf{m}_x\mathbb{E}(\mathbf{x}^\top) + \mathbf{m}_x\mathbf{m}_x^\top \tag{76}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{m}_x^\top + \mathbf{m}_x\mathbf{m}_x^\top \tag{77}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top \tag{78}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top). \tag{79}$$

1.6.3 Scalar covariance identity

Problem. For two random variables x_i, x_j, show that $\sigma_{ij} = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j)$.

Solution. With $m_i = \mathbb{E}(x_i)$, $m_j = \mathbb{E}(x_j)$, expand

$$\begin{aligned}
\sigma_{ij} &= \mathbb{E}\big((x_i - m_i)(x_j - m_j)\big) && (80)\\
&= \iint (x_i - m_i)(x_j - m_j)\, p(x_i, x_j)\,\mathrm{d}x_i\,\mathrm{d}x_j && (81)\\
&= \iint x_i x_j\, p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j - m_j \iint x_i\, p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j \\
&\quad - m_i \iint x_j\, p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j + m_i m_j \iint p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j. && (82)
\end{aligned}$$

Using the marginalization identities (and symmetrically for x_j)

$$\int p(x_i, x_j)\,\mathrm{d}x_i = p(x_j), \qquad \iint p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j = 1, \tag{83}$$

we get

$$\begin{aligned}
m_i \iint x_j\, p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j &= m_i \int x_j\, p(x_j)\,\mathrm{d}x_j && (84)\\
&= m_i m_j, && (85)
\end{aligned}$$

and likewise for the other term, so

$$\begin{aligned}
\sigma_{ij} &= \mathbb{E}(x_i x_j) - m_i m_j - m_i m_j + m_i m_j && (86)\\
&= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j). && (87)
\end{aligned}$$

1.6.4 Independence versus uncorrelatedness

Problem. Resolve the three discussion questions: (1) does independence imply uncorrelatedness? (2) does uncorrelatedness imply independence? (3) for jointly Gaussian variables, are the two equivalent?

Solution. *(1) Independent $\Rightarrow$ uncorrelated.* If $p(x_i, x_j) = p(x_i)p(x_j)$, then

$$\begin{aligned}
\mathbb{E}(x_i x_j) &= \iint x_i x_j\, p(x_i, x_j)\,\mathrm{d}x_i\mathrm{d}x_j && (88)\\
&= \int x_i\, p(x_i)\,\mathrm{d}x_i \cdot \int x_j\, p(x_j)\,\mathrm{d}x_j && (89)\\
&= \mathbb{E}(x_i)\mathbb{E}(x_j), && (90)
\end{aligned}$$

so $\sigma_{ij} = 0$.

(2) Uncorrelated $\not\Rightarrow$ independent. Uncorrelatedness gives only

$$\begin{aligned}
\mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j) &= \iint x_i x_j \big(p(x_i, x_j) - p(x_i)p(x_j)\big)\,\mathrm{d}x_i\mathrm{d}x_j && (91)\\
&= 0, && (92)
\end{aligned}$$

which does *not* force $p(x_i, x_j) = p(x_i)p(x_j)$ in general. Hence uncorrelated does not imply independent.

(3) Jointly Gaussian. For $x_i \sim \mathcal{N}(m_i, \sigma_i^2)$ and $x_j \sim \mathcal{N}(m_j, \sigma_j^2)$, the product of marginals is

$$p(x_i)p(x_j) = \frac{1}{2\pi\sigma_i\sigma_j} \exp\left\{ -\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix} \right\}. \tag{93}$$

The joint Gaussian $[x_i, x_j]^\top \sim \mathcal{N}\big([m_i, m_j]^\top, \mathbf{P}_{xx}\big)$ with $\mathbf{P}_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$ is

$$p(x_i, x_j) = \frac{1}{2\pi\sqrt{|\mathbf{P}_{xx}|}} \exp\left\{ -\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \mathbf{P}_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix} \right\}. \tag{94}$$

If $\sigma_{ij} = \sigma_{ji} = 0$, then $\mathbf{P}_{xx}$ is diagonal and $p(x_i, x_j) = p(x_i)p(x_j)$. Hence, for (jointly) Gaussian variables, *independent* $\Leftrightarrow$ *uncorrelated.*

1.6.5 The χ^2 distribution

Problem. Let $\mathbf{x} = [x_1, \ldots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$ and define $q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Find the mean and variance of q.

Solution. Factor $\mathbf{P}_{xx} = \mathbf{L}\mathbf{L}^\top$ (Cholesky), so $\mathbf{P}_{xx}^{-1} = \mathbf{L}^{-\top}\mathbf{L}^{-1}$, and define the whitened variable $\mathbf{y} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Then

$$\begin{aligned} q &= (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{L}^{-\top}\mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x) && (95) \\ &= \mathbf{y}^\top \mathbf{y} && (96) \\ &= \sum_{i=1}^{n} y_i^2. && (97) \end{aligned}$$

The whitened variable has zero mean and identity covariance,

$$\begin{aligned} \mathbf{m}_y &= \mathbb{E}(\mathbf{y}) && (98) \\ &= \mathbf{L}^{-1}\big(\mathbb{E}(\mathbf{x}) - \mathbf{m}_x\big) && (99) \\ &= \mathbf{0}, && (100) \\ \mathbf{P}_{yy} &= \mathbb{E}(\mathbf{y}\mathbf{y}^\top) && (101) \\ &= \mathbf{L}^{-1}\,\mathbb{E}\big((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top\big)\,\mathbf{L}^{-\top} && (102) \\ &= \mathbf{L}^{-1}\mathbf{P}_{xx}\mathbf{L}^{-\top} && (103) \\ &= \mathbf{L}^{-1}\mathbf{L}\mathbf{L}^\top\mathbf{L}^{-\top} && (104) \\ &= \mathbf{I}_n, && (105) \end{aligned}$$

so $y_i \sim \mathcal{N}(0, 1)$ are independent. Therefore the mean of q is

$$\begin{aligned} \mathbb{E}(q) &= \mathbb{E}(\mathbf{y}^\top \mathbf{y}) && (106) \\ &= \sum_{i=1}^{n} \mathbb{E}(y_i^2) && (107) \\ &= n. && (108) \end{aligned}$$

For the variance, with $\mathbb{E}(y_i^2) = 1$ the cross terms vanish by independence, giving

$$\begin{aligned}
\sigma_{qq}^2 &= \mathbb{E}\big((q-n)^2\big) && (109)\\
&= \mathbb{E}\Big(\big(\textstyle\sum_i (y_i^2 - 1)\big)^2\Big) && (110)\\
&= \sum_i \mathbb{E}\big((y_i^2-1)^2\big) && (111)\\
&= \sum_i \big(\mathbb{E}(y_i^4) - 2\mathbb{E}(y_i^2) + 1\big). && (112)
\end{aligned}$$

The fourth moment of a standard normal follows from integration by parts (with $u = y_i^3$, $\mathrm{d}v = y_i\, p(y_i)\,\mathrm{d}y_i$, $v = -p(y_i)$):

$$\begin{aligned}
\mathbb{E}(y_i^4) &= \int_{-\infty}^{+\infty} y_i^4\, p(y_i)\,\mathrm{d}y_i && (113)\\
&= \underbrace{\big[-y_i^3 p(y_i)\big]_{-\infty}^{+\infty}}_{=0} + \int_{-\infty}^{+\infty} 3y_i^2\, p(y_i)\,\mathrm{d}y_i && (114)\\
&= 3\,\mathbb{E}(y_i^2) && (115)\\
&= 3. && (116)
\end{aligned}$$

Hence each term equals $3 - 2 + 1 = 2$, and

$$\sigma_{qq}^2 = \sum_{i=1}^{n} 2 = 2n. \tag{117}$$

So $q \sim \chi^2(n)$ has mean n and variance $2n$.

2 Lecture 2: Rotation, Lie Groups, SE(3), SE$_2$(3), and Quaternions

Further reading. This lecture draws on several standard references: matrix Lie groups in Barfoot [1, Ch. 7]; the Lie-group treatment of $SO(3)/SE(3)$ in the GTSAM notes [2], Atanasov's ECE276A slides [3], and the micro Lie theory of Solà, Deray & Atchuthan [4]; and quaternion algebra/kinematics in Trawny & Roumeliotis [5, Sec. 1] and Solà [6, Sec. 1–3].

2.1 Preliminaries: useful series

The matrix exponential and the sine/cosine series underpin everything that follows:

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k \tag{118}$$

$$= I + x + \tfrac{1}{2!}x^2 + \tfrac{1}{3!}x^3 + \tfrac{1}{4!}x^4 + \cdots , \tag{119}$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \tag{120}$$

$$= x - \tfrac{1}{3!}x^3 + \tfrac{1}{5!}x^5 - \cdots , \tag{121}$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \tag{122}$$

$$= 1 - \tfrac{1}{2!}x^2 + \tfrac{1}{4!}x^4 - \cdots . \tag{123}$$

For small x (so that $x^2, x^3 \to 0$), the important first-order approximations are $e^x \approx I + x$, $\sin(x) \approx x$, and $\cos(x) \approx 1$.

2.2 Frames, notation, and transformations

We work with coordinate frames $\{A\}$ and $\{B\}$. A point/vector $\mathbf{L}$ expressed in $\{A\}$ is ${}^{A}\mathbf{L}$, and in $\{B\}$ is ${}^{B}\mathbf{L}$. The angle $\theta_{A\to B}$ is the rotation about the z-axis taking $\{A\}$ to $\{B\}$ (Fig. 1).

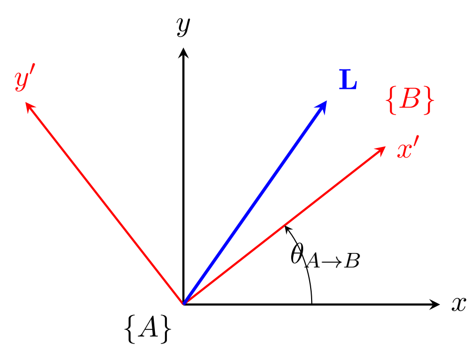


Figure 1: Frame $\{B\}$ is frame $\{A\}$ rotated by $\theta_{A\to B}$ about the z-axis; the vector $\mathbf{L}$ can be expressed in either frame.

The rotation matrix ${}^{A}_{B}\mathbf{R}$ maps a vector from $\{B\}$ to $\{A\}$:

$${}^{A}\mathbf{L} = {}^{A}_{B}\mathbf{R}\,{}^{B}\mathbf{L} \tag{124}$$

$$= {}^{A}_{B}\mathbf{R}(\theta_{A\to B})\,{}^{B}\mathbf{L}. \tag{125}$$

Adding the translation ${}^{A}\mathbf{p}_B$ (the origin of $\{B\}$ expressed in $\{A\}$, Fig. 2) gives the full rigid-body transformation

$$\{A\},\{B\} \;\Rightarrow\; \left\{ {}^{A}_{B}\mathbf{R},\; {}^{A}\mathbf{p}_B \right\} \;\Rightarrow\; {}^{A}_{B}\mathbf{T} = \begin{bmatrix} {}^{A}_{B}\mathbf{R} & {}^{A}\mathbf{p}_B \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}. \tag{126}$$

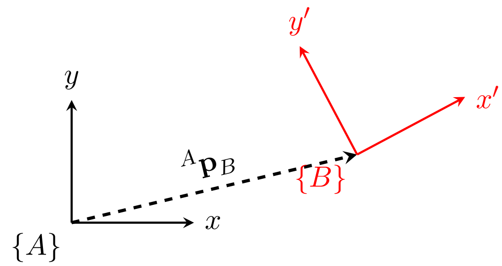


Figure 2: A rigid-body transformation combines the rotation ${}^{A}_{B}\mathbf{R}$ with the translation ${}^{A}\mathbf{p}_B$.

2.3 Defining rotation

2.3.1 Euler angles

With roll θ (about x), pitch r (about y), yaw ψ (about z),

$$\mathbf{R}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \; \mathbf{R}(r) = \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix}, \; \mathbf{R}(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{127}$$

and the yaw–pitch–roll composition is

$$\mathbf{R} = \mathbf{R}(\psi)\,\mathbf{R}(r)\,\mathbf{R}(\theta) \tag{128}$$

$$= \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}. \tag{129}$$

2.3.2 Infinitesimal rotation and the skew-symmetric matrix

For small angles $\delta\theta, \delta r, \delta\psi \to 0$ each elementary rotation linearizes ($\cos \to 1$, $\sin \to$ angle):

$$\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix}, \quad \mathbf{R}(\delta r) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix}, \quad \mathbf{R}(\delta\psi) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{130}$$

Multiplying the last two and dropping all second-order products ($\delta r\,\delta\theta \approx 0$ etc.),

$$\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix} \tag{131}$$

$$= \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}, \tag{132}$$

and then

$$\mathbf{R} = \mathbf{R}(\delta\psi)\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\delta\psi & \delta r \\ \delta\psi & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \tag{133}$$

$$= \mathbf{I} + \begin{bmatrix} 0 & -\delta\psi & \delta r \\ \delta\psi & 0 & -\delta\theta \\ -\delta r & \delta\theta & 0 \end{bmatrix} \tag{134}$$

$$= \mathbf{I} + \lfloor \boldsymbol{\theta} \rfloor_\times , \tag{135}$$

where the small-angle vector is $\boldsymbol{\theta} = [\delta\theta,\ \delta r,\ \delta\psi]^\top$. The matrix $\lfloor \boldsymbol{\theta} \rfloor_\times$ is skew-symmetric, $\lfloor \boldsymbol{\theta} \rfloor_\times = -\lfloor \boldsymbol{\theta} \rfloor_\times^\top$. This motivates writing a finite rotation as $\mathbf{R}(\boldsymbol{\theta}) \approx \mathbf{I} + \lfloor \boldsymbol{\theta} \rfloor_\times \approx \exp(\lfloor \boldsymbol{\theta} \rfloor_\times)$, and defining $\mathrm{Exp}(\boldsymbol{\theta}) \triangleq \exp(\lfloor \boldsymbol{\theta} \rfloor_\times)$. The question is then whether $\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})$.

2.4 Axis–angle and Rodrigues' formula

Let $\mathbf{k}$ be the unit rotation axis and θ the rotation angle, with $\boldsymbol{\theta} = \theta\mathbf{k} = [\theta_1, \theta_2, \theta_3]^\top$, so

$$\mathbf{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}, \quad \lfloor \mathbf{k} \rfloor_\times = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix}, \quad \lfloor \boldsymbol{\theta} \rfloor_\times = \begin{bmatrix} 0 & -\theta_3 & \theta_2 \\ \theta_3 & 0 & -\theta_1 \\ -\theta_2 & \theta_1 & 0 \end{bmatrix}. \tag{136}$$

The skew operator on a unit vector satisfies the cyclic power identities

$$\lfloor \mathbf{k} \rfloor_\times^2 = \mathbf{k}\mathbf{k}^\top - \mathbf{I}_3, \qquad \lfloor \mathbf{k} \rfloor_\times^3 = -\lfloor \mathbf{k} \rfloor_\times , \qquad \lfloor \mathbf{k} \rfloor_\times^4 = -\lfloor \mathbf{k} \rfloor_\times^2 , \tag{137}$$

$$\lfloor \mathbf{k} \rfloor_\times^5 = \lfloor \mathbf{k} \rfloor_\times , \qquad \lfloor \mathbf{k} \rfloor_\times^6 = \lfloor \mathbf{k} \rfloor_\times^2 , \qquad \lfloor \mathbf{k} \rfloor_\times^7 = -\lfloor \mathbf{k} \rfloor_\times . \tag{138}$$

Substituting $\lfloor \boldsymbol{\theta} \rfloor_\times = \theta \lfloor \mathbf{k} \rfloor_\times$ into the exponential series term by term,

$$\begin{aligned} \mathrm{Exp}(\boldsymbol{\theta}) = \exp(\theta \lfloor \mathbf{k} \rfloor_\times) &= \mathbf{I} + \theta \lfloor \mathbf{k} \rfloor_\times + \tfrac{\theta^2}{2!} \lfloor \mathbf{k} \rfloor_\times^2 + \tfrac{\theta^3}{3!} \lfloor \mathbf{k} \rfloor_\times^3 + \tfrac{\theta^4}{4!} \lfloor \mathbf{k} \rfloor_\times^4 \\ &\quad + \tfrac{\theta^5}{5!} \lfloor \mathbf{k} \rfloor_\times^5 + \tfrac{\theta^6}{6!} \lfloor \mathbf{k} \rfloor_\times^6 + \tfrac{\theta^7}{7!} \lfloor \mathbf{k} \rfloor_\times^7 + \cdots \\ &= \mathbf{I} + \theta \lfloor \mathbf{k} \rfloor_\times + \tfrac{\theta^2}{2!} \lfloor \mathbf{k} \rfloor_\times^2 - \tfrac{\theta^3}{3!} \lfloor \mathbf{k} \rfloor_\times - \tfrac{\theta^4}{4!} \lfloor \mathbf{k} \rfloor_\times^2 \\ &\quad + \tfrac{\theta^5}{5!} \lfloor \mathbf{k} \rfloor_\times + \tfrac{\theta^6}{6!} \lfloor \mathbf{k} \rfloor_\times^2 - \tfrac{\theta^7}{7!} \lfloor \mathbf{k} \rfloor_\times + \cdots \\ &= \mathbf{I} + \underbrace{\left(\theta - \tfrac{\theta^3}{3!} + \tfrac{\theta^5}{5!} - \tfrac{\theta^7}{7!} + \cdots\right)}_{\sin\theta} \lfloor \mathbf{k} \rfloor_\times \\ &\quad + \underbrace{\left(\tfrac{\theta^2}{2!} - \tfrac{\theta^4}{4!} + \tfrac{\theta^6}{6!} - \cdots\right)}_{1-\cos\theta} \lfloor \mathbf{k} \rfloor_\times^2 , \end{aligned} \tag{139}$$

which gives **Rodrigues' rotation formula**

$$\boxed{\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta}) = \exp(\theta \lfloor \mathbf{k} \rfloor_\times) = \mathbf{I} + \sin\theta\ \lfloor \mathbf{k} \rfloor_\times + (1 - \cos\theta)\ \lfloor \mathbf{k} \rfloor_\times^2 .} \tag{140}$$

For a small angle $\delta\theta \to 0$, $\mathbf{R}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \lfloor \delta\boldsymbol{\theta} \rfloor_\times$ (small-angle approximation).

2.4.1 Logarithmic map

The log map is the inverse of the exponential map:

$$\log(\mathbf{R}) = \lfloor \boldsymbol{\theta} \rfloor_\times, \quad \mathrm{Log}(\mathbf{R}) = \boldsymbol{\theta}_{3\times 1}, \qquad \theta = \arccos\Big(\tfrac{\mathrm{tr}\,\mathbf{R}-1}{2}\Big), \quad \lfloor \mathbf{k} \rfloor_\times = \frac{\mathbf{R}-\mathbf{R}^\top}{2\sin\theta}. \tag{141}$$

2.5 SO(3) kinematics

The two directions of the maps are: from the vector space $\mathbb{R}^3$ via the Lie algebra to the Lie group, $\boldsymbol{\theta} \to \lfloor \boldsymbol{\theta} \rfloor_\times \to \mathbf{R} = \mathrm{Exp}(\boldsymbol{\theta})$; and back, $\mathbf{R} \to \lfloor \boldsymbol{\theta} \rfloor_\times \to \boldsymbol{\theta} = \mathrm{Log}(\mathbf{R})$.

(1) Angular-velocity kinematics. Differentiating the orthogonality constraint $\mathbf{R}\mathbf{R}^\top = \mathbf{I}_3$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{R}\mathbf{R}^\top) = \mathbf{0} \;\Rightarrow\; \dot{\mathbf{R}}\mathbf{R}^\top + \mathbf{R}\dot{\mathbf{R}}^\top = \mathbf{0} \;\Rightarrow\; \dot{\mathbf{R}}\mathbf{R}^\top = -\mathbf{R}\dot{\mathbf{R}}^\top = -(\dot{\mathbf{R}}\mathbf{R}^\top)^\top, \tag{142}$$

so $\dot{\mathbf{R}}\mathbf{R}^\top$ is skew-symmetric. Writing it as $\lfloor \boldsymbol{\omega} \rfloor_\times$ with $\lfloor \boldsymbol{\omega} \rfloor_\times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$,

$$\dot{\mathbf{R}}\mathbf{R}^\top = \lfloor \boldsymbol{\omega} \rfloor_\times \;\Rightarrow\; \dot{\mathbf{R}} = \lfloor \boldsymbol{\omega} \rfloor_\times \mathbf{R}. \tag{143}$$

(2) Equivalent limit form. With $\mathbf{R}(t+\Delta t) = \mathbf{R}(t)\exp(\lfloor \boldsymbol{\omega} \rfloor_\times \Delta t)$,

$$\begin{aligned} \dot{\mathbf{R}}(t) &= \lim_{\Delta t \to 0} \frac{\mathbf{R}(t+\Delta t) - \mathbf{R}(t)}{\Delta t} && (144)\\ &= \lim_{\Delta t \to 0} \frac{\mathbf{R}(t)\exp(\lfloor \boldsymbol{\omega} \rfloor_\times \Delta t) - \mathbf{R}(t)}{\Delta t} && (145)\\ &= \mathbf{R}(t) \lim_{\Delta t \to 0} \frac{\exp(\lfloor \boldsymbol{\omega} \rfloor_\times \Delta t) - \mathbf{I}}{\Delta t} && (146)\\ &= \mathbf{R}(t)\, \lfloor \boldsymbol{\omega} \rfloor_\times. && (147) \end{aligned}$$

(3) Integration. With $\mathbf{R}_i = \exp(\lfloor \boldsymbol{\omega}_i \rfloor_\times \Delta t_i)$, composing increments gives

$$\begin{aligned} \mathbf{R}_{1\to k} &= \mathbf{R}_k \mathbf{R}_{k-1} \cdots \mathbf{R}_2 \mathbf{R}_1 && (148)\\ &= \exp(\lfloor \boldsymbol{\omega}_k \rfloor_\times \Delta t_k) \cdots \exp(\lfloor \boldsymbol{\omega}_1 \rfloor_\times \Delta t_1). && (149) \end{aligned}$$

(4) Interpolation. For ${}^{G}_{t_1}\mathbf{R}, {}^{G}_{t_2}\mathbf{R}$ and $t \in [t_1, t_2]$, define the relative rotation and its angle, and the interpolation ratio:

$$\begin{aligned} {}^{t_1}_{t_2}\mathbf{R} &= {}^{G}_{t_1}\mathbf{R}^\top\, {}^{G}_{t_2}\mathbf{R}, && (150)\\ {}^{t_1}_{t_2}\boldsymbol{\theta} &= \mathrm{Log}({}^{t_1}_{t_2}\mathbf{R}) = \mathrm{Log}({}^{G}_{t_1}\mathbf{R}^\top\, {}^{G}_{t_2}\mathbf{R}), && (151)\\ \lambda &= \frac{t - t_1}{t_2 - t_1}. && (152) \end{aligned}$$

Then

$$\begin{aligned} {}^{G}_{t}\mathbf{R} &= {}^{G}_{t_1}\mathbf{R}\, {}^{t_1}_{t}\mathbf{R} && (153)\\ &= {}^{G}_{t_1}\mathbf{R}\; \mathrm{Exp}\big(\lambda\, {}^{t_1}_{t_2}\boldsymbol{\theta}\big). && (154) \end{aligned}$$

(5) Right Jacobian. For a rotation $\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})$, perturb the angle by $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ and let $\delta\boldsymbol{\varphi}$ be the corresponding small rotation in $\mathfrak{so}(3)$:

$$\mathbf{R}(\boldsymbol{\theta})\ \mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathbf{R}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \ \Rightarrow\ \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}). \tag{155}$$

Solving for $\delta\boldsymbol{\varphi}$,

$$\mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathrm{Exp}(\boldsymbol{\theta})^{-1}\,\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \ \Rightarrow\ \delta\boldsymbol{\varphi} = \mathrm{Log}\big(\,\mathrm{Exp}(\boldsymbol{\theta})^{-1}\,\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})\big), \tag{156}$$

and the *right Jacobian* is the derivative of $\delta\boldsymbol{\varphi}$ with respect to $\delta\boldsymbol{\theta}$:

$$\mathbf{J}_r(\boldsymbol{\theta}) \triangleq \frac{\partial\,\delta\boldsymbol{\varphi}}{\partial\,\delta\boldsymbol{\theta}} = \lim_{\delta\boldsymbol{\theta}\to 0} \frac{\mathrm{Log}\big(\,\mathrm{Exp}(\boldsymbol{\theta})^{-1}\,\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})\big)}{\delta\boldsymbol{\theta}}. \tag{157}$$

Equivalently, $\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r(\boldsymbol{\theta})\,\delta\boldsymbol{\theta}\big)$, and inverting,

$$\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathrm{Exp}\big(\boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta})\,\delta\boldsymbol{\varphi}\big) \ \Rightarrow\ \mathrm{Log}\big(\,\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\delta\boldsymbol{\varphi})\big) = \boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta})\,\delta\boldsymbol{\varphi}, \tag{158}$$

with the closed form

$$\mathbf{J}_r(\boldsymbol{\theta}) = \mathbf{I} - \frac{1 - \cos\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^2}\,\lfloor\boldsymbol{\theta}\rfloor_\times + \frac{\|\boldsymbol{\theta}\| - \sin\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^3}\,\lfloor\boldsymbol{\theta}\rfloor_\times^2\,. \tag{159}$$

(6) Left Jacobian. Applying the perturbation on the left instead defines the left Jacobian:

$$\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r(\boldsymbol{\theta})\,\delta\boldsymbol{\theta}\big) = \mathrm{Exp}\big(\mathbf{J}_l(\boldsymbol{\theta})\,\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(\boldsymbol{\theta}). \tag{160}$$

Using the adjoint identity $\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\mathbf{x})\,\mathrm{Exp}(\boldsymbol{\theta})^{-1} = \mathrm{Exp}(\mathbf{R}(\boldsymbol{\theta})\mathbf{x})$ on the middle expression,

$$\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r\delta\boldsymbol{\theta}\big) = \mathrm{Exp}\big(\mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(\boldsymbol{\theta}) \ \Rightarrow\ \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta})\,\mathbf{J}_r(\boldsymbol{\theta}). \tag{161}$$

(7) Sign symmetry. From $\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta})$, replacing $\boldsymbol{\theta} \to -\boldsymbol{\theta}$, $\delta\boldsymbol{\theta} \to -\delta\boldsymbol{\theta}$,

$$\mathrm{Exp}(-\boldsymbol{\theta} - \delta\boldsymbol{\theta}) = \mathrm{Exp}\big(-\mathbf{J}_l(-\boldsymbol{\theta})\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(-\boldsymbol{\theta}) = \mathrm{Exp}\big(-\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(-\boldsymbol{\theta}) \ \Rightarrow\ \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \tag{162}$$

(8) Summary. Collecting (5)–(7),

$$\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}\big), \qquad \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r(\boldsymbol{\theta}), \qquad \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \tag{163}$$

(9) Identity $\lfloor\mathbf{R}\boldsymbol{\omega}\rfloor_\times = \mathbf{R}\,\lfloor\boldsymbol{\omega}\rfloor_\times\,\mathbf{R}^\top$. For any vector $\mathbf{v}$, using $\mathbf{R}(\mathbf{u} \times \mathbf{v}) = (\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v})$ and $\mathbf{R}^\top\mathbf{R} = \mathbf{I}$,

$$\begin{aligned}
\lfloor\mathbf{R}\mathbf{u}\rfloor_\times\,(\mathbf{R}\mathbf{v}) &= \mathbf{R}\big(\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{v}\big) && (164)\\
&= \mathbf{R}\big(\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{R}^\top\mathbf{R}\mathbf{v}\big) && (165)\\
&= \mathbf{R}\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{R}^\top\,(\mathbf{R}\mathbf{v}). && (166)
\end{aligned}$$

Since this holds for all $\mathbf{R}\mathbf{v}$, we conclude $\lfloor\mathbf{R}\mathbf{u}\rfloor_\times = \mathbf{R}\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{R}^\top$.

Homework. Prove $\lfloor\mathbf{R}\boldsymbol{\omega}\rfloor_\times = \mathbf{R}\,\lfloor\boldsymbol{\omega}\rfloor_\times\,\mathbf{R}^\top$.

(10) Identity $\mathrm{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R}\,\mathrm{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top$**.** Using $\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^k = \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times^k \mathbf{R}^\top$ (which follows from (9), since $\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^2 = \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times \mathbf{R}^\top \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times \mathbf{R}^\top = \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times^2 \mathbf{R}^\top$, etc.),

$$\begin{aligned}
\mathrm{Exp}(\mathbf{R}\boldsymbol{\varphi}) &= \mathbf{I} + \lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times + \tfrac{1}{2!}\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^2 + \tfrac{1}{3!}\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^3 + \cdots \\
&= \mathbf{R}\mathbf{R}^\top + \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times \mathbf{R}^\top + \tfrac{1}{2!}\mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times^2 \mathbf{R}^\top + \cdots \\
&= \mathbf{R}\big(\mathbf{I} + \lfloor \boldsymbol{\varphi} \rfloor_\times + \tfrac{1}{2!}\lfloor \boldsymbol{\varphi} \rfloor_\times^2 + \tfrac{1}{3!}\lfloor \boldsymbol{\varphi} \rfloor_\times^3 + \cdots\big)\mathbf{R}^\top && (167) \\
&= \mathbf{R}\,\mathrm{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top. && (168)
\end{aligned}$$

Homework. Prove $\mathrm{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R}\,\mathrm{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top$.

2.5.1 Worked applications

(11) Recovering ${}^C_I\mathbf{R}$ from vector pairs. Given camera frame $\{C\}$ and IMU frame $\{I\}$ with ${}^C\mathbf{v}_1 = {}^C_I\mathbf{R}\,{}^I\mathbf{v}_1$ and ${}^C\mathbf{v}_2 = {}^C_I\mathbf{R}\,{}^I\mathbf{v}_2$, the trick is to stack three independent directions (the third from the cross product) into an invertible matrix:

$$\begin{bmatrix} {}^C\mathbf{v}_1 & {}^C\mathbf{v}_2 & {}^C\mathbf{v}_1 \times {}^C\mathbf{v}_2 \end{bmatrix} = {}^C_I\mathbf{R}\begin{bmatrix} {}^I\mathbf{v}_1 & {}^I\mathbf{v}_2 & {}^I\mathbf{v}_1 \times {}^I\mathbf{v}_2 \end{bmatrix}. \tag{169}$$

(12) Given instead relative rotations ${}^{C_1}_{C_2}\mathbf{R} = {}^C_I\mathbf{R}\,{}^{I_1}_{I_2}\mathbf{R}\,{}^C_I\mathbf{R}^\top$, recover ${}^C_I\mathbf{R}$ from two such pairs.

(13) Discussion. For (11)/(12): What is the minimum number of data pairs needed? What requirements must the pairs satisfy? What if more than the minimum is available?

(14) Averaging noisy rotations. Given n measurements $\mathbf{R}_{m_i} = \mathbf{R}\exp(\lfloor \mathbf{n}_i \rfloor_\times)$ with $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}_{3\times1}, \mathbf{P}_i)$, estimate $\mathbf{R}$ and its covariance.

2.6 Groups: SO(3) and SE(3)

SO(3) — special orthogonal group.

$$SO(3) = \big\{\mathbf{R} \in \mathbb{R}^{3\times3} \,\big|\, \mathbf{R}^\top\mathbf{R} = \mathbf{I}_3,\ \det(\mathbf{R}) = 1\big\}. \tag{170}$$

It satisfies the four group axioms: *closure* ($\mathbf{R}_1\mathbf{R}_2 \in SO(3)$), *identity* ($\mathbf{I}_3 \in SO(3)$), *inverse* ($\mathbf{R}^{-1} = \mathbf{R}^\top \in SO(3)$), and *associativity* ($(\mathbf{R}_1\mathbf{R}_2)\mathbf{R}_3 = \mathbf{R}_1(\mathbf{R}_2\mathbf{R}_3)$).

SE(3) — special Euclidean group.

$$SE(3) = \left\{\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \,\middle|\, \mathbf{R} \in SO(3),\ \mathbf{p} \in \mathbb{R}^3\right\}. \tag{171}$$

The four axioms: *closure*

$$\begin{aligned}
\mathbf{T}_1\mathbf{T}_2 &= \begin{bmatrix} \mathbf{R}_1 & \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix}\begin{bmatrix} \mathbf{R}_2 & \mathbf{p}_2 \\ \mathbf{0} & 1 \end{bmatrix} && (172) \\
&= \begin{bmatrix} \mathbf{R}_1\mathbf{R}_2 & \mathbf{R}_1\mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), && (173)
\end{aligned}$$

identity $\begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3\times1} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}$, *inverse*

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \tag{174}$$

and *associativity* $(\mathbf{T}_1\mathbf{T}_2)\mathbf{T}_3 = \mathbf{T}_1(\mathbf{T}_2\mathbf{T}_3)$.

2.6.1 Matrix Lie groups

A *group* is a set with an operation satisfying closure, associativity, identity, and invertibility. A *Lie group* is a group that is also a smooth (differentiable) manifold whose group operation is smooth. A *matrix Lie group* has matrices as its elements. The maps between the vector space, the Lie algebra, and the Lie group are

$$\mathbb{R}^m \xrightarrow{\wedge\text{ (wedge)}} \mathfrak{so}(n) \xrightarrow{\exp} SO(n), \qquad SO(n) \xrightarrow{\log} \mathfrak{so}(n) \xrightarrow{\vee\text{ (vee)}} \mathbb{R}^m, \tag{175}$$

and for short we write $\mathbf{X} = \mathrm{Exp}(\mathbf{x}) = \exp(\mathbf{x}^\wedge)$ and $\mathbf{x} = \mathrm{Log}(\mathbf{X}) = \log(\mathbf{X})^\vee$. For $SO(3)$ specifically, $\boldsymbol{\theta} \in \mathbb{R}^3 \to \lfloor \boldsymbol{\theta} \rfloor_\times = \boldsymbol{\theta}^\wedge \to \mathbf{R} = \exp(\boldsymbol{\theta}^\wedge)$, and $\mathbf{R} \in SO(3) \to \lfloor \boldsymbol{\theta} \rfloor_\times = \log(\mathbf{R}) \to \boldsymbol{\theta} = \log(\mathbf{R})^\vee$.

2.6.2 Why this (manifold) machinery?

It *simplifies* estimation: the core idea is to use a smooth manifold to model parameters that are awkward in plain vector space, which makes optimization, interpolation, integration, and differentiation easier or even possible (Fig. 3). For example, a raw angle $\theta \in [-\pi, \pi]$ has a wrap-around cut; the circle/sphere is smooth, and the tangent line at a point models the small change / error state used in optimization.

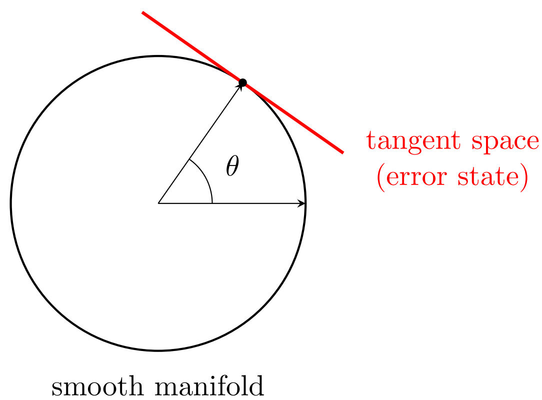


Figure 3: A smooth manifold (here a circle/sphere) avoids the wrap-around of a raw angle; the red tangent line is the local vector space used to model derivatives and error states for optimization.

For $SO(3)$ the left Jacobian also appears directly in the exponential expansion:

$$\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta}) \tag{176}$$

$$= \mathbf{I}_3 + \lfloor \boldsymbol{\theta} \rfloor_\times \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{I}_3 + \boldsymbol{\theta}^\wedge \mathbf{J}_l(\boldsymbol{\theta}), \tag{177}$$

$$\mathbf{J}_l(\boldsymbol{\theta}) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} (\boldsymbol{\theta}^\wedge)^n. \tag{178}$$

2.7 SE(3)

Exp / Log. The SE(3) tangent vector uses $\boldsymbol{\rho}$ for its translation component, $\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^6$, which is *distinct* from the group translation $\mathbf{p}$ in $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$; the two are related by $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$. With the 4×4 algebra element $\boldsymbol{\xi}^{\wedge} = \begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} \\ \mathbf{0}_{1\times 3} & 0 \end{bmatrix}$,

$$\mathbf{T} = \mathrm{Exp}(\boldsymbol{\xi}) = \exp\begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \tag{179}$$

$$= \begin{bmatrix} \exp(\lfloor\boldsymbol{\theta}\rfloor_\times) & \mathbf{J}_l(\boldsymbol{\theta})\,\boldsymbol{\rho} \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \tag{180}$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}, \qquad \mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\,\boldsymbol{\rho}, \tag{181}$$

and inverting,

$$\boldsymbol{\xi} = \mathrm{Log}(\mathbf{T})^{\vee}, \qquad \log(\mathbf{T}) = \log\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \tag{182}$$

$$= \begin{bmatrix} \log(\mathbf{R}) & \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \\ \mathbf{0} & 0 \end{bmatrix} \;\Rightarrow\; \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix}. \tag{183}$$

Discussion. Prove $\exp\left(\begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \right) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}$.

Left/right Jacobians. With $\mathrm{Exp}(\boldsymbol{\xi} + \delta\boldsymbol{\xi}) = \mathrm{Exp}(\boldsymbol{\xi})\,\mathrm{Exp}(\mathbf{J}_R(\boldsymbol{\xi})\delta\boldsymbol{\xi}) = \mathrm{Exp}(\mathbf{J}_L(\boldsymbol{\xi})\delta\boldsymbol{\xi})\,\mathrm{Exp}(\boldsymbol{\xi})$,

$$\mathbf{J}_R(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_R(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_R(\boldsymbol{\xi}) & \mathbf{J}_R(\boldsymbol{\theta}) \end{bmatrix}_{6\times 6}, \quad \mathbf{J}_L(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_L(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_L(\boldsymbol{\xi}) & \mathbf{J}_L(\boldsymbol{\theta}) \end{bmatrix}_{6\times 6}, \tag{184}$$

and to first order, with $\boldsymbol{\xi}^{\curlywedge} = \begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \mathbf{0} \\ \lfloor\boldsymbol{\rho}\rfloor_\times & \lfloor\boldsymbol{\theta}\rfloor_\times \end{bmatrix}_{6\times 6}$,

$$\mathbf{J}_L(\boldsymbol{\xi}) \approx \mathbf{I}_6 + \tfrac{1}{2}\boldsymbol{\xi}^{\curlywedge}, \qquad \mathbf{J}_R(\boldsymbol{\xi}) \approx \mathbf{I}_6 - \tfrac{1}{2}\boldsymbol{\xi}^{\curlywedge}. \tag{185}$$

Homework. Find $\mathbf{Q}_L$ (and $\mathbf{Q}_R$).

Error states: SE(3) vs SO(3) + $\mathbb{R}^3$. Perturb $\mathbf{T}$ on the right by the SE(3) tangent $\delta\boldsymbol{\xi} = [\delta\boldsymbol{\theta}^\top,\ \delta\boldsymbol{\rho}^\top]^\top$. Expanding with $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2}\lfloor\delta\boldsymbol{\theta}\rfloor_\times$ and $\delta\mathbf{R} = \mathrm{Exp}(\delta\boldsymbol{\theta})$,

$$\mathbf{T}\,\delta\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathrm{Exp}(\delta\boldsymbol{\theta}) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \tag{186}$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & (\mathbf{I} + \frac{1}{2}\lfloor\delta\boldsymbol{\theta}\rfloor_\times)\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \tag{187}$$

$$\approx \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \tag{188}$$

$$= \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{R}\delta\boldsymbol{\rho} + \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \tag{189}$$

If instead we treat the pose as $SO(3) + \mathbb{R}^3$ (rotation and translation perturbed separately, $\mathbf{p} \to \mathbf{p} + \delta\mathbf{p}$),

$$\mathbf{T} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \tag{190}$$

$$= \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \mathbf{R}\mathbf{R}^\top\delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \tag{191}$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \mathbf{R}^\top\delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \tag{192}$$

Comparing the two conventions gives the relation between the error states (SE(3) tangent translation $\delta\boldsymbol{\rho}$ vs. naive position error $\delta\mathbf{p}$),

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}_{SO(3)+\mathbb{R}^3} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}_{SE(3)} \Rightarrow \begin{cases} \delta\boldsymbol{\rho} = \mathbf{R}^\top\delta\mathbf{p} \\ \delta\mathbf{R}_{SE(3)} = \delta\mathbf{R} \end{cases}. \tag{193}$$

Adjoint. The adjoint transfers algebra elements across the group:

$$SO(3): \quad \mathbf{R}\,\mathrm{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top = \mathrm{Exp}(\mathbf{R}\boldsymbol{\varphi}) \;\Leftrightarrow\; \mathrm{Ad}_{\mathbf{R}} = \mathbf{R}, \qquad SE(3): \quad \mathbf{T}\,\mathrm{Exp}(\boldsymbol{\xi})\,\mathbf{T}^{-1} = \mathrm{Exp}(\mathrm{Ad}_{\mathbf{T}}\,\boldsymbol{\xi}), \tag{194}$$

with $\mathrm{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \lfloor\mathbf{p}\rfloor_\times \mathbf{R} & \mathbf{R} \end{bmatrix}_{6\times 6}$. The adjoint is itself a Lie group, with $\mathrm{Ad}_{\mathbf{T}} = \exp(\boldsymbol{\xi}^\curlywedge)$ and $\log(\mathrm{Ad}_{\mathbf{T}})^\vee = \boldsymbol{\xi}$.

Homework. Prove $\mathbf{T}\,\mathrm{Exp}(\boldsymbol{\xi})\,\mathbf{T}^{-1} = \mathrm{Exp}(\mathrm{Ad}_{\mathbf{T}}\,\boldsymbol{\xi})$.

2.7.1 More operations

(1) Interpolation. For ${}^G_{t_1}\mathbf{T}, {}^G_{t_2}\mathbf{T}$, $t \in [t_1, t_2]$, $\lambda = \frac{t-t_1}{t_2-t_1}$: in $SE(3)$, with ${}^{t_1}_{t_2}\boldsymbol{\xi} = \mathrm{Log}({}^G_{t_1}\mathbf{T}^{-1}\,{}^G_{t_2}\mathbf{T})$,

$${}^G_t\mathbf{T} = {}^G_{t_1}\mathbf{T}\,\mathrm{Exp}\big(\lambda\,{}^{t_1}_{t_2}\boldsymbol{\xi}\big); \tag{195}$$

in $SO(3) + \mathbb{R}^3$, the rotation interpolates on the manifold and the position linearly:

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R}\,\mathrm{Exp}\big(\lambda\,\mathrm{Log}({}^G_{t_1}\mathbf{R}^\top\,{}^G_{t_2}\mathbf{R})\big), \qquad {}^G\mathbf{p}_t = {}^G\mathbf{p}_{t_1} + \lambda({}^G\mathbf{p}_{t_2} - {}^G\mathbf{p}_{t_1}). \tag{196}$$

(2) Time derivative. For $SE(3)$, with $\mathbf{T} = \mathbf{T}\,\mathrm{Exp}\big([\delta\boldsymbol{\theta};\delta\boldsymbol{\rho}]^\wedge\big)$ and body velocity, expand the limit using $\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) - \mathbf{I}_3 \approx \lfloor\delta\boldsymbol{\theta}\rfloor_\times$ and $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2}\lfloor\delta\boldsymbol{\theta}\rfloor_\times$:

$$\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t} = \lim_{\delta t\to 0}\frac{\mathbf{T}(t+\delta t) - \mathbf{T}(t)}{\delta t} \tag{197}$$

$$= \lim_{\delta t\to 0}\frac{1}{\delta t}\left(\begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho}\\ \mathbf{0} & 1\end{bmatrix} - \begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\right) \tag{198}$$

$$= \lim_{\delta t\to 0}\frac{1}{\delta t}\begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) - \mathbf{I}_3 & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho}\\ \mathbf{0} & 0\end{bmatrix} \tag{199}$$

$$= \lim_{\delta t\to 0}\frac{1}{\delta t}\begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\lfloor\delta\boldsymbol{\theta}\rfloor_\times & \delta\boldsymbol{\rho}\\ \mathbf{0} & 0\end{bmatrix} \tag{200}$$

$$= \begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\lfloor\boldsymbol{\omega}\rfloor_\times & \mathbf{v}\\ \mathbf{0} & 0\end{bmatrix} \tag{201}$$

$$= \begin{bmatrix}\mathbf{R}\lfloor\boldsymbol{\omega}\rfloor_\times & \mathbf{R}\mathbf{v}\\ \mathbf{0} & 0\end{bmatrix}, \tag{202}$$

where $\mathbf{v}$ is the body velocity. For $SO(3)+\mathbb{R}^3$ the same computation with

$$\mathbf{T} = \begin{bmatrix}\mathbf{R}\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) & \mathbf{p}+\delta\mathbf{p}\\ \mathbf{0} & 1\end{bmatrix} \tag{203}$$

gives

$$\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t} = \begin{bmatrix}\mathbf{R}\lfloor\boldsymbol{\omega}\rfloor_\times & {}^{G}\mathbf{v}\\ \mathbf{0} & 0\end{bmatrix}, \tag{204}$$

with ${}^{G}\mathbf{v}$ the global velocity.

(3) Calibration. For IMU frame $\{I\}$ and camera frame $\{C\}$,

$$ {}^{C_1}_{C_2}\mathbf{T} = {}^{C}_{I}\mathbf{T}\,{}^{I_1}_{I_2}\mathbf{T}\,{}^{C}_{I}\mathbf{T}^{-1}, \qquad \mathrm{Exp}({}^{C_1}_{C_2}\boldsymbol{\xi}) = \mathrm{Exp}\big(\mathrm{Ad}_{({}^{C}_{I}\mathbf{T})}\,{}^{I_1}_{I_2}\boldsymbol{\xi}\big) \;\Rightarrow\; {}^{C_1}_{C_2}\boldsymbol{\xi} = \mathrm{Ad}_{({}^{C}_{I}\mathbf{T})}\,{}^{I_1}_{I_2}\boldsymbol{\xi}. \tag{205}$$

Homework. Compute the Jacobians $\partial\,{}^{C_1}_{C_2}\boldsymbol{\xi}/\partial\,{}^{I_1}_{I_2}\boldsymbol{\xi}$ and $\partial\,{}^{C_1}_{C_2}\boldsymbol{\xi}/\partial\,{}^{C}_{I}\boldsymbol{\xi}$ (and the corresponding $SO(3)+\mathbb{R}^3$ versions).

2.8 SE$_2$(3)

Augmenting the pose with a velocity column gives the 5×5 group element

$$\boldsymbol{\Upsilon} = \begin{bmatrix}\mathbf{R} & \mathbf{p} & \mathbf{v}\\ \mathbf{0}_{1\times3} & 1 & 0\\ \mathbf{0}_{1\times3} & 0 & 1\end{bmatrix}, \qquad \boldsymbol{\xi} = \begin{bmatrix}\boldsymbol{\theta}\\ \boldsymbol{\rho}\\ \boldsymbol{\nu}\end{bmatrix} \to \boldsymbol{\xi}^\wedge = \begin{bmatrix}\lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} & \boldsymbol{\nu}\\ \mathbf{0} & 0 & 0\\ \mathbf{0} & 0 & 0\end{bmatrix}_{5\times5}, \tag{206}$$

where (as in SE(3)) the tangent components $\boldsymbol{\rho},\boldsymbol{\nu}$ are distinct from the group position/velocity $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$, $\mathbf{v} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}$. The exponential, logarithm, and adjoint follow the same pattern as $SE(3)$:

$$\mathrm{Exp}(\boldsymbol{\xi}) = \begin{bmatrix}\mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}\\ \mathbf{0} & 1 & 0\\ \mathbf{0} & 0 & 1\end{bmatrix} = \begin{bmatrix}\mathbf{R} & \mathbf{p} & \mathbf{v}\\ \mathbf{0} & 1 & 0\\ \mathbf{0} & 0 & 1\end{bmatrix}, \tag{207}$$

$$\mathrm{Log}(\boldsymbol{\Upsilon})^\vee = \begin{bmatrix}\boldsymbol{\theta}\\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p}\\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{v}\end{bmatrix}, \qquad \mathrm{Ad}_{\boldsymbol{\Upsilon}} = \begin{bmatrix}\mathbf{R} & \mathbf{0} & \mathbf{0}\\ \lfloor\mathbf{p}\rfloor_\times\mathbf{R} & \mathbf{R} & \mathbf{0}\\ \lfloor\mathbf{v}\rfloor_\times\mathbf{R} & \mathbf{0} & \mathbf{R}\end{bmatrix}. \tag{208}$$

2.9 Quaternions

Quaternions are presented last because $SO(3)$, $SE(3)$, and the Lie-group view already cover most needs. There are two main conventions — *Hamilton* (right-handed) and *JPL* (left-handed); we use the **Hamilton** convention. For a thorough treatment see Solà [6] (Hamilton) and Trawny & Roumeliotis [5] (JPL).

A quaternion and its imaginary units satisfy

$$Q = q_w + q_x i + q_y j + q_z k, \qquad i^2 = j^2 = k^2 = ijk = -1, \quad ij = k,\ jk = i,\ ki = j, \tag{209}$$

written as the 4×1 vector $\bar{q} = [q_w,\ q_x,\ q_y,\ q_z]^\top = [q_w,\ \mathbf{q}_v^\top]^\top$.

Product. The quaternion product is bilinear and can be written with left/right matrices:

$$\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p})\,\bar{q} = \mathcal{R}(\bar{q})\,\bar{p}, \tag{210}$$

$$\mathcal{L}(\bar{p}) = p_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{p}_v^\top \\ \mathbf{p}_v & \lfloor \mathbf{p}_v \rfloor_\times \end{bmatrix}, \tag{211}$$

$$\mathcal{R}(\bar{q}) = q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -\lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix}. \tag{212}$$

Identity, unit quaternion, inverse. The identity is $\bar{q} = [1,\ \mathbf{0}_{3\times1}^\top]^\top$. A rotation by angle θ about unit axis $\mathbf{k}$ is the unit quaternion

$$\bar{q} = \begin{bmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\,\mathbf{k} \end{bmatrix}, \qquad \bar{q}^{-1} = \begin{bmatrix} \cos\frac{\theta}{2} \\ -\sin\frac{\theta}{2}\,\mathbf{k} \end{bmatrix}, \qquad \mathbf{R}_{12} = \mathbf{R}_1\mathbf{R}_2 \ \leftrightarrow\ \bar{q}_{12} = \bar{q}_1 \otimes \bar{q}_2. \tag{213}$$

Why $\theta/2$? — rotating a vector. With global frame $\{G\}$ and IMU frame $\{I\}$, rotate ${}^I\mathbf{p}$ via the sandwich product and convert to matrices with $\bar{p} \otimes \bar{q} = \mathcal{R}(\bar{q})\bar{p}$ and $\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p})\bar{q}$:

$$\begin{bmatrix} 0 \\ {}^G\mathbf{p} \end{bmatrix} = {}^G_I\bar{q} \otimes \begin{bmatrix} 0 \\ {}^I\mathbf{p} \end{bmatrix} \otimes {}^G_I\bar{q}^{-1} \tag{214}$$

$$= \mathcal{R}({}^G_I\bar{q}^{-1})\,\mathcal{L}({}^G_I\bar{q}) \begin{bmatrix} 0 \\ {}^I\mathbf{p} \end{bmatrix}. \tag{215}$$

Writing $\bar{q} = [q_w, \mathbf{q}_v^\top]^\top$ and multiplying the two matrices,

$$\begin{aligned} \mathcal{R}(\bar{q}^{-1})\mathcal{L}(\bar{q}) &= \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & \mathbf{q}_v^\top \\ -\mathbf{q}_v & \lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix}\right)\left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & \lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix}\right) \\ &= q_w^2 \mathbf{I}_4 + 2q_w \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & \lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix} + \begin{bmatrix} \mathbf{q}_v^\top \mathbf{q}_v & \mathbf{0} \\ \mathbf{0} & -\mathbf{q}_v\mathbf{q}_v^\top + \lfloor \mathbf{q}_v \rfloor_\times^2 \end{bmatrix}, \end{aligned} \tag{216}$$

whose lower-right 3×3 block (acting on ${}^I\mathbf{p}$) gives

$${}^G\mathbf{p} = \left(q_w^2 \mathbf{I}_3 + 2q_w \lfloor \mathbf{q}_v \rfloor_\times + \mathbf{q}_v\mathbf{q}_v^\top + \lfloor \mathbf{q}_v \rfloor_\times^2\right){}^I\mathbf{p}. \tag{217}$$

Using $\lfloor \mathbf{q}_v \rfloor_\times^2 = \mathbf{q}_v\mathbf{q}_v^\top - (\mathbf{q}_v^\top \mathbf{q}_v)\mathbf{I}_3$ and the unit-norm condition $q_w^2 + \mathbf{q}_v^\top\mathbf{q}_v = 1$,

$$q_w^2 \mathbf{I}_3 + \mathbf{q}_v\mathbf{q}_v^\top + \lfloor \mathbf{q}_v \rfloor_\times^2 = q_w^2 \mathbf{I}_3 + (\mathbf{q}_v^\top\mathbf{q}_v)\mathbf{I}_3 + 2\lfloor \mathbf{q}_v \rfloor_\times^2 \tag{218}$$

$$= \mathbf{I}_3 + 2\lfloor \mathbf{q}_v \rfloor_\times^2, \tag{219}$$

so

$$ {}^{G}\mathbf{p} = \left(\mathbf{I}_3 + 2q_w \lfloor \mathbf{q}_v \rfloor_\times + 2 \lfloor \mathbf{q}_v \rfloor_\times^2 \right) {}^{I}\mathbf{p}. \tag{220} $$

Finally substitute $q_w = \cos\frac{\theta}{2}$, $\mathbf{q}_v = \sin\frac{\theta}{2}\mathbf{k}$, so $2q_w \lfloor \mathbf{q}_v \rfloor_\times = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2} \lfloor \mathbf{k} \rfloor_\times = \sin\theta \lfloor \mathbf{k} \rfloor_\times$ and $2 \lfloor \mathbf{q}_v \rfloor_\times^2 = 2\sin^2\frac{\theta}{2} \lfloor \mathbf{k} \rfloor_\times^2 = (1-\cos\theta) \lfloor \mathbf{k} \rfloor_\times^2$:

$$ {}^{G}\mathbf{p} = \left(\mathbf{I}_3 + \sin\theta \lfloor \mathbf{k} \rfloor_\times + (1-\cos\theta) \lfloor \mathbf{k} \rfloor_\times^2 \right) {}^{I}\mathbf{p} \tag{221} $$
$$ = {}^{G}_{I}\mathbf{R}\, {}^{I}\mathbf{p} \tag{222} $$
$$ = \mathrm{Exp}(\theta\mathbf{k})\, {}^{I}\mathbf{p}, \tag{223} $$

recovering Rodrigues' formula — which is why the half-angle $\theta/2$ appears.

Derivative. With $\delta\boldsymbol{\theta} = \delta\theta\,\mathbf{k}$ and small-angle $\delta\bar{q} = [\cos\frac{\delta\theta}{2},\ \sin\frac{\delta\theta}{2}\mathbf{k}^\top]^\top \approx [1,\ \frac{1}{2}\delta\boldsymbol{\theta}^\top]^\top$,

$$ {}^{G}_{t}\dot{\bar{q}} = \lim_{\Delta t \to 0} \frac{{}^{G}_{t+\Delta t}\bar{q} - {}^{G}_{t}\bar{q}}{\Delta t} \tag{224} $$
$$ = \lim_{\Delta t \to 0} \frac{{}^{G}_{t}\bar{q} \otimes \begin{bmatrix} \cos\frac{\delta\theta}{2} \\ \sin\frac{\delta\theta}{2}\mathbf{k} \end{bmatrix} - {}^{G}_{t}\bar{q} \otimes \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}}{\Delta t} \tag{225} $$
$$ = \lim_{\Delta t \to 0} \frac{{}^{G}_{t}\bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2}\delta\boldsymbol{\theta} \end{bmatrix}}{\Delta t} \tag{226} $$
$$ = {}^{G}_{t}\bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2}\boldsymbol{\omega} \end{bmatrix} \tag{227} $$
$$ = \tfrac{1}{2}\,\mathcal{R}\left(\begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}\right) {}^{G}_{t}\bar{q} \tag{228} $$
$$ = \tfrac{1}{2} \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -\lfloor \boldsymbol{\omega} \rfloor_\times \end{bmatrix} {}^{G}_{t}\bar{q} \tag{229} $$
$$ = \tfrac{1}{2}\,\Omega(\boldsymbol{\omega})\, {}^{G}_{t}\bar{q}, \tag{230} $$

where $\Omega(\boldsymbol{\omega}) \triangleq \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -\lfloor \boldsymbol{\omega} \rfloor_\times \end{bmatrix}$.

Integration. Given ${}^{G}_{t}\bar{q}$, $\boldsymbol{\omega}$, δt, with $\delta\boldsymbol{\theta} = \boldsymbol{\omega}\,\delta t = \delta\theta\,\mathbf{k}$ and $\delta\bar{q} = [\cos\frac{\delta\theta}{2},\ \sin\frac{\delta\theta}{2}\mathbf{k}^\top]^\top$,

$$ {}^{G}_{t+\delta t}\bar{q} = {}^{G}_{t}\bar{q} \otimes \delta\bar{q}. \tag{231} $$

3 Lecture 3: Camera Model, Triangulation, Point/Line/Plane

Topics. (1) camera measurements; (2) essential matrix / triangulation; (3) point / line / plane representations.

3.1 The camera measurement model

A camera produces pixel measurements that depend on the camera pose and the observed scene:

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c, \tag{232}$$

where $\mathbf{z}_c$ are the pixel measurements, $\mathbf{x}$ is the state vector (camera pose and point feature), $\mathbf{h}_c(\cdot)$ is the camera model, and $\mathbf{n}_c$ is the pixel noise. The geometry is shown in Fig. 4.

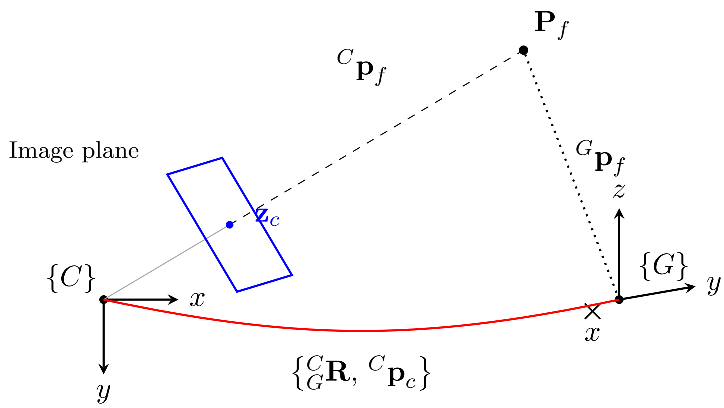


Figure 4: Camera measurement geometry: the feature $\mathbf{P}_f$ projects through the image plane to the pixel $\mathbf{z}_c$ in the camera frame $\{C\}$; the global frame $\{G\}$ and the extrinsic $\{{}^{C}_{G}\mathbf{R},\, {}^{C}\mathbf{p}_c\}$ relate the two views of the same point (${}^{C}\mathbf{p}_f$ vs. ${}^{G}\mathbf{p}_f$).

Simplest pinhole model. With the feature in the camera frame ${}^{C}\mathbf{p}_f = [x,\, y,\, z]^\top$, the normalized projection is

$$\mathbf{z}_n = \begin{bmatrix} u_n \\ v_n \end{bmatrix} = \begin{bmatrix} x/z \\ y/z \end{bmatrix}, \qquad \mathbf{z}_n = \mathbf{h}_p({}^{C}\mathbf{p}_f) + \mathbf{n}_c, \quad {}^{C}\mathbf{p}_f = {}^{C}_{G}\mathbf{R}^\top\big({}^{G}\mathbf{p}_f - {}^{G}\mathbf{p}_c\big). \tag{233}$$

Linearizing the measurement, $\tilde{\mathbf{z}}_n = \mathbf{H}_c\, \tilde{\mathbf{x}} + \mathbf{n}_c$, with the residual

$$\tilde{\mathbf{z}}_n = \mathbf{z}_n - \mathbf{h}_p({}^{C}\hat{\mathbf{p}}_f) = \begin{bmatrix} u_n \\ v_n \end{bmatrix} - \begin{bmatrix} \hat{x}/\hat{z} \\ \hat{y}/\hat{z} \end{bmatrix}. \tag{234}$$

The state and its error state are

$$\mathbf{x} : \big\{{}^{C}_{G}\mathbf{R},\, {}^{G}\mathbf{p}_c,\, {}^{G}\mathbf{p}_f\big\}, \quad {}^{C}_{G}\mathbf{R} = {}^{C}_{G}\hat{\mathbf{R}}\ \mathrm{Exp}(\delta\boldsymbol{\theta}), \quad {}^{G}\mathbf{p}_c = {}^{G}\hat{\mathbf{p}}_c + {}^{G}\tilde{\mathbf{p}}_c, \quad {}^{G}\mathbf{p}_f = {}^{G}\hat{\mathbf{p}}_f + {}^{G}\tilde{\mathbf{p}}_f, \tag{235}$$

with error state $\tilde{\mathbf{x}} = [\,\delta\boldsymbol{\theta}^\top,\, {}^{G}\tilde{\mathbf{p}}_c^\top,\, {}^{G}\tilde{\mathbf{p}}_f^\top\,]^\top$.

3.2 The measurement Jacobian

By the chain rule, factoring every block through ${}^C\tilde{\mathbf{p}}_f$,

$$\begin{aligned}
\tilde{\mathbf{z}}_n &= \mathbf{H}_c\,\tilde{\mathbf{x}} + \mathbf{n}_c &(236)\\
&= \begin{bmatrix} \dfrac{\partial \tilde{\mathbf{z}}_n}{\partial \delta\boldsymbol{\theta}} & \dfrac{\partial \tilde{\mathbf{z}}_n}{\partial {}^G\tilde{\mathbf{p}}_c} & \dfrac{\partial \tilde{\mathbf{z}}_n}{\partial {}^G\tilde{\mathbf{p}}_f} \end{bmatrix} \tilde{\mathbf{x}} + \mathbf{n}_c &(237)\\
&= \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^C\tilde{\mathbf{p}}_f} \begin{bmatrix} \dfrac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \delta\boldsymbol{\theta}} & \dfrac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_c} & \dfrac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_f} \end{bmatrix} \tilde{\mathbf{x}} + \mathbf{n}_c. &(238)
\end{aligned}$$

The projection Jacobian, writing ${}^C\mathbf{p}_f = [c_x,\, c_y,\, c_z]^\top$, is

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix}, \qquad \frac{\partial \mathbf{z}_n}{\partial {}^C\mathbf{p}_f} = \begin{bmatrix} \dfrac{1}{c_z} & 0 & -\dfrac{c_x}{c_z^2} \\ 0 & \dfrac{1}{c_z} & -\dfrac{c_y}{c_z^2} \end{bmatrix} = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix}. \tag{239}$$

Perturbing ${}^C\mathbf{p}_f$. From ${}^C\mathbf{p}_f = {}^C_G\mathbf{R}^\top({}^G\mathbf{p}_f - {}^G\mathbf{p}_c)$ with ${}^C_G\mathbf{R} = {}^C_G\hat{\mathbf{R}}\,\mathrm{Exp}(\delta\boldsymbol{\theta})$ and $\mathrm{Exp}(\delta\boldsymbol{\theta})^\top \approx \mathbf{I} - \lfloor \delta\boldsymbol{\theta} \rfloor_\times$,

$$\begin{aligned}
{}^C\hat{\mathbf{p}}_f + {}^C\tilde{\mathbf{p}}_f &= \big({}^C_G\hat{\mathbf{R}}\,\mathrm{Exp}(\delta\boldsymbol{\theta})\big)^\top \big({}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c - {}^G\tilde{\mathbf{p}}_c\big) &(240)\\
&= \big(\mathbf{I} - \lfloor \delta\boldsymbol{\theta} \rfloor_\times\big)\, {}^C_G\hat{\mathbf{R}}^\top \big({}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c - {}^G\tilde{\mathbf{p}}_c\big) &(241)\\
&\approx {}^C_G\hat{\mathbf{R}}^\top \big({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c\big) - \lfloor \delta\boldsymbol{\theta} \rfloor_\times\, {}^C_G\hat{\mathbf{R}}^\top \big({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c\big) \\
&\quad + {}^C_G\hat{\mathbf{R}}^\top \big({}^G\tilde{\mathbf{p}}_f - {}^G\tilde{\mathbf{p}}_c\big). &(242)
\end{aligned}$$

Using $-\lfloor \delta\boldsymbol{\theta} \rfloor_\times\, \mathbf{a} = \lfloor \mathbf{a} \rfloor_\times\, \delta\boldsymbol{\theta}$ with $\mathbf{a} = {}^C\hat{\mathbf{p}}_f = {}^C_G\hat{\mathbf{R}}^\top({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)$ and matching hat/tilde terms,

$$\begin{aligned}
{}^C\hat{\mathbf{p}}_f + {}^C\tilde{\mathbf{p}}_f &= \underbrace{{}^C_G\hat{\mathbf{R}}^\top \big({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c\big)}_{{}^C\hat{\mathbf{p}}_f} + \Big\lfloor {}^C_G\hat{\mathbf{R}}^\top({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) \Big\rfloor_\times \delta\boldsymbol{\theta} \\
&\quad - {}^C_G\hat{\mathbf{R}}^\top\, {}^G\tilde{\mathbf{p}}_c + {}^C_G\hat{\mathbf{R}}^\top\, {}^G\tilde{\mathbf{p}}_f, &(243)
\end{aligned}$$

so that, with ${}^C\hat{\mathbf{p}}_f = {}^C_G\hat{\mathbf{R}}^\top({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)$, the three blocks are

$$\frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \delta\boldsymbol{\theta}} = \big\lfloor {}^C\hat{\mathbf{p}}_f \big\rfloor_\times, \qquad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_c} = -{}^C_G\hat{\mathbf{R}}^\top, \qquad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_f} = {}^C_G\hat{\mathbf{R}}^\top. \tag{244}$$

Therefore $\tilde{\mathbf{z}}_n = \mathbf{H}_c\,\tilde{\mathbf{x}} + \mathbf{n}_c$ with

$$\mathbf{H}_c = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix} \begin{bmatrix} \lfloor {}^C\hat{\mathbf{p}}_f \rfloor_\times & -{}^C_G\hat{\mathbf{R}}^\top & {}^C_G\hat{\mathbf{R}}^\top \end{bmatrix}. \tag{245}$$

3.3 A few important discussions

(1) SLAM. The pose $\{{}^C_G\mathbf{R}, {}^G\mathbf{p}_c\}$ is the localization of the camera, and ${}^G\mathbf{p}_f$ is a 3D feature of the environment. Solving them together means we solve localization and mapping together $\Rightarrow$ *Simultaneous Localization And Mapping* (SLAM).

(2) PnP / P3P — known feature, solve pose. If ${}^G\mathbf{p}_f$ is known, given (u, v), how do we solve $\{{}^C_G\mathbf{R}, {}^C\mathbf{p}_c\}$? Here $\mathbf{x} : \{{}^C_G\mathbf{R}, {}^C\mathbf{p}_c\}$ is 6-dof, while each measurement $\mathbf{z}_n = \mathbf{h}({}^C_G\mathbf{R}, {}^C\mathbf{p}_c) + \mathbf{n}_c$ contributes 2 dof, so we need at least 3 points ${}^C\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$ to solve $\{{}^C\mathbf{R}, {}^C\mathbf{p}_c\}$.

- 3 pairs ${}^C\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$, **P3P** (perspective-3-point), i.e. 2+1 points: use 3D↔3D correspondences to determine the same 3D pose.
- n pairs of 3D↔3D correspondences, **PnP** (perspective-n-point).
- **RANSAC** (random sample consensus) for outlier rejection.

This is the *structure* → *motion* direction (known structure, solve motion).

(3) Triangulation — known poses, solve feature. If $\{{}^C_G\mathbf{R}, {}^C\mathbf{p}_c\}$ is known, given (u_1, v_1), (u_2, v_2), solve $\mathbf{P}_f$ (Fig. 5). Each pixel gives a unit bearing and an unknown range r_i:

$$(u_1, v_1) \to {}^{C_1}\mathbf{b}_1 = \frac{[u_1, v_1, 1]^\top}{\|[u_1, v_1, 1]^\top\|}, \quad (u_2, v_2) \to {}^{C_2}\mathbf{b}_2 = \frac{[u_2, v_2, 1]^\top}{\|[u_2, v_2, 1]^\top\|}, \tag{246}$$

$${}^{C_1}\mathbf{p}_f = {}^{C_1}\mathbf{b}_1\, r_1, \quad {}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{b}_2\, r_2, \tag{247}$$

with r_1, r_2 unknown. From ${}^{C_1}\mathbf{p}_f = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f + {}^{C_1}\mathbf{p}_{C_2}$ we obtain ${}^{C_1}\mathbf{b}_1\, r_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + {}^{C_1}\mathbf{p}_{C_2}$.

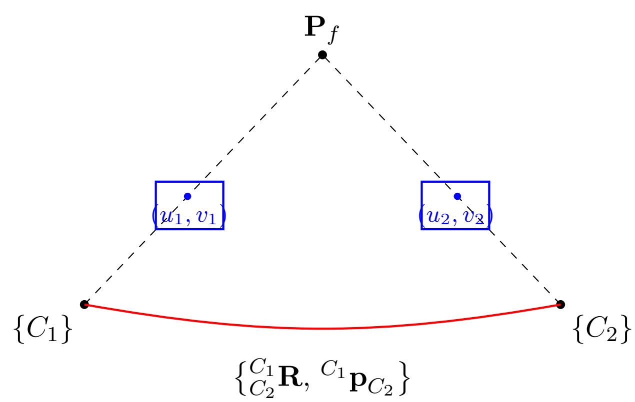


Figure 5: Two-view triangulation: bearings ${}^{C_1}\mathbf{b}_1, {}^{C_2}\mathbf{b}_2$ from pixels $(u_1, v_1), (u_2, v_2)$ and the known relative pose $\{{}^{C_1}_{C_2}\mathbf{R}, {}^{C_1}\mathbf{p}_{C_2}\}$ fix the feature $\mathbf{P}_f$ up to the two ranges r_1, r_2.

Way 1 (stacked linear system). Collect the two ranges:

$${}^{C_1}\mathbf{b}_1\, r_1 - {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 = {}^{C_1}\mathbf{p}_{C_2} \;\Rightarrow\; \underbrace{\begin{bmatrix} {}^{C_1}\mathbf{b}_1 & -{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2 \end{bmatrix}}_{3\times 2} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = {}^{C_1}\mathbf{p}_{C_2}, \tag{248}$$

a 3×2 system; solving r_1, r_2 gives ${}^{C_1}\mathbf{p}_f, {}^{C_2}\mathbf{p}_f$.

> **Degenerate case.** If the motion is pure rotation (no translation), then ${}^{C_1}\mathbf{b}_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2$ and ${}^{C_1}\mathbf{p}_{C_2} = \mathbf{0}_{3\times 1}$, so $r_1 = r_2$ and there is no way to solve r. For a monocular camera, pure rotation is dangerous.

Way 2 (left null space). With ${}^{C_1}\mathbf{b}_1\, r_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + {}^{C_1}\mathbf{p}_{C_2}$, collect into $\mathbf{B}^\perp = [\,\mathbf{b}^{\perp_1}\ \ \mathbf{b}^{\perp_2}\,]$ two directions orthogonal to ${}^{C_1}\mathbf{b}_1$ (${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_1} = 0$, ${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_2} = 0$); projecting removes r_1:

$$\mathbf{B}^{\perp\top}\,{}^{C_1}\mathbf{b}_1\, r_1 = \mathbf{B}^{\perp\top}\big({}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + {}^{C_1}\mathbf{p}_{C_2}\big) = \mathbf{0}_{2\times 1} \;\Rightarrow\; \underbrace{\mathbf{B}^{\perp\top}\,{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2}_{2\times 1}\, r_2 = -\underbrace{\mathbf{B}^{\perp\top}\,{}^{C_1}\mathbf{p}_{C_2}}_{2\times 1}. \tag{249}$$

(4) Monocular triangulation is up to scale. If $\{{}^{C}_{G}\mathbf{R}\}$ is known but for ${}^{C_1}\mathbf{p}_{C_2}$ we only know the direction $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}$ (with $\|{}^{C_1}\mathbf{p}_{C_2}\|$ unknown), can we triangulate ${}^{C}\mathbf{p}_f$? Let $\|{}^{C_1}\mathbf{p}_{C_2}\| = \rho$, then ${}^{C_1}\mathbf{p}_f = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f + {}^{C_1}\mathbf{p}_{C_2}$ gives

$$
{}^{C_1}\mathbf{b}_1\, r_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + \rho\,\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}. \tag{250}
$$

With $\mathbf{A} = [\mathbf{v}^{\perp_1}\ \ \mathbf{v}^{\perp_2}]^\top$ built from $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_1} = 0$, $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_2} = 0$, projecting kills the translation term:

$$
\mathbf{A}\,{}^{C_1}\mathbf{b}_1\, r_1 = \mathbf{A}\,{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + \rho\,\underbrace{\mathbf{A}\,\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}}_{\mathbf{0}} \tag{251}
$$

$$
\Rightarrow\quad \begin{bmatrix}\mathbf{A}\,{}^{C_1}\mathbf{b} & -\mathbf{A}\,{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}\end{bmatrix}\begin{bmatrix} r_1 \\ r_2\end{bmatrix} = \mathbf{0}. \tag{252}
$$

Hence r_1, r_2 are up to scale: a monocular camera does not have scale. The above uses *motion to infer structure.*

(5) Essential matrix — solve the motion. Given $\{u_1, v_1\}, \{u_2, v_2\}$, can we solve $\{{}^{C}_{G}\mathbf{R}, {}^{C}\mathbf{p}_c\}$? The three vectors ${}^{C_1}\mathbf{p}_f$, ${}^{C_2}\mathbf{p}_f$, ${}^{C_1}\mathbf{p}_{C_2}$ are coplanar (Fig. 6), so ${}^{C_1}\mathbf{p}_{C_2} \times {}^{C_1}\mathbf{p}_f \perp {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f$:

$$
\left(\lfloor {}^{C_1}\mathbf{p}_{C_2}\rfloor_\times\, {}^{C_1}\mathbf{p}_f\right)^\top {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f = 0 \ \Rightarrow\ {}^{C_1}\mathbf{p}_f^\top\ \lfloor {}^{C_1}\mathbf{p}_{C_2}\rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f = 0. \tag{253}
$$

With ${}^{C_1}\mathbf{p}_f = {}^{C_1}\mathbf{b}\, r_1$, ${}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{b}\, r_2$,

$$
\boxed{{}^{C_1}\mathbf{b}^\top\ \lfloor {}^{C_1}\mathbf{p}_{C_2}\rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b} = 0, \qquad \mathbf{E} = \lfloor {}^{C_1}\mathbf{p}_{C_2}\rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\quad \text{(essential matrix)}.} \tag{254}
$$

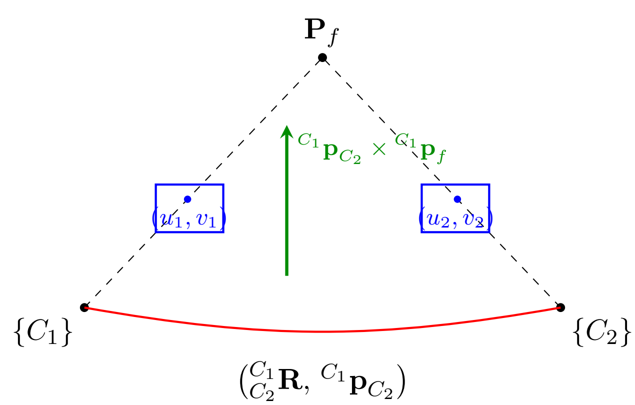


Figure 6: Epipolar geometry: ${}^{C_1}\mathbf{p}_f$, ${}^{C_2}\mathbf{p}_f$, and the baseline ${}^{C_1}\mathbf{p}_{C_2}$ are coplanar, giving the essential-matrix constraint ${}^{C_1}\mathbf{b}^\top \mathbf{E}\,{}^{C_2}\mathbf{b} = 0$.

Why is $\mathbf{E}$ 5-dof? Each pair $(u_1, v_1), (u_2, v_2)$ provides 1 constraint, so we need at least **5 pairs of points** to solve $\mathbf{E} \Rightarrow {}^{C_1}_{C_2}\mathbf{R}$ (3 dof) and $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}$ (2 dof, direction only). When selecting pairs of points, RANSAC (random sample consensus) comes in. Finally:

$$
\text{normalized coordinates:}\ \ {}^{C_1}\mathbf{b}^\top \lfloor {}^{C_1}\mathbf{p}_{C_2}\rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b} = 0\ \ \text{(essential matrix)}, \tag{255}
$$

$$
\text{pixel coordinates:}\ \ \text{fundamental matrix.} \tag{256}
$$

This is the *motion from structure* direction.

3.4 The full intrinsic calibration model

The calibration geometry is the same as Fig. 4 (repeated here as Fig. 7). The camera model is a composition of four functions.

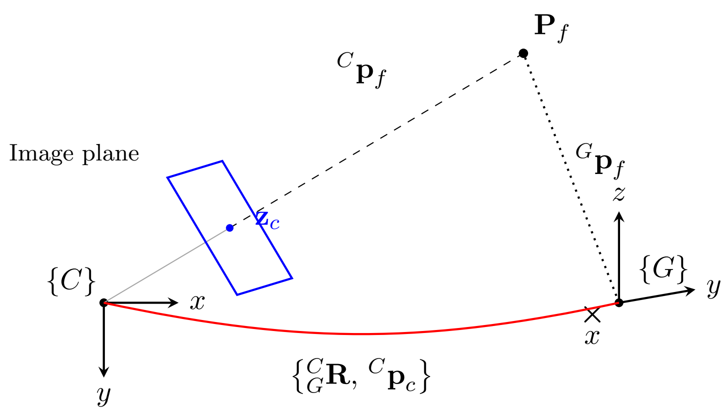


Figure 7: The intrinsic calibration sits inside the same projection geometry: the 3D feature is transformed into $\{C\}$, projected, distorted, and mapped to pixels.

Projection and transform.

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix} \triangleq \mathbf{h}_p({}^{C}\mathbf{p}_f), \qquad {}^{C}\mathbf{p}_f = {}^{C}_{G}\mathbf{R}^\top({}^{G}\mathbf{p}_f - {}^{G}\mathbf{p}_c) \triangleq \mathbf{h}_t({}^{C}_{G}\mathbf{R}, {}^{G}\mathbf{p}_c, {}^{G}\mathbf{p}_f), \tag{257}$$

$$\mathbf{z}_n = \mathbf{h}_p\big(\mathbf{h}_t({}^{C}_{G}\mathbf{R}, {}^{G}\mathbf{p}_c, {}^{G}\mathbf{p}_f)\big) + \mathbf{n}. \tag{258}$$

Distortion. Radial-tangential (for example); fisheye uses the equidistant model. With $r^2 = u_n^2 + v_n^2$ and distortion parameters $\mathbf{x}_d = [k_1, k_2, p_1, p_2]^\top$,

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = \begin{bmatrix} u_n \\ v_n \end{bmatrix} \left(1 + k_1 r^2 + k_2 r^4\right) + \begin{bmatrix} 2p_1 u_n v_n + p_2\left(r^2 + 2u_n^2\right) \\ p_1\left(r^2 + 2v_n^2\right) + 2p_2 u_n v_n \end{bmatrix}, \qquad \mathbf{z}_d = \begin{bmatrix} u_d \\ v_d \end{bmatrix} = \mathbf{h}_d(\mathbf{z}_n). \tag{259}$$

Intrinsics / homography. With $\mathbf{x}_{in} = [f_u, f_v, c_u, c_v]$,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix}, \qquad \mathbf{z} = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} f_u & 0 \\ 0 & f_v \end{bmatrix} \begin{bmatrix} u_d \\ v_d \end{bmatrix} + \begin{bmatrix} c_u \\ c_v \end{bmatrix} \triangleq \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}). \tag{260}$$

Full composition.

$$\begin{aligned} \mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c &= \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}) + \mathbf{n}_c && (261)\\ &= \mathbf{h}_k\big(\mathbf{h}_d(\mathbf{z}_n, \mathbf{x}_d), \mathbf{x}_{in}\big) + \mathbf{n}_c && (262)\\ &= \mathbf{h}_k\big(\mathbf{h}_d(\mathbf{h}_p({}^{C}\mathbf{p}_f), \mathbf{x}_d), \mathbf{x}_{in}\big) + \mathbf{n}_c && (263)\\ &= \mathbf{h}_k\Big(\mathbf{h}_d\big(\mathbf{h}_p(\mathbf{h}_t({}^{C}_{G}\mathbf{R}, {}^{G}\mathbf{p}_c, {}^{G}\mathbf{p}_f)), \mathbf{x}_d\big), \mathbf{x}_{in}\Big) + \mathbf{n}_c, && (264) \end{aligned}$$

where $\mathbf{h}_k(\cdot)$ is the camera-intrinsics homography, $\mathbf{h}_d(\cdot)$ the distortion function, $\mathbf{h}_p(\cdot)$ the projection function, $\mathbf{h}_t(\cdot)$ the transform function, $\mathbf{x}_d$ the distortion parameters, and $\mathbf{x}_{in} = \{f_u, f_v, c_u, c_v\}$ the intrinsics. The Jacobians for calibration follow by the chain rule.

3.5 The extrinsic (camera–IMU) calibration

The camera and IMU sensor frames are rigidly connected (Fig. 8).

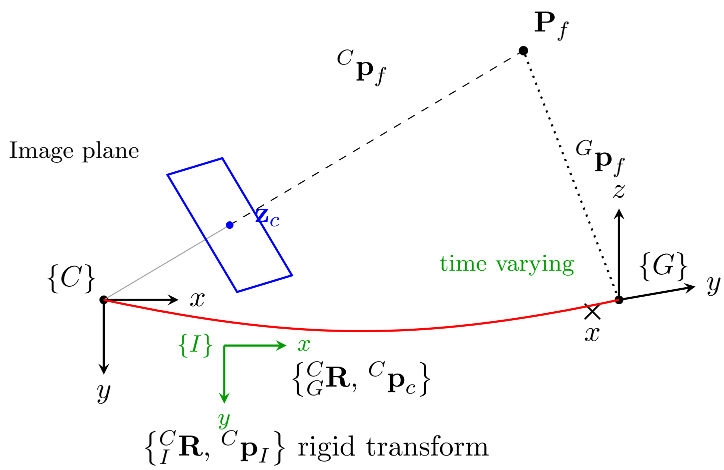


Figure 8: Extrinsic calibration: the IMU frame $\{I\}$ is rigidly fixed to the camera $\{C\}$ via $\{{}^{C}_{I}\mathbf{R}, {}^{C}\mathbf{p}_I\}$, while the camera-to-global transform is time varying.

$$\begin{bmatrix} {}^{C}\mathbf{p}_f \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^{C}_{I}\mathbf{R} & {}^{C}\mathbf{p}_I \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}}_{{}^{C}_{I}\mathbf{T}} \underbrace{\begin{bmatrix} {}^{G}_{I}\mathbf{R} & {}^{G}\mathbf{p}_I \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}^{-1}}_{{}^{G}_{I}\mathbf{T}^{-1}} \begin{bmatrix} {}^{G}\mathbf{p}_f \\ 1 \end{bmatrix}, \tag{265}$$

$${}^{C}\mathbf{p}_f = {}^{C}_{I}\mathbf{R}\,{}^{I}\mathbf{p}_f + {}^{C}\mathbf{p}_I = {}^{C}_{I}\mathbf{R}\big({}^{I}_{G}\mathbf{R}({}^{G}\mathbf{p}_f - {}^{G}\mathbf{p}_I)\big) + {}^{C}\mathbf{p}_I. \tag{266}$$

Time offset. How do we estimate the time offset between camera and IMU? With $t_I = t_c + t_d$,

$$\begin{bmatrix} {}^{C_{t_c}}\mathbf{p}_f \\ 1 \end{bmatrix} = {}^{C}_{I}\mathbf{T} \cdot \big({}^{G}_{I(t+t_d)}\mathbf{T}\big)^{-1} \begin{bmatrix} {}^{G}\mathbf{p}_f \\ 1 \end{bmatrix}, \tag{267}$$

$${}^{G}_{I(t+t_d)}\dot{\mathbf{T}} = \begin{bmatrix} {}^{G}_{I}\mathbf{R} & {}^{G}\mathbf{p}_I \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \lfloor {}^{I}\boldsymbol{\omega} \rfloor_\times & {}^{I}\mathbf{v} \\ \mathbf{0}_{1\times 3} & 0 \end{bmatrix}. \tag{268}$$

which is the Jacobian for t_d. This is then the full calibration for the vision system.

3.6 Point / line / plane representation

3.6.1 Point representation

In $\mathbb{R}^3$ (closest point),

$$\mathbf{P}_f = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} = z \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix}. \tag{269}$$

Inverse depth. Store the inverse of the depth, $\lambda \triangleq 1/z$:

$$\mathbf{x}_f = \begin{bmatrix} x \\ y \\ 1/z \end{bmatrix} = \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix}. \tag{270}$$

What is the motivation to use inverse depth? Transform the feature into a second view and divide by z:

$$^{C_2}\mathbf{p}_f = {}^{C_2}_{C_1}\mathbf{R}\big({}^{C_1}\mathbf{p}_f - {}^{C_1}\mathbf{p}_{C_2}\big) = {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\Bigg) = {}^{C_2}_{C_1}\mathbf{R}\Bigg(z\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\Bigg), \tag{271}$$

$$\frac{^{C_2}\mathbf{p}_f}{z} = {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - \frac{1}{z}{}^{C_1}\mathbf{p}_{C_2}\Bigg) \Rightarrow \begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} = {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda\,{}^{C_1}\mathbf{p}_{C_2}\Bigg). \tag{272}$$

If $z \to \infty$, the feature is far away, $z \approx {}^{C_2}z \gg {}^{C_2}x, {}^{C_2}y$, then the left side tends to the bearing measurement in $\{C_2\}$:

$$\begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} \approx \underbrace{\begin{bmatrix} u_{n_2} \\ v_{n_2} \\ 1 \end{bmatrix}}_{\text{measurements in } C_2} \propto {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda\,{}^{C_1}\mathbf{p}_{C_2}\Bigg). \tag{273}$$

The Jacobian with respect to the inverse-depth state $\tilde{\mathbf{x}} = [\tilde{x}, \tilde{y}, \tilde{\lambda}]$ is

$$\frac{\partial[u_{n_2}; v_{n_2}]}{\partial \tilde{\mathbf{x}}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} \mathbf{H}_\theta & \mathbf{H}_p & \mathbf{H}_{p_f} \end{bmatrix}, \tag{274}$$

with the three blocks

$$\mathbf{H}_\theta = \Big\lfloor {}^{C_2}_{C_1}\mathbf{R}\big([x\lambda;\ y\lambda;\ 1] - \lambda\,{}^{C_1}\mathbf{p}_{C_2}\big)\Big\rfloor_\times, \qquad \mathbf{H}_p = \underbrace{-\lambda\,{}^{C_2}_{C_1}\mathbf{R}}_{\lambda\to 0}, \tag{275}$$

$$\mathbf{H}_{p_f} = {}^{C_2}_{C_1}\mathbf{R}\big[\ \underbrace{\lambda\mathbf{e}_1}_{0}\ \ \underbrace{\lambda\mathbf{e}_2}_{0}\ \ [x;\ y;\ 0] - {}^{C_1}\mathbf{p}_{C_2}\ \big]. \tag{276}$$

If the feature is far away, then the rotation is improved, but the Jacobian for translation will not be improved (the $-\lambda\,{}^{C_2}_{C_1}\mathbf{R}$ and $\lambda\mathbf{e}_1, \lambda\mathbf{e}_2$ columns vanish as $\lambda \to 0$).

3.6.2 Plane representation

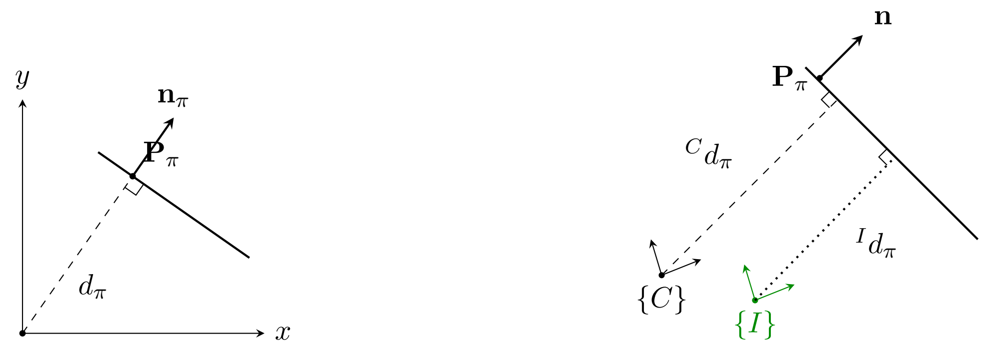
A plane is parameterized by its normal direction and distance to the origin (Fig. 9a):

$$\boldsymbol{\pi} = \begin{bmatrix} \mathbf{n}_\pi \\ d_\pi \end{bmatrix}_{4\times 1}, \qquad \mathbf{n}_\pi : \text{normal direction}, \quad d_\pi : \text{distance to origin}. \tag{277}$$

A plane is 3-dof but $\boldsymbol{\pi}$ has 4 parameters. The *closest-point representation* packs both into one vector,

$$\mathbf{P}_\pi = d_\pi\,\mathbf{n}_\pi = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \tag{278}$$

the closest point from the plane to the origin.



(a) plane: $\mathbf{n}_\pi$, d_π, foot $\mathbf{P}_\pi$

(b) transfer the plane $\{C\} \to \{I\}$

Figure 9: Plane representation (a) and its transfer between the camera and IMU frames (b).

Transfer $\{C\} \to \{I\}$. For ${}^C\mathbf{p}_\pi \leftrightarrow {}^I\mathbf{p}_\pi$ (Fig. 9b),

$$\begin{cases} {}^I\mathbf{n}_\pi = {}^I_C\mathbf{R}\, {}^C\mathbf{n}_\pi & \Rightarrow \text{normal direction} \\ {}^I d_\pi = {}^C d_\pi - {}^C\mathbf{n}_\pi^\top \cdot {}^C\mathbf{p}_I & \Rightarrow \text{origin to plane} \end{cases} \tag{279}$$

Hence, using ${}^I d_\pi = {}^C d_\pi - {}^C\mathbf{p}_I^\top\, {}^C\mathbf{n}_\pi$ and ${}^C\mathbf{p}_I = -{}^C_I\mathbf{R}\, {}^I\mathbf{p}_C$ (so $-{}^C\mathbf{p}_I^\top = {}^I\mathbf{p}_C^\top\, {}^I_C\mathbf{R}$),

$$\begin{bmatrix} {}^I\mathbf{n}_\pi \\ {}^I d_\pi \end{bmatrix} = \begin{bmatrix} {}^I_C\mathbf{R} & \mathbf{0}_{3\times 1} \\ -{}^C\mathbf{p}_I^\top & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{n}_\pi \\ {}^C d_\pi \end{bmatrix} = \begin{bmatrix} {}^I_C\mathbf{R} & \mathbf{0}_{3\times 1} \\ {}^I\mathbf{p}_C^\top\, {}^I_C\mathbf{R} & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{n}_\pi \\ {}^C d_\pi \end{bmatrix}. \tag{280}$$

The transformation chain is therefore

$${}^G\mathbf{p}_\pi \to ({}^C\mathbf{n}_\pi, {}^C d_\pi) \to ({}^I_C\mathbf{R}, {}^I\mathbf{p}_C) \to ({}^I\mathbf{n}_\pi, {}^I d_\pi) \to {}^I\mathbf{p}_\pi. \tag{281}$$

3.6.3 Line representation

How do we represent a 3D line (Fig. 10)? For a line L: $\mathbf{v}_e$ is the 3D direction, d_L is the distance from the origin to the line, and $\mathbf{n}_e$ is the normal direction of the plane (through the line and the origin).

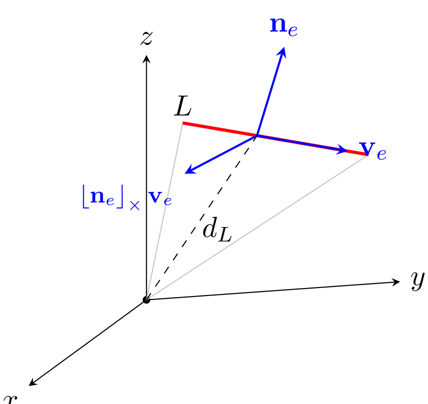


Figure 10: 3D line L: direction $\mathbf{v}_e$, plane normal $\mathbf{n}_e$, third axis $\lfloor \mathbf{n}_e \rfloor_\times \mathbf{v}_e$, and perpendicular distance d_L from the origin.

We can represent a line with $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$, where

$$\mathbf{R}_L = \begin{bmatrix}\mathbf{n}_e & \mathbf{v}_e & \lfloor \mathbf{n}_e \rfloor_\times \mathbf{v}_e\end{bmatrix} \text{ (a rotation matrix)}, \qquad d_L \text{ (a distance scalar)}. \tag{282}$$

So $L : \{\mathbf{R}_L, d_L\} : \{\bar{\mathbf{q}}_L, d_L\}$, packed as the 4D vector $\mathbf{P}_L = \bar{\mathbf{q}}_L \cdot d_L$, with the error states in Euclidean space. Then

$$\mathbf{P}_L = \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \bar{\mathbf{q}}_L \cdot d_L = \hat{\bar{\mathbf{q}}}_L \otimes \delta\mathbf{q}_L \cdot (\hat{d}_L + \tilde{d}_L) \tag{283}$$

$$\Leftrightarrow \; \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \hat{\bar{\mathbf{q}}}_L \otimes \begin{bmatrix} 1 \\ \frac{1}{2}\delta\boldsymbol{\theta}_L \end{bmatrix} (\hat{d}_L + \tilde{d}_L) \tag{284}$$

$$= \hat{\bar{\mathbf{q}}}_L \otimes \begin{bmatrix} \hat{d}_L + \tilde{d}_L \\ \frac{1}{2}\hat{d}_L \, \delta\boldsymbol{\theta}_L \end{bmatrix} \tag{285}$$

$$= \hat{\bar{\mathbf{q}}}_L \otimes \left(\begin{bmatrix} \hat{d}_L \\ \mathbf{0}_{3\times 1} \end{bmatrix} + \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2}\hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta\boldsymbol{\theta}_L \end{bmatrix} \right), \tag{286}$$

so that, with $\lfloor \hat{\bar{\mathbf{q}}}_L \rfloor_L$ the left quaternion-product matrix,

$$\boxed{\tilde{\mathbf{P}}_L = \lfloor \hat{\bar{\mathbf{q}}}_L \rfloor_L \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2}\hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta\boldsymbol{\theta}_L \end{bmatrix}.} \tag{287}$$

A few discussions on lines with cameras. (1) How to project a 3D line to a 2D line in the image? Given a line $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$ and camera parameters f_u, f_v, c_u, c_v (Fig. 11a):

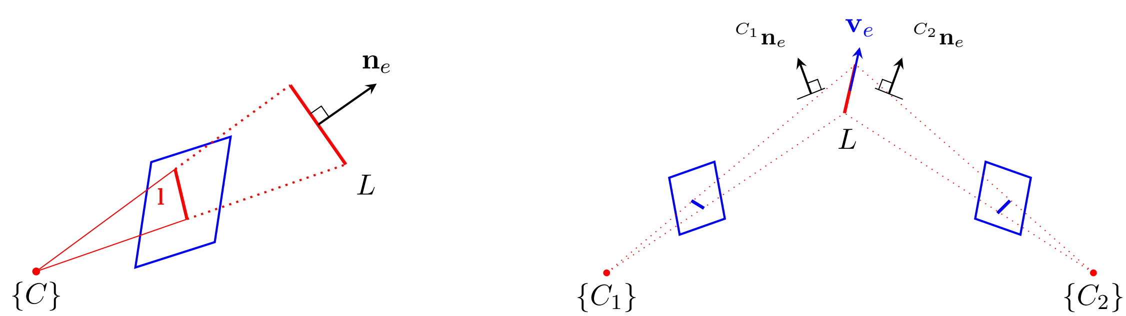
$$\begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix} \sim \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{p}_f}, \qquad \underbrace{\mathbf{p}}_{\text{pixel on image}} \sim \mathbf{K}\,\mathbf{p}_f. \tag{288}$$

Every $\mathbf{p}_f$ on the line lies in the plane through the line and the origin, so $\mathbf{p}_f^\top \mathbf{n}_e = 0$. With $\mathbf{p}_f \sim \mathbf{K}^{-1}\mathbf{p}$,

$$(\mathbf{K}^{-1}\mathbf{p})^\top \mathbf{n}_e = 0 \;\Rightarrow\; \mathbf{p}^\top \mathbf{K}^{-\top} \mathbf{n}_e = 0 \;\Rightarrow\; \mathbf{n}_e^\top \mathbf{K}^{-1} \mathbf{p} = 0 \;\Rightarrow\; (\mathbf{K}^{-\top}\mathbf{n}_e)^\top \mathbf{p} = 0, \tag{289}$$

so the 2D image line is

$$\mathbf{l} = \mathbf{K}^{-\top}\mathbf{n}_e, \qquad \mathbf{p} \text{ is every pixel on the line.} \tag{290}$$



(a) project 3D line $\rightarrow$ image line $\mathbf{l} = \mathbf{K}^{-\top}\mathbf{n}_e$ (b) transfer a line $\{C_1\} \rightarrow \{C_2\}$

Figure 11: Lines with cameras: projecting a 3D line to a 2D image line (a) and transferring a line between two camera frames (b).

(2) How to transfer a line from frame $\{C_1\}$ to $\{C_2\}$? (Fig. 11b) The line L is given by ${}^{C_1}\mathbf{n}_e$, ${}^{C_1}\mathbf{v}_e$, ${}^{C_1}d_L$ in $\{C_1\}$ and by ${}^{C_2}\mathbf{n}_e$, ${}^{C_2}\mathbf{v}_e$, ${}^{C_2}d_L$ in $\{C_2\}$. Assume ${}^{C_1}\mathbf{p}_f$ is a point on L; then its moment about the origin is

$$\left\lfloor {}^{C_1}\mathbf{p}_f \right\rfloor_\times {}^{C_1}\mathbf{v}_e = {}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L. \tag{291}$$

The direction transforms by rotation only, and the point by the full rigid transform:

$${}^{C_2}\mathbf{v}_e = {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e, \qquad {}^{C_2}\mathbf{p}_f = {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1}. \tag{292}$$

Hence the moment in $\{C_2\}$ is

$$\left\lfloor {}^{C_2}\mathbf{p}_f \right\rfloor_\times {}^{C_2}\mathbf{v}_e = \left\lfloor {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L. \tag{293}$$

Using $\lfloor \mathbf{R}\mathbf{a} \rfloor_\times \mathbf{R} = \mathbf{R} \lfloor \mathbf{a} \rfloor_\times$ on the first term,

$${}^{C_2}_{C_1}\mathbf{R} \left\lfloor {}^{C_1}\mathbf{p}_f \right\rfloor_\times {}^{C_1}\mathbf{v}_e + \left\lfloor {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L, \tag{294}$$

and substituting the moment identity ${}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L = \left\lfloor {}^{C_1}\mathbf{p}_f \right\rfloor_\times {}^{C_1}\mathbf{v}_e$,

$$\begin{bmatrix} {}^{C_2}_{C_1}\mathbf{R} & \left\lfloor {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix} = {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L. \tag{295}$$

Stacking the moment and the direction (${}^{C_2}\mathbf{v}_e = {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e$) gives the full line transfer:

$$\boxed{\begin{bmatrix} {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L \\ {}^{C_2}\mathbf{v}_e \end{bmatrix} = \begin{bmatrix} {}^{C_2}_{C_1}\mathbf{R} & \left\lfloor {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \\ \mathbf{0}_3 & {}^{C_2}_{C_1}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix}.} \tag{296}$$

4 Lecture 4: Information, Marginalization, and Estimator Design

Topics. (1) the matrix inversion lemma; (2) covariance vs. information; (3) marginalization and the Schur complement; (4) why the information matrix reflects problem structure; (5) a typical visual-odometry / SLAM pipeline; (6) many ways to handle $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$; (7) designing a common estimator — deriving the EKF from MAP by marginalization.

4.1 The matrix inversion lemma

For a 2×2 block matrix, the inverse can be written block by block. In the general form,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{E} & \mathbf{F} \\ \mathbf{G} & \mathbf{H} \end{bmatrix}, \qquad \begin{cases} \mathbf{E} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1}, \\ \mathbf{F} = -\mathbf{E}\mathbf{B}\mathbf{D}^{-1}, \\ \mathbf{G} = -\mathbf{D}^{-1}\mathbf{C}\mathbf{E}, \\ \mathbf{H} = \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{E}\mathbf{B}\mathbf{D}^{-1}. \end{cases} \tag{297}$$

For the symmetric case ($\mathbf{C} = \mathbf{B}^\top$, so the inverse is symmetric too),

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{E} & \mathbf{F} \\ \mathbf{F}^\top & \mathbf{H} \end{bmatrix}, \qquad \begin{aligned} &\mathbf{E} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^\top)^{-1}, \\ &\mathbf{F} = -\mathbf{E}\mathbf{B}\mathbf{D}^{-1}, \\ &\mathbf{H} = \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{B}^\top\mathbf{E}\mathbf{B}\mathbf{D}^{-1} = (\mathbf{D} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B})^{-1}. \end{aligned} \tag{298}$$

The term $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^\top$ (resp. $\mathbf{D} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B}$) is the *Schur complement*. This is one of the most important matrix operations for the derivations that follow.

4.2 Covariance and information

Recall the measurement model from Lecture 1, $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$ with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ and $\mathbf{R}$ the noise covariance. The MAP / weighted-least-squares estimate is

$$\max_{\mathbf{x}} \ p(\mathbf{z} \mid \mathbf{x}) \ \Leftrightarrow \ \min_{\mathbf{x}} \ \|\mathbf{z} - \mathbf{h}(\mathbf{x})\|^2_{\mathbf{R}}. \tag{299}$$

Given a linearization point $\hat{\mathbf{x}}$, $\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$, so with the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$,

$$\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}. \tag{300}$$

The normal equations and their solution are

$$\boxed{\underbrace{\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}}_{\text{information}}\,\tilde{\mathbf{x}} = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}, \qquad \tilde{\mathbf{x}} = \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}.} \tag{301}$$

To find the covariance of the corrected estimate $\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, substitute $\tilde{\mathbf{z}} = \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$:

$$\begin{aligned} \mathbf{x}^\oplus &= \hat{\mathbf{x}} + \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} && (302) \\ &= \hat{\mathbf{x}} + \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\left(\mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}\right) && (303) \\ &= \mathbf{x} + \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{n}. && (304) \end{aligned}$$

Hence $\mathbf{x}-\mathbf{x}^{\oplus}=-(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{n}$, and

$$\begin{aligned}\mathbf{P}_{\mathbf{x}^{\oplus}\mathbf{x}^{\oplus}} &= \mathbb{E}\big[(\mathbf{x}-\mathbf{x}^{\oplus})(\mathbf{x}-\mathbf{x}^{\oplus})^\top\big] \\ &= \big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\,\mathbb{E}(\mathbf{n}\mathbf{n}^\top)\,\mathbf{R}^{-1}\mathbf{H}\big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1} \qquad (305)\\ &= \big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}, \qquad (306)\end{aligned}$$

using $\mathbb{E}(\mathbf{n}\mathbf{n}^\top)=\mathbf{R}$. Therefore information and covariance are inverses of each other:

$$\boxed{\text{information}\quad \boldsymbol{\Sigma}=\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}, \qquad \text{covariance}\quad \mathbf{P}=\big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}, \qquad \boldsymbol{\Sigma}=\mathbf{P}^{-1},\ \ \mathbf{P}=\boldsymbol{\Sigma}^{-1}.} \qquad (307)$$

This is fundamental for what follows:

- for **optimization / least squares**, we keep track of the information matrix $\boldsymbol{\Sigma}$;
- for **filtering (EKF)**, we keep track of the covariance $\mathbf{P}$.

4.3 Marginalization

Let $\mathbf{x}\sim\mathcal{N}(\hat{\mathbf{x}},\mathbf{P}_{\mathbf{xx}})$, where $\hat{\mathbf{x}}$ is the best estimate and $\mathbf{P}_{\mathbf{xx}}$ the covariance. Partition the state as $\mathbf{x}=[\mathbf{x}_A^\top,\mathbf{x}_B^\top]^\top$, with $\mathbf{x}_A$ and $\mathbf{x}_B$ both part of $\mathbf{x}$:

$$\mathbf{x}\sim\mathcal{N}\left(\begin{bmatrix}\hat{\mathbf{x}}_A\\ \hat{\mathbf{x}}_B\end{bmatrix},\begin{bmatrix}\mathbf{P}_{AA} & \mathbf{P}_{AB}\\ \mathbf{P}_{BA} & \mathbf{P}_{BB}\end{bmatrix}\right), \qquad \mathbf{P}^\top=\mathbf{P}\ \text{(SPD)}, \quad \mathbf{P}_{AB}=\mathbf{P}_{BA}^\top. \qquad (308)$$

Covariance form (easy). The marginal of either block is read off directly:

$$\mathbf{x}_A\sim\mathcal{N}(\hat{\mathbf{x}}_A,\mathbf{P}_{AA}), \qquad \mathbf{x}_B\sim\mathcal{N}(\hat{\mathbf{x}}_B,\mathbf{P}_{BB}). \qquad (309)$$

Marginalization in covariance form is straightforward: just keep the corresponding diagonal block.

Information form (Schur complement). For optimization we work with the information system $(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H})\tilde{\mathbf{x}}=\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}$, i.e.

$$\boldsymbol{\Sigma}\,\tilde{\mathbf{x}}=\mathbf{b}, \qquad \begin{bmatrix}\boldsymbol{\Sigma}_{AA} & \boldsymbol{\Sigma}_{AB}\\ \boldsymbol{\Sigma}_{AB}^\top & \boldsymbol{\Sigma}_{BB}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}_A\\ \tilde{\mathbf{x}}_B\end{bmatrix}=\begin{bmatrix}\mathbf{b}_A\\ \mathbf{b}_B\end{bmatrix}, \qquad (310)$$

where $\boldsymbol{\Sigma}$ is the information matrix and $\mathbf{b}$ the residual vector. We seek $\boldsymbol{\Sigma}'_{AA},\mathbf{b}'_A$ (and $\boldsymbol{\Sigma}'_{BB},\mathbf{b}'_B$) such that $\boldsymbol{\Sigma}'_{AA}\tilde{\mathbf{x}}_A=\mathbf{b}'_A$ and $\boldsymbol{\Sigma}'_{BB}\tilde{\mathbf{x}}_B=\mathbf{b}'_B$ hold after eliminating the other block.

Solving $\boldsymbol{\Sigma}\tilde{\mathbf{x}}=\mathbf{b}$ via the block inverse,

$$\begin{bmatrix}\tilde{\mathbf{x}}_A\\ \tilde{\mathbf{x}}_B\end{bmatrix}=\begin{bmatrix}\boldsymbol{\Sigma}_{AA} & \boldsymbol{\Sigma}_{AB}\\ \boldsymbol{\Sigma}_{AB}^\top & \boldsymbol{\Sigma}_{BB}\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{b}_A\\ \mathbf{b}_B\end{bmatrix}=\begin{bmatrix}\mathbf{P}_{AA} & \mathbf{P}_{AB}\\ \mathbf{P}_{AB}^\top & \mathbf{P}_{BB}\end{bmatrix}\begin{bmatrix}\mathbf{b}_A\\ \mathbf{b}_B\end{bmatrix}=\begin{bmatrix}\mathbf{P}_{AA}\mathbf{b}_A+\mathbf{P}_{AB}\mathbf{b}_B\\ \mathbf{P}_{AB}^\top\mathbf{b}_A+\mathbf{P}_{BB}\mathbf{b}_B\end{bmatrix}, \qquad (311)$$

with the blocks of the inverse given by the matrix inversion lemma,

$$\mathbf{P}_{AA}=\big(\boldsymbol{\Sigma}_{AA}-\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\boldsymbol{\Sigma}_{AB}^\top\big)^{-1}, \qquad \mathbf{P}_{AB}=-\mathbf{P}_{AA}\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}, \qquad (312)$$

$$\mathbf{P}_{BB}=\boldsymbol{\Sigma}_{BB}^{-1}+\boldsymbol{\Sigma}_{BB}^{-1}\boldsymbol{\Sigma}_{AB}^\top\mathbf{P}_{AA}\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}=\big(\boldsymbol{\Sigma}_{BB}-\boldsymbol{\Sigma}_{AB}^\top\boldsymbol{\Sigma}_{AA}^{-1}\boldsymbol{\Sigma}_{AB}\big)^{-1}. \qquad (313)$$

For the A block,

$$\tilde{\mathbf{x}}_A=\mathbf{P}_{AA}\mathbf{b}_A+\mathbf{P}_{AB}\mathbf{b}_B=\mathbf{P}_{AA}\mathbf{b}_A-\mathbf{P}_{AA}\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\mathbf{b}_B=\mathbf{P}_{AA}\big(\mathbf{b}_A-\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\mathbf{b}_B\big), \qquad (314)$$

so $\mathbf{P}_{AA}^{-1}\tilde{\mathbf{x}}_A = \mathbf{b}_A - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\mathbf{b}_B$. Using the identity $\mathbf{P}_{AB} = -\mathbf{P}_{AA}\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1} = -\boldsymbol{\Sigma}_{AA}^{-1}\boldsymbol{\Sigma}_{AB}\mathbf{P}_{BB}$, the B block gives symmetrically

$$\tilde{\mathbf{x}}_B = \mathbf{P}_{AB}^{\top}\mathbf{b}_A + \mathbf{P}_{BB}\mathbf{b}_B = \mathbf{P}_{BB}\big(\mathbf{b}_B - \boldsymbol{\Sigma}_{AB}^{\top}\boldsymbol{\Sigma}_{AA}^{-1}\mathbf{b}_A\big), \tag{315}$$

so $\mathbf{P}_{BB}^{-1}\tilde{\mathbf{x}}_B = \mathbf{b}_B - \boldsymbol{\Sigma}_{AB}^{\top}\boldsymbol{\Sigma}_{AA}^{-1}\mathbf{b}_A$. Substituting the Schur-complement expressions for $\mathbf{P}_{AA}^{-1}$ and $\mathbf{P}_{BB}^{-1}$:

$$\boxed{\begin{aligned} &\text{marginalize } \mathbf{x}_B,\ \text{keep } \mathbf{x}_A: && \big(\boldsymbol{\Sigma}_{AA} - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\boldsymbol{\Sigma}_{AB}^{\top}\big)\tilde{\mathbf{x}}_A = \mathbf{b}_A - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\mathbf{b}_B, && (316)\\ &\text{marginalize } \mathbf{x}_A,\ \text{keep } \mathbf{x}_B: && \big(\boldsymbol{\Sigma}_{BB} - \boldsymbol{\Sigma}_{AB}^{\top}\boldsymbol{\Sigma}_{AA}^{-1}\boldsymbol{\Sigma}_{AB}\big)\tilde{\mathbf{x}}_B = \mathbf{b}_B - \boldsymbol{\Sigma}_{AB}^{\top}\boldsymbol{\Sigma}_{AA}^{-1}\mathbf{b}_A. && (317)\end{aligned}}$$

Both the new information block and the new residual are Schur complements of the eliminated variable. This is exactly how variables are dropped from a sliding window while preserving their influence on the kept states.

4.4 Why we care about the information matrix

Take visual measurements as an example. A pixel observation depends on a camera pose and a feature,

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n} = \mathbf{h}_c(\mathbf{x}_c, \mathbf{x}_f) + \mathbf{n}, \qquad \tilde{\mathbf{z}}_c = \mathbf{H}_c\tilde{\mathbf{x}}_c + \mathbf{H}_f\tilde{\mathbf{x}}_f + \mathbf{n}. \tag{318}$$

Consider three poses $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ observing two features f_1, f_2 (Fig. 12). Stacking the four measurements as $\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$, with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and state $\mathbf{x} = [\mathbf{x}_A^{\top}, \mathbf{x}_B^{\top}, \mathbf{x}_C^{\top}, f_1^{\top}, f_2^{\top}]^{\top}$:

$$\begin{bmatrix}\tilde{\mathbf{z}}_{A1}\\ \tilde{\mathbf{z}}_{B1}\\ \tilde{\mathbf{z}}_{B2}\\ \tilde{\mathbf{z}}_{C2}\end{bmatrix} = \underbrace{\begin{bmatrix}\mathbf{H}_A & \mathbf{0} & \mathbf{0} & \mathbf{H}_{A1} & \mathbf{0}\\ \mathbf{0} & \mathbf{H}_B & \mathbf{0} & \mathbf{H}_{B1} & \mathbf{0}\\ \mathbf{0} & \mathbf{H}_B & \mathbf{0} & \mathbf{0} & \mathbf{H}_{B2}\\ \mathbf{0} & \mathbf{0} & \mathbf{H}_C & \mathbf{0} & \mathbf{H}_{C2}\end{bmatrix}}_{\mathbf{H}} \begin{bmatrix}\tilde{\mathbf{x}}_A\\ \tilde{\mathbf{x}}_B\\ \tilde{\mathbf{x}}_C\\ \tilde{f}_1\\ \tilde{f}_2\end{bmatrix} + \mathbf{n}. \tag{319}$$

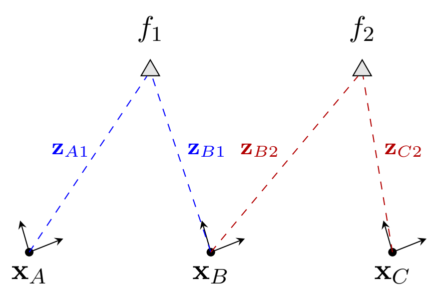


Figure 12: Visual measurement graph: poses $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ observe features f_1, f_2. Each ray $\mathbf{z}_{(\cdot)}$ couples one pose to one feature, which determines the sparsity of $\mathbf{H}$ and of $\boldsymbol{\Sigma} = \mathbf{H}^{\top}\mathbf{H}$.

The information matrix $\boldsymbol{\Sigma} = \mathbf{H}^{\top}\mathbf{H}$ has a block-sparse structure (a block (i, j) is nonzero only when states i and j are co-observed in some measurement), shown in Fig. 13. Partitioning into poses $p = \{A, B, C\}$ and features $f = \{f_1, f_2\}$,

$$\boldsymbol{\Sigma} = \mathbf{H}^{\top}\mathbf{H} \triangleq \begin{bmatrix}\boldsymbol{\Sigma}_{pp} & \boldsymbol{\Sigma}_{pf}\\ \boldsymbol{\Sigma}_{pf}^{\top} & \boldsymbol{\Sigma}_{ff}\end{bmatrix}. \tag{320}$$

$$
\begin{array}{c|ccc|cc|}
 & A & B & C & f_1 & f_2 \\
\hline
A & \otimes & 0 & 0 & \otimes & 0 \\
B & 0 & \otimes & 0 & \otimes & \otimes \\
C & 0 & 0 & \otimes & 0 & \otimes \\
\hline
f_1 & \otimes & \otimes & 0 & \otimes & 0 \\
f_2 & 0 & \otimes & \otimes & 0 & \otimes \\
\hline
\end{array}
\qquad
\begin{array}{l}
\boldsymbol{\Sigma}_{pp} \\[2em]
\boldsymbol{\Sigma}_{ff}
\end{array}
$$

Figure 13: Sparsity of $\boldsymbol{\Sigma} = \mathbf{H}^\top\mathbf{H}$ ($\otimes$: nonzero block). The pose–pose block $\boldsymbol{\Sigma}_{pp}$ is block-diagonal, while $\boldsymbol{\Sigma}_{pf}$ records exactly which pose sees which feature. The information matrix reflects the problem structure.

4.5 A typical visual-odometry / SLAM pipeline

How do the previous two lectures fit together into a working monocular system? A common pipeline incrementally grows poses $C_1, C_2, \ldots$ and features $f_1, f_2, \ldots$ (Fig. 14):

(1) **5-point algorithm** on C_1, C_2: recover ${}^{C_1}_{C_2}\mathbf{R}$ and ${}^{C_1}\mathbf{p}^*_{C_2}$ (up to scale).
(2) **Feature triangulation**: with the recovered relative pose, triangulate $f_1, f_2, \ldots$.
(3) **P3P / PnP** for C_3: with known features, solve ${}^{C_1}_{C_3}\mathbf{R}$, ${}^{C_1}\mathbf{p}_{C_3}$ (then triangulate new features).
(4) **5-point or P3P/PnP** for C_4: solve ${}^{C_1}_{C_4}\mathbf{R}$, ${}^{C_1}\mathbf{p}_{C_4}$.
(5) **Bundle adjustment (BA)** over C_1, C_2, C_3 and $f_1, f_2, f_3, \ldots$ (joint refinement of poses and structure).
(6) **BA** extended to C_1, C_2, C_3, C_4 and $f_1, f_2, f_3, \ldots$ as the window grows.
(7) **Marginalization**: when there are too many poses $(C_1, \ldots, C_n)$ and features $(f_1, \ldots, f_m)$, marginalize old states $C_1, f_1, \ldots$ to keep the system light-weight.

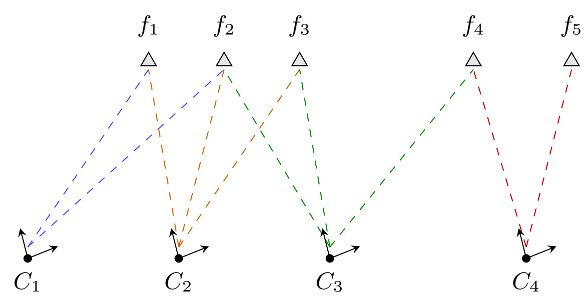


Figure 14: A monocular VO/SLAM pipeline grows poses $C_1, \ldots, C_4$ and features $f_1, \ldots, f_5$: bootstrap with the 5-point algorithm + triangulation, extend new poses with P3P/PnP, refine with bundle adjustment, and marginalize old states to bound cost. Combined with the previous two lectures, this builds up visual SLAM.

4.6 Designing estimators: many ways to handle $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$

Given $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, there are many ways to linearize and solve, each leading to a different family of algorithms.

(1) Iterated re-linearization (GN / LM). The typical solution $\min_{\mathbf{x}} \|\mathbf{z} - \mathbf{h}(\mathbf{x})\|^2_{\mathbf{R}}$. Given a linearization $\hat{\mathbf{x}}$, $\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$; update $\mathbf{x}^{\oplus} = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, then *redo* the linearization $\tilde{\mathbf{z}} = \mathbf{H}^{\oplus}\tilde{\mathbf{x}} + \mathbf{n}$ (Gauss–Newton or Levenberg–Marquardt).

(2) Linearize once + residual correction (FEJ). What if we linearize only once, at a fixed $\hat{\mathbf{x}}^0$? With $\mathbf{H}^0 = \mathbf{H}|_{\hat{\mathbf{x}}^0}$,

$$\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}^0) + \mathbf{H}^0(\mathbf{x} - \hat{\mathbf{x}}^0) + \mathbf{n} \tag{321}$$

$$\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^0) = \mathbf{H}^0\big(\mathbf{x} - \hat{\mathbf{x}} + \underbrace{\hat{\mathbf{x}} - \hat{\mathbf{x}}^0}_{\Delta\mathbf{x}}\big) + \mathbf{n} = \mathbf{H}^0\tilde{\mathbf{x}} + \mathbf{H}^0\Delta\mathbf{x} + \mathbf{n}, \tag{322}$$

so moving the known offset to the left gives the *residual correction*

$$\underbrace{\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^0)}_{\tilde{\mathbf{z}}^0} - \mathbf{H}^0\Delta\mathbf{x} = \mathbf{H}^0\tilde{\mathbf{x}} + \mathbf{n}. \tag{323}$$

Keeping the Jacobian fixed at the first estimate is **FEJ** (first-estimate Jacobians) [7], used e.g. in OKVIS [8]; it preserves the correct observability/nullspace structure.

(3) Partial linearization (preintegration / iSAM). For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, linearize only in $\mathbf{x}_B$ (about $\hat{\mathbf{x}}_B^0$) while keeping $\mathbf{x}_A$ exact:

$$\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}_A, \hat{\mathbf{x}}_B^0) + \mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{H}_B^0\tilde{\mathbf{x}}_B + \mathbf{H}_B^0\Delta\mathbf{x}_B + \mathbf{n}, \tag{324}$$

$$\underbrace{\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}_A, \hat{\mathbf{x}}_B^0)}_{\tilde{\mathbf{z}}^0} - \mathbf{H}_B^0\Delta\mathbf{x}_B = \begin{bmatrix}\mathbf{H}_A & \mathbf{H}_B^0\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B\end{bmatrix} + \mathbf{n}. \tag{325}$$

This is the idea behind IMU **pre-integration** [9] and **ACI** (analytic combined integration) [10]; **iSAM** [11] uses a similar partial-relinearization idea.

(4) Nullspace projection (MSCKF / structureless). For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, $\tilde{\mathbf{z}} = \mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{H}_B\tilde{\mathbf{x}}_B + \mathbf{n}$. If we find $\mathbf{N}$ with $\mathbf{N}^\top\mathbf{H}_B = \mathbf{0}$, then projecting onto the left nullspace removes $\mathbf{x}_B$:

$$\mathbf{N}^\top\tilde{\mathbf{z}} = \mathbf{N}^\top\mathbf{H}_A\tilde{\mathbf{x}}_A + \cancel{\mathbf{N}^\top\mathbf{H}_B\tilde{\mathbf{x}}_B} + \mathbf{N}^\top\mathbf{n}. \tag{326}$$

This is the **MSCKF** [12] (multi-state constraint Kalman filter), a.k.a. the smart / structure-less vision factor: the features are eliminated without ever being added to the state.

(5) Invariant errors. Instead of $\mathbf{x} = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, put $\mathbf{x}$ on a group $SE(3)$ / $SE_2(3)$ and use a right-invariant error,

$$\mathbf{x} = \hat{\mathbf{x}} \boxplus \tilde{\mathbf{x}} = \mathrm{Exp}(\tilde{\mathbf{x}}) \cdot \hat{\mathbf{x}} \;\Rightarrow\; \text{invariant SLAM / VIO.} \tag{327}$$

(6) Decoupled (right-)invariant errors. For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, mix error types per block:

$$\begin{cases}\mathbf{x}_A = \hat{\mathbf{x}}_A \boxplus \tilde{\mathbf{x}}_A = \mathrm{Exp}(\tilde{\mathbf{x}}_A)\cdot\hat{\mathbf{x}}_A & \text{(group / invariant error)} \\ \mathbf{x}_B = \hat{\mathbf{x}}_B + \tilde{\mathbf{x}}_B & \text{(vector error)}\end{cases} \tag{328}$$

which gives decoupled right-invariant SLAM / VIO.

> Use your imagination to deal with $\{\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n},\ \tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}\}$ to design different algorithms.

4.7 Designing a common system: from MAP to the EKF

Consider the canonical prior + motion + measurement system

$$\begin{cases} \mathbf{x}_0 \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0) & \to \text{prior} \\ \mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) & \to \text{motion model} \\ \mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n}_1, \quad \mathbf{n}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) & \to \text{measurement model} \end{cases} \tag{329}$$

The MAP estimate factorizes by Bayes' rule,

$$\max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{z}) = \max_{\mathbf{x}} \frac{p(\mathbf{z} \mid \mathbf{x})\, p(\mathbf{x})}{p(\mathbf{z})} = \max_{\mathbf{x}} \frac{p(\mathbf{z}_1 \mid \mathbf{x}_1)\, p(\mathbf{x}_1 \mid \mathbf{x}_0)\, p(\mathbf{x}_0)}{p(\mathbf{z})}, \tag{330}$$

which is equivalent to the nonlinear least-squares cost

$$\min_{\mathbf{x}} \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|^2_{\mathbf{R}} + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|^2_{\mathbf{Q}} + \|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2_{\mathbf{P}_0}. \tag{331}$$

There are two ways to solve it:

1. **Batch:** minimize all three terms jointly $\Rightarrow \hat{\mathbf{x}}_1, \mathbf{P}_1$.
2. **Incremental:**
   (i) handle the prior + motion terms $\|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2_{\mathbf{P}_0} + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|^2_{\mathbf{Q}}$;
   (ii) marginalize $\mathbf{x}_0$, keep $\mathbf{x}_1$, giving $(\hat{\mathbf{x}}_{1|0}, \mathbf{P}_{1|0})$;
   (iii) add the measurement $\left\|\mathbf{x}_1 - \hat{\mathbf{x}}_{1|0}\right\|^2_{\mathbf{P}_{1|0}} + \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|^2_{\mathbf{R}} \Rightarrow \hat{\mathbf{x}}_1, \mathbf{P}_1$.

The incremental route is exactly the EKF predict/update, as we now derive.

Propagation (marginalize $\mathbf{x}_0$). Linearize the motion model, $\mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0 \Rightarrow \hat{\mathbf{x}}_1 + \tilde{\mathbf{x}}_1 = \mathbf{f}(\hat{\mathbf{x}}_0) + \mathbf{F}\tilde{\mathbf{x}}_0 + \mathbf{w}_0$, and define the motion residual

$$\mathbf{r}_{\mathbf{x}} \triangleq \hat{\mathbf{x}}_1 - \mathbf{f}(\hat{\mathbf{x}}_0) = \mathbf{F}\tilde{\mathbf{x}}_0 - \tilde{\mathbf{x}}_1 + \mathbf{w}_0, \qquad \mathbf{w}_0 = \mathbf{r}_{\mathbf{x}} - \begin{bmatrix}\mathbf{F} & -\mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix}. \tag{332}$$

The prior + motion cost (writing $\mathbf{Q}_0 = \mathbf{Q}$) is

$$\min \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0} + \left\| \mathbf{r}_{\mathbf{x}} - \begin{bmatrix}\mathbf{F} & -\mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix} \right\|^2_{\mathbf{Q}_0}. \tag{333}$$

Collecting quadratic and linear terms, the cost C is

$$C = \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix}^\top \begin{bmatrix}\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix} - 2 \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix}^\top \begin{bmatrix}\mathbf{F}^\top \\ -\mathbf{I}\end{bmatrix} \mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} + \mathbf{r}_{\mathbf{x}}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}}. \tag{334}$$

Setting $\partial C/\partial[\tilde{\mathbf{x}}_0; \tilde{\mathbf{x}}_1] = \mathbf{0}$ gives the information system

$$\begin{bmatrix}\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix} = \begin{bmatrix}\mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} \\ -\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}}\end{bmatrix}. \tag{335}$$

Marginalizing $\mathbf{x}_0$ (Schur complement onto the $\mathbf{x}_1$ block) and applying the matrix inversion lemma,

$$\begin{aligned} \left[\mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\right]\tilde{\mathbf{x}}_1 &= -\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} + \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}}, \\ \underbrace{\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}}_{\mathbf{P}_{1|0}^{-1}}\tilde{\mathbf{x}}_1 &= -\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}\mathbf{r}_{\mathbf{x}}. \end{aligned} \tag{336}$$

This is the EKF **propagation**: the predicted covariance and mean are

$$\boxed{\mathbf{P}_{1|0} = \mathbf{F}\mathbf{P}_0\mathbf{F}^\top + \mathbf{Q}_0, \qquad \hat{\mathbf{x}}_{1|0} = \mathbf{f}(\hat{\mathbf{x}}_0) \ \ \text{(i.e. } \mathbf{r}_{\mathbf{x}} = \mathbf{0}\text{)}.} \tag{337}$$

Update (add the measurement). Now include the measurement $\mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n} \Rightarrow \tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}}_1 + \mathbf{n}$. The full cost

$$\|\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}_1\|^2_{\mathbf{R}} + \left\| \mathbf{r}_{\mathbf{x}} - \begin{bmatrix} \mathbf{F} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} \right\|^2_{\mathbf{Q}_0} + \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0} \tag{338}$$

yields the augmented information system (the measurement adds $\mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}$ to the $\mathbf{x}_1$ block and $\mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}}$ to its residual),

$$\begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top \mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top \mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^\top \mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} \\ -\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} + \mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} \end{bmatrix}. \tag{339}$$

Marginalizing $\mathbf{x}_0$ again and recognizing $\mathbf{P}_{1|0}^{-1} = (\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top)^{-1}$,

$$\underbrace{\left(\mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H} + \mathbf{P}_{1|0}^{-1}\right)}_{\boldsymbol{\Sigma} = \mathbf{P}_{1|1}^{-1}} \tilde{\mathbf{x}}_1 = \mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} - \underbrace{\mathbf{P}_{1|0}^{-1}\mathbf{r}_{\mathbf{x}}}_{\to \mathbf{0}}. \tag{340}$$

After propagation $\mathbf{r}_{\mathbf{x}} = \mathbf{0}$, so the second term drops. Applying the matrix inversion lemma to the updated covariance,

$$\mathbf{P}_{1|1} = \left(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\right)^{-1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}, \tag{341}$$

and for the mean correction, using the same lemma to convert information $\to$ covariance form,

$$\begin{aligned}
\tilde{\mathbf{x}}_1 &= \left(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} && (342)\\
&= \Big(\mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}\Big)\mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} && (343)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top \Big(\mathbf{R}^{-1} - \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top \mathbf{R}^{-1}\Big)\tilde{\mathbf{z}} && (344)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\Big(\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)\mathbf{R}^{-1} - \mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top \mathbf{R}^{-1}\Big)\tilde{\mathbf{z}} && (345)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\tilde{\mathbf{z}}. && (346)
\end{aligned}$$

Finally, with the Kalman gain $\mathbf{K} = \mathbf{P}_{1|0}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R})^{-1}$, we recover the EKF **update**:

$$\boxed{\tilde{\mathbf{x}}_1 = \mathbf{P}_{1|0}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\tilde{\mathbf{z}}, \qquad \mathbf{P}_{1|1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}.} \tag{347}$$

The extended Kalman filter is thus the incremental (marginalize-then-update) solution of the same MAP problem that batch least squares solves jointly — the two views are connected by the matrix inversion lemma and the Schur complement.

4.8 The relationship between Schur complement and nullspace projection

The Schur-complement marginalization (§4.3) and the nullspace projection (§4.6, item 4, the MSCKF) both eliminate $\mathbf{x}_B$ — it turns out they are the *same* operation. Take $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, with $\tilde{\mathbf{z}} = \mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{H}_B\tilde{\mathbf{x}}_B + \mathbf{n}$.

(1) Information form. The normal equations are

$$\begin{bmatrix} \mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_A & \mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_B \\ \mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_A & \mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_B \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \end{bmatrix} = \begin{bmatrix} \mathbf{H}_A^\top \\ \mathbf{H}_B^\top \end{bmatrix} \mathbf{R}^{-1}\tilde{\mathbf{z}}. \tag{348}$$

Marginalizing $\mathbf{x}_B$ and keeping $\mathbf{x}_A$ via the Schur complement gives

$$\boxed{\begin{aligned} &\underbrace{\left(\mathbf{H}_A^\top\mathbf{R}^{-1}\mathbf{H}_A - \mathbf{H}_A^\top\mathbf{R}^{-1}\mathbf{H}_B(\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_B)^{-1}\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_A\right)}_{\text{information}}\tilde{\mathbf{x}}_A \\ &\qquad = \underbrace{\mathbf{H}_A^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{H}_A^\top\mathbf{R}^{-1}\mathbf{H}_B(\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_B)^{-1}\mathbf{H}_B^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}}_{\text{residual}}. \end{aligned}} \tag{Eq. 1}$$

(2) Nullspace form. Take a (thin) QR-type factorization of $\mathbf{H}_B$,

$$\mathbf{H}_B = \begin{bmatrix} \mathbf{Q}_e & \mathbf{Q}_N \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{0} \end{bmatrix} = \mathbf{Q}_e\mathbf{U} + \mathbf{Q}_N \cdot \mathbf{0} = \mathbf{Q}_e\mathbf{U}, \tag{349}$$

with $\mathbf{U}$ upper triangular and $[\mathbf{Q}_e\ \mathbf{Q}_N]$ orthogonal, so $\mathbf{Q}_e\mathbf{Q}_e^\top + \mathbf{Q}_N\mathbf{Q}_N^\top = \mathbf{I}$, $\mathbf{Q}_e^\top\mathbf{Q}_e = \mathbf{I}$, $\mathbf{Q}_N^\top\mathbf{Q}_N = \mathbf{I}$, and the left nullspace satisfies $\mathbf{Q}_N^\top\mathbf{H}_B = \mathbf{Q}_N^\top\mathbf{Q}_e\mathbf{U} = \mathbf{0}$. Projecting the residual onto $\mathbf{Q}_N$ removes $\mathbf{x}_B$:

$$\mathbf{Q}_N^\top\tilde{\mathbf{z}} = \mathbf{Q}_N^\top\mathbf{H}_A\tilde{\mathbf{x}}_A + \cancel{\mathbf{Q}_N^\top\mathbf{H}_B\tilde{\mathbf{x}}_B} + \mathbf{Q}_N^\top\mathbf{n} = \mathbf{Q}_N^\top\mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{Q}_N^\top\mathbf{n}, \tag{350}$$

with projected noise covariance $\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N$. The resulting normal equations are

$$\boxed{(\mathbf{Q}_N^\top\mathbf{H}_A)^\top(\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N)^{-1}(\mathbf{Q}_N^\top\mathbf{H}_A)\,\tilde{\mathbf{x}}_A = (\mathbf{Q}_N^\top\mathbf{H}_A)^\top(\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N)^{-1}\mathbf{Q}_N^\top\tilde{\mathbf{z}}.} \tag{Eq. 2}$$

Are Eq. 1 and Eq. 2 equivalent? Consider the isotropic case $\mathbf{R} = \sigma^2\mathbf{I}$. With $\mathbf{H}_B = \mathbf{Q}_e\mathbf{U}$,

$$\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_B = (\mathbf{Q}_e\mathbf{U})^\top\frac{\mathbf{I}}{\sigma^2}\mathbf{Q}_e\mathbf{U} = \frac{\mathbf{U}^\top\mathbf{U}}{\sigma^2}, \tag{351}$$

so the projector that appears in the Schur complement of Eq. 1 is

$$\begin{aligned} \mathbf{H}_A^\top\mathbf{R}^{-1}\mathbf{H}_B(\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_B)^{-1}\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_A &= \mathbf{H}_A^\top\frac{\mathbf{I}}{\sigma^2}\mathbf{Q}_e\mathbf{U}\,\sigma^2(\mathbf{U}^\top\mathbf{U})^{-1}\mathbf{U}^\top\mathbf{Q}_e^\top\frac{\mathbf{I}}{\sigma^2}\mathbf{H}_A \\ &= \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_e\mathbf{Q}_e^\top\mathbf{H}_A. \end{aligned} \tag{352}$$

Hence the Eq. 1 information becomes, using $\mathbf{Q}_e\mathbf{Q}_e^\top = \mathbf{I} - \mathbf{Q}_N\mathbf{Q}_N^\top$,

$$\frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{H}_A - \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_e\mathbf{Q}_e^\top\mathbf{H}_A = \frac{1}{\sigma^2}\mathbf{H}_A^\top\left(\mathbf{I} - \mathbf{Q}_e\mathbf{Q}_e^\top\right)\mathbf{H}_A = \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_N\mathbf{Q}_N^\top\mathbf{H}_A. \tag{353}$$

The Eq. 2 information, using $\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N = \sigma^2\mathbf{Q}_N^\top\mathbf{Q}_N = \sigma^2\mathbf{I}$, is

$$(\mathbf{Q}_N^\top\mathbf{H}_A)^\top(\sigma^2\mathbf{I})^{-1}(\mathbf{Q}_N^\top\mathbf{H}_A) = \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_N\mathbf{Q}_N^\top\mathbf{H}_A, \tag{354}$$

which is identical (the residuals match the same way). Therefore, for isotropic noise, Schur-complement marginalization and left-nullspace projection produce exactly the same information and residual — the MSCKF's structureless update is precisely the marginalization of the feature.

5 Lecture 5: The IMU Model and IMU Integration

Topics. (1) the IMU measurement model (gyroscope, accelerometer, biases, noise); (2) continuous vs. discrete noise; (3) the IMU kinematics as a first-order ODE; (4) the big picture: IMU integration in a visual-inertial navigation system; (5) exact integration from t_k to t_{k+1}; (6) computing $\Delta\mathbf{R}, \Delta\mathbf{p}, \Delta\mathbf{v}$ — preintegration and three methods; (7) mean and covariance propagation with $SO(3) + \mathbb{R}^3$ errors; (8) the linearization using $SE_2(3)$.

5.1 The IMU model

The IMU (inertial measurement unit) is one of the most important sensors for robotics. We use two frames:

- $\{I\}$: the IMU frame (attached to the sensor, see Fig. 15);
- $\{G\}$: the global frame.

The IMU provides two readings: the **gyroscope** reads angular velocity and the **accelerometer** reads acceleration,

$$\boxed{\begin{aligned} {}^I\boldsymbol{\omega}_m &= {}^I\boldsymbol{\omega} + \mathbf{b}_g + \mathbf{n}_g, \\ {}^I\mathbf{a}_m &= {}^I\mathbf{a} + \mathbf{b}_a + \mathbf{n}_a, \end{aligned}} \tag{355, 356}$$

where

- ${}^I\boldsymbol{\omega}$ is the true angular-rate reading and ${}^I\mathbf{a}$ the true acceleration reading;
- $\mathbf{b}_g$ is the gyro bias, driven by a random walk $\dot{\mathbf{b}}_g = \mathbf{n}_{wg}$; $\mathbf{b}_a$ is the accel bias, driven by a random walk $\dot{\mathbf{b}}_a = \mathbf{n}_{wa}$;
- $\mathbf{n}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{I}_3)$ and $\mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2\mathbf{I}_3)$ are Gaussian noises.

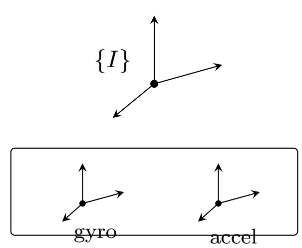


Figure 15: The IMU carries its own frame $\{I\}$; onboard, the gyroscope reads the angular velocity and the accelerometer reads the acceleration.

Another way of modeling the accelerometer makes gravity explicit:

$${}^I\mathbf{a}_m = \left({}^I\mathbf{a}_b - {}^I_G\mathbf{R}\,{}^G\mathbf{g}\right) + \mathbf{b}_a + \mathbf{n}_a, \qquad {}^G\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix}, \tag{357}$$

where ${}^I\mathbf{a}_b$ is the body acceleration and ${}^G\mathbf{g}$ is gravity in $\{G\}$.

We use (355) and (356) as our measurements. Inverting them gives the true readings:

$$\text{true gyro reading:} \quad {}^I\boldsymbol{\omega} = {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g, \tag{358}$$

$$\text{true accel reading:} \quad {}^I\mathbf{a} = {}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a. \tag{359}$$

The IMU state. When we model the IMU, we use the following parameters:

- $\{{}^G_I\mathbf{R}, {}^G\mathbf{p}_I\}$: IMU orientation and position — 6 DoF;
- ${}^G\mathbf{v}_I$: IMU velocity — 3 DoF;
- $\mathbf{b}_g$: gyro bias — 3 DoF; $\mathbf{b}_a$: accel bias — 3 DoF.

Then our state vector for the IMU is

$$\mathbf{x}_I = \{{}^G_I\mathbf{R},\ {}^G\mathbf{p}_I,\ {}^G\mathbf{v}_I,\ \mathbf{b}_g,\ \mathbf{b}_a\}, \qquad 15 \text{ DoF}. \tag{360}$$

Some places use the quaternion ${}^G_I\mathbf{R} \Leftrightarrow {}^G_I\bar{q}$ (Lecture 2); then

$$\mathbf{x}_I = \begin{bmatrix} {}^G_I\bar{q}^\top & {}^G\mathbf{p}_I^\top & {}^G\mathbf{v}_I^\top & \mathbf{b}_g^\top & \mathbf{b}_a^\top \end{bmatrix}^\top, \qquad 16 \times 1 \ \Rightarrow\ 15 \text{ DoF}. \tag{361}$$

5.2 Continuous and discrete noise

For the IMU we specify *continuous-time* noise, $\mathbf{n}_c \sim \mathcal{N}(\mathbf{0}, \sigma_c^2\mathbf{I}_3)$, characterized by

$$\mathbb{E}\big[\mathbf{n}_c(t)\big] = \mathbf{0}_{3\times1}, \qquad \mathbb{E}\big[\mathbf{n}_c(t+\tau)\,\mathbf{n}_c^\top(t)\big] = \sigma_c^2\mathbf{I}_3\,\delta(\tau), \tag{362}$$

where $\delta(\tau)$ is the impulse function: $\delta(\tau) = 1$ if $\tau = 0$ and $\delta(\tau) = 0$ if $\tau \neq 0$.

Consider a random-walk bias driven by this noise, $\dot{\mathbf{b}}(t) = \mathbf{n}_c(t)$. Integrating from t_k to t_{k+1},

$$\int_{t_k}^{t_{k+1}} \dot{\mathbf{b}}(t)\,dt = \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt = \mathbf{b}_{k+1} - \mathbf{b}_k \ \ \Rightarrow\ \ \Delta\mathbf{b} \triangleq \mathbf{b}_{k+1} - \mathbf{b}_k = \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt. \tag{363}$$

Its mean is $\mathbb{E}(\Delta\mathbf{b}) = \mathbb{E}\big(\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt\big) = \mathbf{0}_{3\times1}$, and its covariance is

$$\begin{aligned}
\mathrm{Cov}(\Delta\mathbf{b}) \triangleq \mathbf{P}_{\Delta\mathbf{b}} = \mathbb{E}\big(\Delta\mathbf{b}\,\Delta\mathbf{b}^\top\big) &= \mathbb{E}\left(\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt \cdot \int_{t_k}^{t_{k+1}} \mathbf{n}_c^\top(t)\,dt\right) && (364)\\
&= \mathbb{E}\left(\int_{t_k}^{t_{k+1}}\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,\mathbf{n}_c^\top(s)\,dt\,ds\right) && (365)\\
&= \int_{t_k}^{t_{k+1}} \mathbb{E}\big(\mathbf{n}_c(t)\,\mathbf{n}_c^\top(t)\big)\,dt && (366)\\
&= \int_{t_k}^{t_{k+1}} \sigma_c^2\mathbf{I}_3\,dt = \delta t\,\sigma_c^2\,\mathbf{I}_3, && (367)
\end{aligned}$$

where $\delta t = t_{k+1} - t_k$ (the impulse $\delta(\cdot)$ collapses the double integral). Hence we can **discretize** $\mathbf{n}_c$ in $[t_k, t_{k+1}]$ as

$$\boxed{\mathbf{n}_d \sim \mathcal{N}\left(\mathbf{0},\ \frac{\sigma_c^2}{\delta t}\,\mathbf{I}_3\right), \qquad \delta t = t_{k+1} - t_k.} \tag{368}$$

Then $\Delta\mathbf{b} = \int_{t_k}^{t_{k+1}} \mathbf{n}_c\,dt = \mathbf{n}_d \cdot \delta t$, and the discrete covariance is consistent with the continuous one:

$$\mathrm{Cov}(\Delta\mathbf{b})_d = \mathbb{E}\big(\mathbf{n}_d\mathbf{n}_d^\top\big)\,\delta t^2 = \delta t^2 \cdot \frac{\sigma_c^2}{\delta t}\,\mathbf{I}_3 = \sigma_c^2\,\delta t\,\mathbf{I}_3 = \mathrm{Cov}(\Delta\mathbf{b})_c.\checkmark \tag{369}$$

5.3 The IMU kinematics: a first-order ODE

The IMU state evolves by the first-order ordinary differential equation

$$\begin{cases} {}^G_I\dot{\mathbf{R}} = {}^G_I\mathbf{R} \left\lfloor {}^I\boldsymbol{\omega} \right\rfloor_\times = {}^G_I\mathbf{R} \left\lfloor {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g \right\rfloor_\times, \\ {}^G\dot{\mathbf{p}}_I = {}^G\mathbf{v}_I, \\ {}^G\dot{\mathbf{v}}_I = {}^G\mathbf{a} + {}^G\mathbf{g} = {}^G_I\mathbf{R}\left({}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a\right) + {}^G\mathbf{g}, \\ {}^I\dot{\mathbf{b}}_g = \mathbf{n}_{wg}, \qquad {}^I\dot{\mathbf{b}}_a = \mathbf{n}_{wa}. \end{cases} \tag{370}$$

Stacking, we obtain the compact form

$$\boxed{\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I), \qquad \mathbf{n}_I = \begin{bmatrix} \mathbf{n}_g \\ \mathbf{n}_a \\ \mathbf{n}_{wg} \\ \mathbf{n}_{wa} \end{bmatrix}, \quad \begin{matrix} \mathbf{n}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{I}_3), & \mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2\mathbf{I}_3), \\ \mathbf{n}_{wg} \sim \mathcal{N}(\mathbf{0}, \sigma_{wg}^2\mathbf{I}_3), & \mathbf{n}_{wa} \sim \mathcal{N}(\mathbf{0}, \sigma_{wa}^2\mathbf{I}_3), \end{matrix}} \tag{371}$$

and, per the previous subsection, the discretized noise over one interval is

$$\mathbf{n}_{Id} = \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \\ \mathbf{n}_{wgd} \\ \mathbf{n}_{wad} \end{bmatrix}, \qquad \sigma_{gd}^2 = \frac{\sigma_g^2}{\delta t}, \quad \sigma_{ad}^2 = \frac{\sigma_a^2}{\delta t}, \quad \sigma_{wgd}^2 = \frac{\sigma_{wg}^2}{\delta t}, \quad \sigma_{wad}^2 = \frac{\sigma_{wa}^2}{\delta t}. \tag{372}$$

The IMU model is done. There are various IMU models, but the basic idea is the same.

5.4 IMU integration: the big picture

Recall the typical linear system from Lecture 1 and put it side by side with our sensor models (Fig. 16):

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} & \to \text{state evolution} \to \text{IMU model:} \quad \dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I), \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} & \to \text{observation / measurements} \to \text{vision / camera model:} \quad \mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c. \end{cases} \tag{373}$$

Together they form a **VINS** (visual-inertial navigation system). These two models are the focus of our tutorials: how to process the camera measurement $\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c$ was covered in Lectures 3 and 4; we will learn how to process the IMU in this Lecture 5.

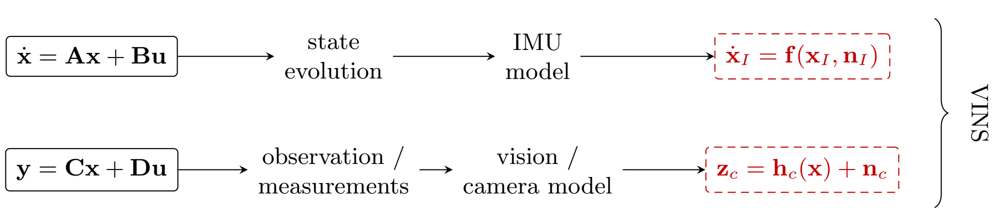


Figure 16: The big picture: the Lecture-1 linear system maps onto the two VINS models — the IMU model plays the state evolution (this lecture) and the camera model plays the observation (Lectures 3–4).

The problem formulation. Given $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k)$ and the IMU readings $({}^{I}\boldsymbol{\omega}_{m_k}, {}^{I}\mathbf{a}_{m_k})$, solve

$$\boxed{\mathbf{x}_{k+1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k+1}, \mathbf{P}_{k+1}) \quad \Rightarrow \quad \text{two targets: } \hat{\mathbf{x}}_{k+1} \text{ and } \mathbf{P}_{k+1}.} \tag{374}$$

That is, the ODE $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ carries the *current state distribution* $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k)$ into the *next state distribution* $\mathbf{x}_{k+1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k+1}, \mathbf{P}_{k+1})$.

How to solve it? From $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ we can get the *integrated function*

$$\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}\big(\mathbf{x}_k,\ {}^{I}\boldsymbol{\omega}_m,\ {}^{I}\mathbf{a}_m,\ \mathbf{n}_k\big), \qquad \begin{cases} \mathbf{x}_{k+1} = \mathbf{x}_{I_{k+1}}, \\ \mathbf{x}_k = \mathbf{x}_{I_k}, \\ \mathbf{n}_k = \mathbf{n}_{I_k} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{I_k}). \end{cases} \tag{375}$$

Hence the **mean** is propagated with zero noise,

$$\hat{\mathbf{x}}_{k+1} = \mathbf{f}_{\text{int}}\big(\hat{\mathbf{x}}_k,\ {}^{I}\boldsymbol{\omega}_m,\ {}^{I}\mathbf{a}_m,\ \mathbf{0}\big) \quad \rightarrow \quad \text{solved } \hat{\mathbf{x}}_{k+1}. \tag{376}$$

Then, linearizing $\mathbf{f}_{\text{int}}$, we get the error-state propagation with the discretized noise $\mathbf{n}_{kd}$,

$$\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\, \tilde{\mathbf{x}}_k + \mathbf{G}_k\, \mathbf{n}_{kd}, \tag{377}$$

and therefore the **covariance**

$$\mathbf{P}_{k+1} = \mathbb{E}\big(\tilde{\mathbf{x}}_{k+1}\tilde{\mathbf{x}}_{k+1}^{\top}\big) = \boldsymbol{\Phi}_{(k+1,k)}\, \mathbf{P}_k\, \boldsymbol{\Phi}_{(k+1,k)}^{\top} + \mathbf{G}_k\, \mathbf{R}_{Id}\, \mathbf{G}_k^{\top} \quad \rightarrow \quad \text{solved } \mathbf{P}_{k+1}. \tag{378}$$

Given the big picture, let's explore the detailed derivations.

5.5 Exact integration from t_k to t_{k+1}

The input. Split each true reading into its estimate and error,

$$ {}^{I}\boldsymbol{\omega} = {}^{I}\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g = \underbrace{\big({}^{I}\boldsymbol{\omega}_m - \hat{\mathbf{b}}_g\big)}_{{}^{I}\hat{\boldsymbol{\omega}}} + \underbrace{\big(-\tilde{\mathbf{b}}_g - \mathbf{n}_g\big)}_{{}^{I}\tilde{\boldsymbol{\omega}}} = {}^{I}\hat{\boldsymbol{\omega}} + {}^{I}\tilde{\boldsymbol{\omega}}, \tag{379}$$

$$ {}^{I}\mathbf{a} = {}^{I}\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a = \underbrace{\big({}^{I}\mathbf{a}_m - \hat{\mathbf{b}}_a\big)}_{{}^{I}\hat{\mathbf{a}}} + \underbrace{\big(-\tilde{\mathbf{b}}_a - \mathbf{n}_a\big)}_{{}^{I}\tilde{\mathbf{a}}} = {}^{I}\hat{\mathbf{a}} + {}^{I}\tilde{\mathbf{a}}. \tag{380}$$

We integrate from k to $k+1$, with $\delta t = t_{k+1} - t_k$.

For rotation.

$$ {}^{G}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R} \cdot {}^{I_k}_{I_{k+1}}\mathbf{R}, \qquad \Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R}, \tag{381}$$

where $\Delta\mathbf{R}$ is the rotation integration from ${}^{I}\boldsymbol{\omega}$.

For velocity. From ${}^G\dot{\mathbf{v}}_{I_\tau} = {}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}$,

$$\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{v}}_{I_\tau}\,d\tau = \int_{t_k}^{t_{k+1}} \left({}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}\right) d\tau. \tag{382}$$

The left-hand side is $\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{v}}_{I_\tau}\,d\tau = {}^G\mathbf{v}_{I_{k+1}} - {}^G\mathbf{v}_{I_k}$, and the right-hand side is

$$\begin{aligned}
\int_{t_k}^{t_{k+1}} \left({}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}\right) d\tau &= \int_{t_k}^{t_{k+1}} {}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\,d\tau + {}^G\mathbf{g}\,\delta t &(383)\\
&= \int_{t_k}^{t_{k+1}} {}^G_{I_k}\mathbf{R}\,{}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\,d\tau + {}^G\mathbf{g}\,\delta t &(384)\\
&= {}^G_{I_k}\mathbf{R}\underbrace{\int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\,d\tau}_{\Delta\mathbf{v}} + {}^G\mathbf{g}\,\delta t. &(385)
\end{aligned}$$

Then

$${}^G\mathbf{v}_{I_{k+1}} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{v} + {}^G\mathbf{g}\,\delta t, \qquad \Delta\mathbf{v} \triangleq \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\,d\tau, \tag{386}$$

where $\Delta\mathbf{v}$ is the integration of ${}^I\boldsymbol{\omega}$ and ${}^I\mathbf{a}$.

For position. From ${}^G\dot{\mathbf{p}}_{I_\tau} = {}^G\mathbf{v}_{I_\tau}$,

$$\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{p}}_{I_\tau}\,d\tau = \int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_\tau}\,d\tau. \tag{387}$$

Note that, by the velocity result applied up to time τ (with $\delta\tau = \tau - t_k$),

$${}^G\mathbf{v}_{I_\tau} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds + {}^G\mathbf{g}\,\delta\tau. \tag{388}$$

The left-hand side is ${}^G\mathbf{p}_{I_{k+1}} - {}^G\mathbf{p}_{I_k}$, and the right-hand side becomes

$$\begin{aligned}
\int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_\tau}\,d\tau &= \int_{t_k}^{t_{k+1}} \left({}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds + {}^G\mathbf{g}\,\delta\tau\right) d\tau &(389)\\
&= \int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_k}\,d\tau + \int_{t_k}^{t_{k+1}} {}^G\mathbf{g}\,\delta\tau\,d\tau + {}^G_{I_k}\mathbf{R}\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds\,d\tau &(390)\\
&= {}^G\mathbf{v}_{I_k}\,\delta t + \frac{1}{2}{}^G\mathbf{g}\,\delta t^2 + {}^G_{I_k}\mathbf{R}\underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds\,d\tau}_{\Delta\mathbf{p}}. &(391)
\end{aligned}$$

Then

$${}^G\mathbf{p}_{I_{k+1}} = {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k}\,\delta t + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{p} + \frac{1}{2}{}^G\mathbf{g}\,\delta t^2, \qquad \Delta\mathbf{p} \triangleq \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds\,d\tau, \tag{392}$$

where $\Delta\mathbf{p}$ is the *double* integration of ${}^I\boldsymbol{\omega}$ and ${}^I\mathbf{a}$.

For the biases. From $\dot{\mathbf{b}}_g = \mathbf{n}_{wg}$,

$$\int_{t_k}^{t_{k+1}} \dot{\mathbf{b}}_g \, d\tau = \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau \quad \Rightarrow \quad \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau, \tag{393}$$

and for the accel bias, $\mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} \, d\tau$.

Finally. Collecting everything:

$$\boxed{\begin{aligned}
{}^{G}_{I_{k+1}}\mathbf{R} &= {}^{G}_{I_k}\mathbf{R} \cdot \Delta\mathbf{R}, & \Delta\mathbf{R} &= {}^{I_k}_{I_{k+1}}\mathbf{R}, \\
{}^{G}\mathbf{p}_{I_{k+1}} &= {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\,\delta t + {}^{G}_{I_k}\mathbf{R}\,\Delta\mathbf{p} + \tfrac{1}{2}{}^{G}\mathbf{g}\,\delta t^2, & \Delta\mathbf{p} &= \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds\,d\tau, \\
{}^{G}\mathbf{v}_{I_{k+1}} &= {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\,\Delta\mathbf{v} + {}^{G}\mathbf{g}\,\delta t, & \Delta\mathbf{v} &= \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\,d\tau, \\
\mathbf{b}_{g_{k+1}} &= \mathbf{b}_{g_k} + \textstyle\int_{t_k}^{t_{k+1}} \mathbf{n}_{wg}\,d\tau, & \Delta\mathbf{b}_g &= \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg}\,d\tau, \\
\mathbf{b}_{a_{k+1}} &= \mathbf{b}_{a_k} + \textstyle\int_{t_k}^{t_{k+1}} \mathbf{n}_{wa}\,d\tau, & \Delta\mathbf{b}_a &= \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa}\,d\tau.
\end{aligned}} \tag{394}$$

This is exactly the integrated function $\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, {}^{I}\boldsymbol{\omega}_m, {}^{I}\mathbf{a}_m, \mathbf{n}_k)$.

> **To this step, no assumption or approximation is made — every step is accurate.** The accuracy of the integration is decided by how $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$ are computed from ${}^{I}\boldsymbol{\omega}$ and ${}^{I}\mathbf{a}$.

5.6 Computing $\Delta\mathbf{R}, \Delta\mathbf{p}, \Delta\mathbf{v}$: preintegration and three methods

The way to compute $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$ given $\boldsymbol{\omega}_m$ and $\mathbf{a}_m$ defines the algorithms. A few discussions follow.

(1) Connection to preintegration. $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$ are also called the *pre-integrated measurements*: they are only affected by ${}^{I}\boldsymbol{\omega}_m$, ${}^{I}\mathbf{a}_m$ and $\mathbf{b}_g$, $\mathbf{b}_a$ (not by the global state). Inverting (394),

$$\Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} \qquad = {}^{G}_{I_k}\mathbf{R}^\top\, {}^{G}_{I_{k+1}}\mathbf{R}, \tag{395}$$

$$\Delta\mathbf{p} \triangleq \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\,ds\,d\tau \qquad = {}^{G}_{I_k}\mathbf{R}^\top\big({}^{G}\mathbf{p}_{I_{k+1}} - {}^{G}\mathbf{p}_{I_k} - {}^{G}\mathbf{v}_{I_k}\,\delta t - \tfrac{1}{2}{}^{G}\mathbf{g}\,\delta t^2\big), \tag{396}$$

$$\Delta\mathbf{v} \triangleq \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\,d\tau \qquad = {}^{G}_{I_k}\mathbf{R}^\top\big({}^{G}\mathbf{v}_{I_{k+1}} - {}^{G}\mathbf{v}_{I_k} - {}^{G}\mathbf{g}\,\delta t\big), \tag{397}$$

$$\Delta\mathbf{b}_g \triangleq \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg}\,d\tau \qquad = \mathbf{b}_{g_{k+1}} - \mathbf{b}_{g_k}, \tag{398}$$

$$\Delta\mathbf{b}_a \triangleq \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa}\,d\tau \qquad = \mathbf{b}_{a_{k+1}} - \mathbf{b}_{a_k}. \tag{399}$$

So the IMU integration/propagation and pre-integration are connected: they compute the same quantities.

(2) Discrete preintegration on the manifold. For example, IMU preintegration on manifold (Forster, Carlone, et al.) [9] and VINS-Mono (Tong Qin, Shaojie Shen) [13]. Assume ${}^{I_k}\boldsymbol{\omega}$ and ${}^{I_k}\mathbf{a}$ are constant in $[t_k, t_{k+1}]$ *and* $\delta t = t_{k+1} - t_k$ is small, so ${}^{I_k}_{I_\tau}\mathbf{R} \approx \mathbf{I}$ as $\delta t \to 0$:

$$\Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} \approx \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega} \cdot \delta t\right), \tag{400}$$

$$\Delta\mathbf{p} = \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\, {}^{I_s}\mathbf{a}\, ds\, d\tau \approx \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \mathbf{I}_3 \cdot {}^{I_k}\mathbf{a}\, ds\, d\tau = \frac{1}{2}\, {}^{I_k}\mathbf{a}\, \delta t^2, \tag{401}$$

$$\Delta\mathbf{v} = \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\, {}^{I_\tau}\mathbf{a}\, d\tau \approx {}^{I_k}\mathbf{a}\, \delta t. \tag{402}$$

Then the propagation becomes

$$\begin{cases} {}^{G}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R} \cdot \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\, \delta t\right), \\ {}^{G}\mathbf{p}_{I_{k+1}} = {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\, \delta t + \frac{1}{2}\, {}^{G}_{I_k}\mathbf{R}\, {}^{I_k}\mathbf{a}\, \delta t^2 + \frac{1}{2}\, {}^{G}\mathbf{g}\, \delta t^2, \\ {}^{G}\mathbf{v}_{I_{k+1}} = {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\, {}^{I_k}\mathbf{a}\, \delta t + {}^{G}\mathbf{g}\, \delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg}\, d\tau, \qquad \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa}\, d\tau. \end{cases} \tag{403}$$

This is easy and simple, but it might lose some accuracy for fast rotation and low sampling rate (δt larger).

(3) Analytic integration: ACI^2 and CPI. ACI^2 — analytic combined IMU integration (Yulin/Chuchu) [14]; CPI — continuous preintegration (Kevin/Patrick) [15, 10]. If we assume only that ${}^{I_k}\boldsymbol{\omega}$ and ${}^{I_k}\mathbf{a}$ are constant in $[t_k, t_{k+1}]$ (without the small-δt approximation), then ${}^{I_k}_{I_\tau}\mathbf{R} = \operatorname{Exp}({}^{I_k}\boldsymbol{\omega}\, \delta\tau)$ and

$$\Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} = \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega} \cdot \delta t\right), \tag{404}$$

$$\Delta\mathbf{p} = \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\, {}^{I_s}\mathbf{a}\, ds\, d\tau = \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\, \delta s\right) {}^{I_k}\mathbf{a}\, ds\, d\tau, \tag{405}$$

$$\Delta\mathbf{v} = \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\, {}^{I_\tau}\mathbf{a}\, d\tau = \int_{t_k}^{t_{k+1}} \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\, \delta\tau\right) {}^{I_k}\mathbf{a}\, d\tau, \tag{406}$$

which admit closed-form (analytic) solutions.

(4) Two ways to solve for $\hat{\mathbf{x}}_{k+1}$ and $\mathbf{P}_{k+1}$. We can use different ways to solve for $\hat{\mathbf{x}}_{k+1}$ and $\mathbf{P}_{k+1}$.

Way 1 — discretize, then linearize. Integrate the nonlinear ODE first, then linearize the integrated function (the route taken in this lecture):

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \quad\rightarrow\quad \mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_k) \quad\rightarrow\quad \begin{cases} \hat{\mathbf{x}}_{k+1} = \mathbf{f}_{\text{int}}(\hat{\mathbf{x}}_k, \mathbf{0}), \\ \tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\, \tilde{\mathbf{x}}_k + \mathbf{G}_k\, \mathbf{n}_{kd}. \end{cases} \tag{407}$$

Way 2 — linearize, then integrate. Linearize the continuous ODE first to get the continuous error-state equation, then integrate it:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \quad\Rightarrow\quad \dot{\tilde{\mathbf{x}}} = \mathbf{F}\, \tilde{\mathbf{x}} + \mathbf{J}\, \mathbf{n}_I, \tag{408}$$

whose solution over $[t_k, t_{k+1}]$ is

$$\tilde{\mathbf{x}}_{k+1} = \mathbf{\Phi}_{(k+1,k)}\,\tilde{\mathbf{x}}_k + \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{n}_I\,d\tau, \qquad \mathbf{\Phi}_{(k+1,k)} = \exp\left(\int_{t_k}^{t_{k+1}} \mathbf{F}\,dt\right). \tag{409}$$

Different systems treat this state-transition matrix differently:

- **VINS-Mono** [13]: take $\mathbf{F}$ constant and $\delta t \to 0$, so $\mathbf{\Phi}_{(k+1,k)} = \exp\left(\int_{t_k}^{t_{k+1}} \mathbf{F}\,dt\right) \approx \mathbf{I} + \mathbf{F}\,\delta t$;
- **OC-VINS** (Joel Hesch) [16] / **CPI** (Kevin/Patrick) [15] / **ACI**^{2} (Yulin/Chuchu) [14]: keep $\mathbf{\Phi}_{(k+1,k)} = \exp\left(\int_{t_k}^{t_{k+1}} \mathbf{F}\,dt\right)$, evaluated with ${}^I\boldsymbol{\omega}_m$, ${}^I\mathbf{a}_m$ constant in $[t_k, t_{k+1}]$.

How to address the noise term of (409),

$$\mathbf{n}_{\text{int}} \triangleq \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{n}_I\,d\tau\,? \tag{410}$$

(1) Pull the *discrete noise* $\mathbf{n}_{Id}$ out of the integral:

$$\mathbf{n}_{\text{int}} = \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,d\tau \cdot \mathbf{n}_{Id}, \tag{411}$$

$$\Rightarrow\ \text{Cov}(\mathbf{n}_{\text{int}}) = \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\mathbf{J}(\tau)\,d\tau \cdot \mathbf{Q}_{Id} \cdot \left(\int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\mathbf{J}(\tau)\,d\tau\right)^{\top}. \tag{412}$$

(2) Keep the noise inside the integral:

$$\text{Cov}(\mathbf{n}_{\text{int}}) = \mathbb{E}\left(\int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{n}_{Id}\,\mathbf{n}_{Id}^{\top}\,\mathbf{J}^{\top}(\tau)\,\mathbf{\Phi}_{(k+1,\tau)}^{\top}\,d\tau\right) \tag{413}$$

$$= \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{Q}_{Id}\,\mathbf{J}^{\top}(\tau)\,\mathbf{\Phi}_{(k+1,\tau)}^{\top}\,d\tau. \tag{414}$$

Either way, the *linearization* itself can be carried out with different error parametrizations, all 15 DoF for $\mathbf{x}_I$:

$$\boxed{\begin{cases} SO(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 & \Rightarrow\ 15 \text{ DoF for } \mathbf{x}_I, \\ SE_2(3) + \mathbb{R}^3 + \mathbb{R}^3 & \Rightarrow\ 15 \text{ DoF for } \mathbf{x}_I, \\ SE(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 & \Rightarrow\ 15 \text{ DoF for } \mathbf{x}_I. \end{cases}} \tag{415}$$

We now use the analytic form from (3) for the detailed mean and covariance propagation, working out the first two parametrizations: $SO(3) + \mathbb{R}^{12}$ in §5.7 and $SE_2(3) + \mathbb{R}^6$ in the final subsection.

5.7 Mean and covariance propagation with $SO(3) + \mathbb{R}^3$ errors

The input, discretized. As in §5.5, but discretizing the noises over the interval ($\mathbf{n}_g \to \mathbf{n}_{gd}$, $\mathbf{n}_a \to \mathbf{n}_{ad}$):

$${}^I\boldsymbol{\omega} = {}^I\hat{\boldsymbol{\omega}} + {}^I\tilde{\boldsymbol{\omega}}, \quad \begin{cases} {}^I\hat{\boldsymbol{\omega}} = {}^I\boldsymbol{\omega}_m - \hat{\mathbf{b}}_g \\ {}^I\tilde{\boldsymbol{\omega}} = -\tilde{\mathbf{b}}_g - \mathbf{n}_{gd} \end{cases} \qquad {}^I\mathbf{a} = {}^I\hat{\mathbf{a}} + {}^I\tilde{\mathbf{a}}, \quad \begin{cases} {}^I\hat{\mathbf{a}} = {}^I\mathbf{a}_m - \hat{\mathbf{b}}_a \\ {}^I\tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_a - \mathbf{n}_{ad} \end{cases} \tag{416}$$

Let's integrate $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$, $\Delta\mathbf{b}_g$, $\Delta\mathbf{b}_a$ with these perturbed inputs.

Perturbing $\Delta\mathbf{R}$. With $\hat{\boldsymbol{\theta}} \triangleq {}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t$ and the right Jacobian $\mathbf{J}_r$ from Lecture 2,

$$\begin{aligned}\Delta\mathbf{R} = \mathrm{Exp}\left({}^{I_k}\boldsymbol{\omega}\,\delta t\right) &= \mathrm{Exp}\left(({}^{I_k}\hat{\boldsymbol{\omega}} + {}^{I_k}\tilde{\boldsymbol{\omega}})\,\delta t\right) && (417)\\ &= \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\,\mathrm{Exp}\left(\mathbf{J}_r(\hat{\boldsymbol{\theta}})\,{}^{I_k}\tilde{\boldsymbol{\omega}}\,\delta t\right) && (418)\\ &\approx \mathrm{Exp}\left(\hat{\boldsymbol{\theta}}\right)\left(\mathbf{I} + \left\lfloor \mathbf{J}_r(\hat{\boldsymbol{\theta}})\,\delta t\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right\rfloor_\times\right). && (419)\end{aligned}$$

Perturbing $\Delta\mathbf{p}$. Substituting the split inputs and expanding to first order (using $\lfloor\mathbf{x}\rfloor_\times\mathbf{y} = -\lfloor\mathbf{y}\rfloor_\times\mathbf{x}$ and dropping second-order error terms),

$$\begin{aligned}\Delta\mathbf{p} &= \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \mathrm{Exp}\left(({}^{I_k}\hat{\boldsymbol{\omega}} + {}^{I_k}\tilde{\boldsymbol{\omega}})\,\delta s\right) ds\,d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) && (420)\\ &= \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)\left(\mathbf{I} + \left\lfloor \mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s)\,\delta s\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right\rfloor_\times\right) ds\,d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) && (421)\\ &= \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right) ds\,d\tau\;\;{}^{I_k}\hat{\mathbf{a}}}_{\Delta\hat{\mathbf{p}}} + \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right) ds\,d\tau}_{\boldsymbol{\Xi}_2}\;\;{}^{I_k}\tilde{\mathbf{a}}\\ &\quad - \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)\left\lfloor{}^{I_k}\hat{\mathbf{a}}\right\rfloor_\times\,\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)\delta s\,ds\,d\tau}_{\boldsymbol{\Xi}_4}\;\;{}^{I_k}\tilde{\boldsymbol{\omega}} && (422)\\ &= \Delta\hat{\mathbf{p}} + \boldsymbol{\Xi}_2\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_4\,{}^{I_k}\tilde{\boldsymbol{\omega}}. && (423)\end{aligned}$$

Perturbing $\Delta\mathbf{v}$. By the same expansion,

$$\begin{aligned}\Delta\mathbf{v} &= \int_{t_k}^{t_{k+1}} \mathrm{Exp}\left(({}^{I_k}\hat{\boldsymbol{\omega}} + {}^{I_k}\tilde{\boldsymbol{\omega}})\,\delta\tau\right) d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) && (424)\\ &= \int_{t_k}^{t_{k+1}} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)\left(\mathbf{I} + \left\lfloor \mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau)\,\delta\tau\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right\rfloor_\times\right) d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) && (425)\\ &= \underbrace{\int_{t_k}^{t_{k+1}} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right) d\tau\;\;{}^{I_k}\hat{\mathbf{a}}}_{\Delta\hat{\mathbf{v}}} + \underbrace{\int_{t_k}^{t_{k+1}} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right) d\tau}_{\boldsymbol{\Xi}_1}\;\;{}^{I_k}\tilde{\mathbf{a}}\\ &\quad - \underbrace{\int_{t_k}^{t_{k+1}} \mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)\left\lfloor{}^{I_k}\hat{\mathbf{a}}\right\rfloor_\times\,\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)\delta\tau\,d\tau}_{\boldsymbol{\Xi}_3}\;\;{}^{I_k}\tilde{\boldsymbol{\omega}} && (426)\\ &= \Delta\hat{\mathbf{v}} + \boldsymbol{\Xi}_1\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_3\,{}^{I_k}\tilde{\boldsymbol{\omega}}. && (427)\end{aligned}$$

The perturbed propagation. Then we have

$$\begin{cases} {}^{G}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R}\cdot\mathrm{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\cdot\mathrm{Exp}\left(\mathbf{J}_r(\hat{\boldsymbol{\theta}})\,\delta t\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right),\\ {}^{G}\mathbf{p}_{I_{k+1}} = {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\,\delta t + {}^{G}_{I_k}\mathbf{R}\left(\Delta\hat{\mathbf{p}} + \boldsymbol{\Xi}_2\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_4\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right) + \frac{1}{2}{}^{G}\mathbf{g}\,\delta t^2,\\ {}^{G}\mathbf{v}_{I_{k+1}} = {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\left(\Delta\hat{\mathbf{v}} + \boldsymbol{\Xi}_1\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_3\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right) + {}^{G}\mathbf{g}\,\delta t,\\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \mathbf{n}_{wgd}\,\delta t,\\ \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \mathbf{n}_{wad}\,\delta t. \end{cases} \qquad (428)$$

First: the mean propagation. Setting all errors and noises to zero,

$$\boxed{\begin{cases} {}^{G}_{I_{k+1}}\hat{\mathbf{R}} = {}^{G}_{I_k}\hat{\mathbf{R}} \cdot \operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right), \\ {}^{G}\hat{\mathbf{p}}_{I_{k+1}} = {}^{G}\hat{\mathbf{p}}_{I_k} + {}^{G}\hat{\mathbf{v}}_{I_k}\,\delta t + {}^{G}_{I_k}\hat{\mathbf{R}}\,\Delta\hat{\mathbf{p}} + \frac{1}{2}{}^{G}\mathbf{g}\,\delta t^2, \\ {}^{G}\hat{\mathbf{v}}_{I_{k+1}} = {}^{G}\hat{\mathbf{v}}_{I_k} + {}^{G}_{I_k}\hat{\mathbf{R}}\,\Delta\hat{\mathbf{v}} + {}^{G}\mathbf{g}\,\delta t, \\ \hat{\mathbf{b}}_{g_{k+1}} = \hat{\mathbf{b}}_{g_k}, \qquad \hat{\mathbf{b}}_{a_{k+1}} = \hat{\mathbf{b}}_{a_k}. \end{cases}} \tag{429}$$

Hence $\hat{\mathbf{x}}_k \to \hat{\mathbf{x}}_{k+1}$: solved.

Second: the linearization for covariance propagation. Use $SO(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3$ for example, i.e., a right rotation perturbation ${}^{G}_{I}\mathbf{R} = {}^{G}_{I}\hat{\mathbf{R}}\,\operatorname{Exp}(\delta\boldsymbol{\theta})$ and additive errors elsewhere. For the rotation,

$${}^{G}_{I_{k+1}}\hat{\mathbf{R}}\,\operatorname{Exp}(\delta\boldsymbol{\theta}_{k+1}) = {}^{G}_{I_k}\hat{\mathbf{R}}\,\operatorname{Exp}(\delta\boldsymbol{\theta}_k)\,\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\,\operatorname{Exp}\left(\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t)\,\delta t\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right), \tag{430}$$

so, moving $\operatorname{Exp}(\delta\boldsymbol{\theta}_k)$ across $\operatorname{Exp}({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t) = \Delta\hat{\mathbf{R}}$ with the adjoint identity,

$$\delta\boldsymbol{\theta}_{k+1} = {}^{I_k}_{I_{k+1}}\hat{\mathbf{R}}^{\top}\,\delta\boldsymbol{\theta}_k + \mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\delta t\,{}^{I_k}\tilde{\boldsymbol{\omega}}. \tag{431}$$

For the position and velocity (substituting ${}^{G}_{I_k}\mathbf{R} = {}^{G}_{I_k}\hat{\mathbf{R}}\,\operatorname{Exp}(\delta\boldsymbol{\theta}_k) \approx {}^{G}_{I_k}\hat{\mathbf{R}}(\mathbf{I} + \lfloor\delta\boldsymbol{\theta}_k\rfloor_\times)$ and keeping first-order terms),

$${}^{G}\tilde{\mathbf{p}}_{I_{k+1}} = {}^{G}\tilde{\mathbf{p}}_{I_k} + {}^{G}\tilde{\mathbf{v}}_{I_k}\,\delta t - {}^{G}_{I_k}\hat{\mathbf{R}}\,\lfloor\Delta\hat{\mathbf{p}}\rfloor_\times\,\delta\boldsymbol{\theta}_k + {}^{G}_{I_k}\hat{\mathbf{R}}\left(\boldsymbol{\Xi}_2\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_4\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right), \tag{432}$$

$${}^{G}\tilde{\mathbf{v}}_{I_{k+1}} = {}^{G}\tilde{\mathbf{v}}_{I_k} - {}^{G}_{I_k}\hat{\mathbf{R}}\,\lfloor\Delta\hat{\mathbf{v}}\rfloor_\times\,\delta\boldsymbol{\theta}_k + {}^{G}_{I_k}\hat{\mathbf{R}}\left(\boldsymbol{\Xi}_1\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_3\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right), \tag{433}$$

$$\tilde{\mathbf{b}}_{g_{k+1}} = \tilde{\mathbf{b}}_{g_k} + \mathbf{n}_{wgd}\,\delta t, \qquad \tilde{\mathbf{b}}_{a_{k+1}} = \tilde{\mathbf{b}}_{a_k} + \mathbf{n}_{wad}\,\delta t, \tag{434}$$

where the input errors are

$${}^{I_k}\tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_a - \mathbf{n}_{ad}, \qquad {}^{I_k}\tilde{\boldsymbol{\omega}} = -\tilde{\mathbf{b}}_g - \mathbf{n}_{gd}. \tag{435}$$

Hence, stacking the error state $\tilde{\mathbf{x}} = [\delta\boldsymbol{\theta}^{\top}, {}^{G}\tilde{\mathbf{p}}_I^{\top}, {}^{G}\tilde{\mathbf{v}}_I^{\top}, \tilde{\mathbf{b}}_g^{\top}, \tilde{\mathbf{b}}_a^{\top}]^{\top}$,

$$\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\,\tilde{\mathbf{x}}_k + \mathbf{G}_k\,\mathbf{n}_{Id}, \qquad \mathbf{n}_{Id} = \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \\ \mathbf{n}_{wgd} \\ \mathbf{n}_{wad} \end{bmatrix}, \tag{436}$$

with the state-transition matrix

$$\boldsymbol{\Phi}_{(k+1,k)} = \begin{bmatrix} {}^{I_k}_{I_{k+1}}\hat{\mathbf{R}}^{\top} & \mathbf{0}_3 & \mathbf{0}_3 & -\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\delta t & \mathbf{0}_3 \\ -{}^{G}_{I_k}\hat{\mathbf{R}}\,\lfloor\Delta\hat{\mathbf{p}}\rfloor_\times & \mathbf{I}_3 & \mathbf{I}_3\,\delta t & {}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_4 & -{}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_2 \\ -{}^{G}_{I_k}\hat{\mathbf{R}}\,\lfloor\Delta\hat{\mathbf{v}}\rfloor_\times & \mathbf{0}_3 & \mathbf{I}_3 & {}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_3 & -{}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_1 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \tag{437}$$

and the noise Jacobian

$$\mathbf{G}_k = \begin{bmatrix} -\mathbf{J}_r\big({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\big)\,\delta t & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ {}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_4 & -{}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_2 & \mathbf{0}_3 & \mathbf{0}_3 \\ {}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_3 & -{}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_1 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t \end{bmatrix}. \tag{438}$$

Hence the covariance propagation

$$\boxed{\mathbf{P}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\,\mathbf{P}_k\,\boldsymbol{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k\,\mathbf{Q}_d\,\mathbf{G}_k^\top \quad \rightarrow \quad \text{solved.}} \tag{439}$$

5.8 The linearization using $SE_2(3)$

Now the linearization using $SE_2(3)$ (Lecture 2). The states live on $SE_2(3) + \mathbb{R}^3 + \mathbb{R}^3$:

$$\mathbf{x}_I = (\mathbf{X}_n, \mathbf{x}_b), \qquad \mathbf{X}_n = \big({}^G_I\mathbf{R}, {}^G\mathbf{p}_I, {}^G\mathbf{v}_I\big) = \begin{bmatrix} {}^G_I\mathbf{R} & {}^G\mathbf{p}_I & {}^G\mathbf{v}_I \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}, \qquad \mathbf{x}_b = (\mathbf{b}_g, \mathbf{b}_a). \tag{440}$$

The error states. Define a left (group) error on $\mathbf{X}_n$ and additive errors on the biases,

$$\mathbf{x}_I = \hat{\mathbf{x}}_I \boxplus \delta\mathbf{x}_I = \big(\operatorname{Exp}(\delta\mathbf{x}_n)\,\hat{\mathbf{X}}_n,\ \hat{\mathbf{x}}_b + \tilde{\mathbf{x}}_b\big), \tag{441}$$

which, writing $\delta\mathbf{x}_n = [\delta\boldsymbol{\theta}^\top, \delta\mathbf{p}^\top, \delta\mathbf{v}^\top]^\top$ and using the $SE_2(3)$ exponential (with the left Jacobian $\mathbf{J}_l$),

$$\begin{cases} {}^G_I\mathbf{R} = \operatorname{Exp}(\delta\boldsymbol{\theta})\,{}^G_I\hat{\mathbf{R}}, \\ {}^G\mathbf{p}_I = \operatorname{Exp}(\delta\boldsymbol{\theta})\,{}^G\hat{\mathbf{p}}_I + \mathbf{J}_l(\delta\boldsymbol{\theta})\,\delta\mathbf{p}, \\ {}^G\mathbf{v}_I = \operatorname{Exp}(\delta\boldsymbol{\theta})\,{}^G\hat{\mathbf{v}}_I + \mathbf{J}_l(\delta\boldsymbol{\theta})\,\delta\mathbf{v}, \\ \mathbf{b}_g = \hat{\mathbf{b}}_g + \delta\mathbf{b}_g, \qquad \mathbf{b}_a = \hat{\mathbf{b}}_a + \delta\mathbf{b}_a. \end{cases} \tag{442}$$

The propagation as a group factorization. The propagation equations

$$\begin{cases} {}^G_{I_{k+1}}\mathbf{R} = {}^G_{I_k}\mathbf{R}\cdot\Delta\mathbf{R}, \\ {}^G\mathbf{p}_{I_{k+1}} = {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k}\,\delta t + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{p} + \frac{1}{2}{}^G\mathbf{g}\,\delta t^2, \\ {}^G\mathbf{v}_{I_{k+1}} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{v} + {}^G\mathbf{g}\,\delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \mathbf{n}_{wgd}\,\delta t, \qquad \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \mathbf{n}_{wad}\,\delta t \end{cases} \tag{443}$$

factor into a product of three $SE_2(3)$ elements,

$$\boxed{\mathbf{X}_{n_{k+1}} = \boldsymbol{\Gamma}_k\cdot\boldsymbol{\Psi}(\mathbf{X}_{n_k})\cdot\mathbf{Y}_k,} \tag{444}$$

with

$$\boldsymbol{\Gamma}_k = \begin{bmatrix} \mathbf{I}_3 & \frac{1}{2}{}^G\mathbf{g}\,\delta t^2 & {}^G\mathbf{g}\,\delta t \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}, \qquad \mathbf{Y}_k = \begin{bmatrix} \Delta\mathbf{R} & \Delta\mathbf{p} & \Delta\mathbf{v} \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}, \tag{445}$$

$$\boldsymbol{\Psi}(\mathbf{X}_{n_k}) = \begin{bmatrix} {}^G_{I_k}\mathbf{R} & {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k}\,\delta t & {}^G\mathbf{v}_{I_k} \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}. \tag{446}$$

The properties of $\boldsymbol{\Psi}(\cdot)$. The map $\boldsymbol{\Psi}$ satisfies

$$\begin{cases} \boldsymbol{\Psi}(\mathbf{X}_{n_i} \cdot \mathbf{X}_{n_j}) = \boldsymbol{\Psi}(\mathbf{X}_{n_i}) \cdot \boldsymbol{\Psi}(\mathbf{X}_{n_j}), \\ \boldsymbol{\Psi}\big(\operatorname{Exp}(\delta \mathbf{x}_n)\big) \approx \operatorname{Exp}\left(\mathbf{F}\, \delta \mathbf{x}_n\right), \\ \boldsymbol{\Psi}\big(\operatorname{Exp}(\delta \mathbf{x}_n)\, \hat{\mathbf{X}}_n\big) = \boldsymbol{\Psi}\big(\operatorname{Exp}(\delta \mathbf{x}_n)\big) \cdot \boldsymbol{\Psi}(\hat{\mathbf{X}}_n) = \operatorname{Exp}\left(\mathbf{F}\, \delta \mathbf{x}_n\right) \boldsymbol{\Psi}(\hat{\mathbf{X}}_n), \end{cases} \qquad \mathbf{F} \triangleq \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{I}_3\, \delta t \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}. \tag{447}$$

The linearization. Hence we can do the linearization as

$$\begin{aligned} \mathbf{X}_{n_{k+1}} = \operatorname{Exp}\left(\delta \mathbf{x}_{n_{k+1}}\right) \hat{\mathbf{X}}_{n_{k+1}} &= \boldsymbol{\Gamma}_k\, \boldsymbol{\Psi}(\mathbf{X}_{n_k})\, \mathbf{Y}_k && (448) \\ &= \boldsymbol{\Gamma}_k\, \boldsymbol{\Psi}\big(\operatorname{Exp}(\delta \mathbf{x}_{n_k})\, \hat{\mathbf{X}}_{n_k}\big)\, \hat{\mathbf{Y}}_k \cdot \tilde{\mathbf{Y}}_k && (449) \\ &\approx \boldsymbol{\Gamma}_k\, \operatorname{Exp}\left(\mathbf{F}\, \delta \mathbf{x}_{n_k}\right) \boldsymbol{\Psi}(\hat{\mathbf{X}}_{n_k})\, \hat{\mathbf{Y}}_k\, \tilde{\mathbf{Y}}_k && (450) \\ &= \operatorname{Exp}\left(\operatorname{Ad}_{\boldsymbol{\Gamma}_k} \mathbf{F}\, \delta \mathbf{x}_{n_k}\right) \underbrace{\boldsymbol{\Gamma}_k\, \boldsymbol{\Psi}(\hat{\mathbf{X}}_{n_k})\, \hat{\mathbf{Y}}_k}_{\hat{\mathbf{X}}_{n_{k+1}}}\, \tilde{\mathbf{Y}}_k && (451) \\ &= \operatorname{Exp}\left(\operatorname{Ad}_{\boldsymbol{\Gamma}_k} \mathbf{F}\, \delta \mathbf{x}_{n_k}\right) \hat{\mathbf{X}}_{n_{k+1}}\, \tilde{\mathbf{Y}}_k && (452) \\ &= \operatorname{Exp}\left(\boldsymbol{\Phi}_{nn}\, \delta \mathbf{x}_{n_k}\right) \hat{\mathbf{X}}_{n_{k+1}}\, \tilde{\mathbf{Y}}_k, \qquad \boxed{\boldsymbol{\Phi}_{nn} \triangleq \operatorname{Ad}_{\boldsymbol{\Gamma}_k} \mathbf{F}} && (453) \end{aligned}$$

where the preintegration factor splits as $\mathbf{Y}_k = \hat{\mathbf{Y}}_k\, \tilde{\mathbf{Y}}_k$,

$$\mathbf{Y}_k = \hat{\mathbf{Y}}_k\, \tilde{\mathbf{Y}}_k = \begin{bmatrix} \Delta\hat{\mathbf{R}} & \Delta\hat{\mathbf{p}} & \Delta\hat{\mathbf{v}} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\tilde{\mathbf{R}} & \Delta\hat{\mathbf{R}}^\top \Delta\tilde{\mathbf{p}} & \Delta\hat{\mathbf{R}}^\top \Delta\tilde{\mathbf{v}} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix} = \hat{\mathbf{Y}}_k \cdot \operatorname{Exp}\left(\begin{bmatrix} \delta\Delta\boldsymbol{\theta} \\ \Delta\hat{\mathbf{R}}^\top \Delta\tilde{\mathbf{p}} \\ \Delta\hat{\mathbf{R}}^\top \Delta\tilde{\mathbf{v}} \end{bmatrix} \right), \tag{454}$$

with $\delta\Delta\boldsymbol{\theta} = \operatorname{Log}(\Delta\tilde{\mathbf{R}})$ (to first order). From the perturbations of §5.7,

$$\begin{aligned} \begin{bmatrix} \delta\Delta\boldsymbol{\theta} \\ \Delta\hat{\mathbf{R}}^\top \Delta\tilde{\mathbf{p}} \\ \Delta\hat{\mathbf{R}}^\top \Delta\tilde{\mathbf{v}} \end{bmatrix} &= \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top \end{bmatrix} \begin{bmatrix} \delta\Delta\boldsymbol{\theta} \\ \Delta\tilde{\mathbf{p}} \\ \Delta\tilde{\mathbf{v}} \end{bmatrix} && (455) \\ &= \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top \end{bmatrix} \begin{bmatrix} -\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\, \delta t & \mathbf{0}_3 \\ \boldsymbol{\Xi}_4 & -\boldsymbol{\Xi}_2 \\ \boldsymbol{\Xi}_3 & -\boldsymbol{\Xi}_1 \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix} && (456) \\ &= \begin{bmatrix} -\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\, \delta t & \mathbf{0}_3 \\ \Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_4 & -\Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_2 \\ \Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_3 & -\Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_1 \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix}, && (457) \end{aligned}$$

with $\Delta\hat{\boldsymbol{\theta}} = {}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t$. Finally, move $\tilde{\mathbf{Y}}_k$ across $\hat{\mathbf{X}}_{n_{k+1}}$ with the adjoint:

$$\begin{aligned} \operatorname{Exp}\left(\delta \mathbf{x}_{n_{k+1}}\right) &= \operatorname{Exp}\left(\boldsymbol{\Phi}_{nn}\, \delta \mathbf{x}_{n_k}\right) \hat{\mathbf{X}}_{n_{k+1}}\, \tilde{\mathbf{Y}}_k\, \hat{\mathbf{X}}_{n_{k+1}}^{-1} && (458) \\ &= \operatorname{Exp}\left(\boldsymbol{\Phi}_{nn}\, \delta \mathbf{x}_{n_k}\right) \operatorname{Exp}\left(\operatorname{Ad}_{\hat{\mathbf{X}}_{n_{k+1}}} \begin{bmatrix} -\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\, \delta t & \mathbf{0}_3 \\ \Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_4 & -\Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_2 \\ \Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_3 & -\Delta\hat{\mathbf{R}}^\top \boldsymbol{\Xi}_1 \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix} \right), && (459) \end{aligned}$$

so that

$$\delta \mathbf{x}_{n_{k+1}} = \mathbf{\Phi}_{nn}\,\delta \mathbf{x}_{n_k} + \mathbf{\Phi}_{nb}\begin{bmatrix}\tilde{\mathbf{b}}_g + \mathbf{n}_{gd}\\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad}\end{bmatrix}, \qquad \boxed{\mathbf{\Phi}_{nb} \triangleq \mathrm{Ad}_{\hat{\mathbf{X}}_{n_{k+1}}}\begin{bmatrix}-\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\,\delta t & \mathbf{0}_3\\ \Delta\hat{\mathbf{R}}^\top\,\mathbf{\Xi}_4 & -\Delta\hat{\mathbf{R}}^\top\,\mathbf{\Xi}_2\\ \Delta\hat{\mathbf{R}}^\top\,\mathbf{\Xi}_3 & -\Delta\hat{\mathbf{R}}^\top\,\mathbf{\Xi}_1\end{bmatrix}} \tag{460}$$

The stacked system. Then

$$\begin{cases}\delta \mathbf{x}_{n_{k+1}} = \mathbf{\Phi}_{nn}\,\delta \mathbf{x}_{n_k} + \mathbf{\Phi}_{nb}\begin{bmatrix}\tilde{\mathbf{b}}_g\\ \tilde{\mathbf{b}}_a\end{bmatrix} + \mathbf{\Phi}_{nb}\begin{bmatrix}\mathbf{n}_{gd}\\ \mathbf{n}_{ad}\end{bmatrix},\\ \tilde{\mathbf{b}}_{g_{k+1}} = \tilde{\mathbf{b}}_{g_k} + \mathbf{n}_{wgd}\,\delta t, \qquad \tilde{\mathbf{b}}_{a_{k+1}} = \tilde{\mathbf{b}}_{a_k} + \mathbf{n}_{wad}\,\delta t,\end{cases} \tag{461}$$

which stacks into $\tilde{\mathbf{x}}_{k+1} = \mathbf{\Phi}_{(k+1,k)}\tilde{\mathbf{x}}_k + \mathbf{G}_k\,\mathbf{n}_{Id}$ with

$$\mathbf{\Phi}_{(k+1,k)} = \begin{bmatrix}\mathbf{\Phi}_{nn} & \mathbf{\Phi}_{nb}\\ \mathbf{0}_{6\times 9} & \mathbf{I}_6\end{bmatrix} = \begin{bmatrix}\mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{\Phi}_{14} & \mathbf{0}_3\\ \frac{1}{2}\left\lfloor {}^G\mathbf{g}\right\rfloor_\times\,\delta t^2 & \mathbf{I}_3 & \mathbf{I}_3\,\delta t & \mathbf{\Phi}_{24} & \mathbf{\Phi}_{25}\\ \left\lfloor {}^G\mathbf{g}\right\rfloor_\times\,\delta t & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{\Phi}_{34} & \mathbf{\Phi}_{35}\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\end{bmatrix}, \tag{462}$$

where the first three block-columns are exactly $\mathbf{\Phi}_{nn} = \mathrm{Ad}_{\mathbf{\Gamma}_k}\,\mathbf{F}$ and the $\mathbf{\Phi}_{i4}, \mathbf{\Phi}_{i5}$ blocks are the columns of $\mathbf{\Phi}_{nb}$ ($\mathbf{\Phi}_{15} = \mathbf{0}_3$ since the accel error does not enter the rotation), and

$$\mathbf{G}_k = \begin{bmatrix}\mathbf{\Phi}_{nb} & \mathbf{0}_{9\times 6}\\ \mathbf{0}_{6\times 6} & \mathbf{I}_6\,\delta t\end{bmatrix} = \begin{bmatrix}\mathbf{G}_{11} & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3\\ \mathbf{G}_{21} & \mathbf{G}_{22} & \mathbf{0}_3 & \mathbf{0}_3\\ \mathbf{G}_{31} & \mathbf{G}_{32} & \mathbf{0}_3 & \mathbf{0}_3\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t & \mathbf{0}_3\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t\end{bmatrix}, \tag{463}$$

with $[\mathbf{G}_{i1}, \mathbf{G}_{i2}]$ the blocks of $\mathbf{\Phi}_{nb}$. Hence, as before,

$$\boxed{\tilde{\mathbf{x}}_{k+1} = \mathbf{\Phi}_{(k+1,k)}\,\tilde{\mathbf{x}}_k + \mathbf{G}_k\,\mathbf{n}_{Id} \quad \Rightarrow \quad \mathbf{P}_{k+1} = \mathbf{\Phi}_{(k+1,k)}\,\mathbf{P}_k\,\mathbf{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k\,\mathbf{Q}_d\,\mathbf{G}_k^\top \quad \rightarrow \quad \text{solved.}} \tag{464}$$

Note that $\mathbf{\Phi}_{nn}$ in (462) depends only on gravity and δt — not on the state estimate — which is the hallmark of the (right-)invariant error design mentioned in Lecture 4.

6 Lecture 6: Kalman Filter, Observability Analysis, and KF-Based VINS

Topics. (1) the MAP problem behind the Kalman filter; (2) linearization of the MAP problem; (3) Solution 1: solve it incrementally — the (E)KF; (4) Solution 2: solve it by batch; (5) discussion: the EKF, MLE/MAP, and the VINS system; (6) visual-inertial initialization; (7) the EKF summary for the VINS; (8) observability analysis.

6.1 The MAP problem behind the Kalman filter

The Kalman filter (KF) is a *linear* estimator based on the *Gaussian* assumption. Before deriving it, let's first start with a MAP problem. Given

$$\boxed{\begin{cases} \mathbf{x}_0 \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0) & & \rightarrow \text{ prior,} \\ \mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0, & \mathbf{w}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_0) & \rightarrow \text{ motion model (IMU),} \\ \mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n}_1, & \mathbf{n}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) & \rightarrow \text{ measurement model (vision),} \end{cases}} \tag{465}$$

the target is the posterior distribution of the new state:

$$\mathbf{x}_1 \sim \mathcal{N}(\hat{\mathbf{x}}_1, \mathbf{P}_1). \tag{466}$$

How to compute it? With $\mathbf{x} = \{\mathbf{x}_0, \mathbf{x}_1\}$ and $\mathbf{z} = \mathbf{z}_1$, apply Bayes' rule and factorize:

$$\max_{\mathbf{x}} \; p(\mathbf{x}|\mathbf{z}) = \max_{\mathbf{x}} \; \frac{p(\mathbf{z}|\mathbf{x})\, p(\mathbf{x})}{p(\mathbf{z})} = \max_{\mathbf{x}} \; \frac{p(\mathbf{z}_1|\mathbf{x}_1, \mathbf{x}_0)\, p(\mathbf{x}_1, \mathbf{x}_0)}{p(\mathbf{z}_1)} \tag{467}$$

$$= \max_{\mathbf{x}} \; \frac{p(\mathbf{z}_1|\mathbf{x}_1, \mathbf{x}_0)\, p(\mathbf{x}_1|\mathbf{x}_0)\, p(\mathbf{x}_0)}{p(\mathbf{z}_1)} \tag{468}$$

$$= \max_{\mathbf{x}} \; \frac{p(\mathbf{z}_1|\mathbf{x}_1)\, p(\mathbf{x}_1|\mathbf{x}_0)\, p(\mathbf{x}_0)}{p(\mathbf{z}_1)}, \tag{469}$$

where in the last step $\mathbf{z}_1$ only depends on $\mathbf{x}_1$, so we drop $\mathbf{x}_0$ from the measurement factor. Taking the negative log-likelihood of the three Gaussian factors (and recalling $\|\mathbf{e}\|^2_{\mathbf{P}} \triangleq \mathbf{e}^\top \mathbf{P}^{-1} \mathbf{e}$), the MAP problem is equivalent to

$$\boxed{\min_{\mathbf{x}} \; \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|^2_{\mathbf{R}} + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|^2_{\mathbf{Q}_0} + \|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2_{\mathbf{P}_0}.} \tag{470}$$

6.2 Linearization

Stack the state and choose the linearization point:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \end{bmatrix}, \qquad \text{linearization point: } \hat{\mathbf{x}} = \begin{bmatrix} \hat{\mathbf{x}}_0 \\ \hat{\mathbf{x}}_1 \end{bmatrix}, \qquad \tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}. \tag{471}$$

Motion model. From $\mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0$, linearizing $\mathbf{f}$ at $\hat{\mathbf{x}}_0$ with Jacobian $\mathbf{F}$,

$$\hat{\mathbf{x}}_1 + \tilde{\mathbf{x}}_1 \approx \mathbf{f}(\hat{\mathbf{x}}_0) + \mathbf{F}\,(\mathbf{x}_0 - \hat{\mathbf{x}}_0) + \mathbf{w}_0 = \mathbf{f}(\hat{\mathbf{x}}_0) + \mathbf{F}\,\tilde{\mathbf{x}}_0 + \mathbf{w}_0. \tag{472}$$

Define the *motion residual* $\mathbf{r}_x$:

$$\mathbf{r}_x \triangleq \hat{\mathbf{x}}_1 - \mathbf{f}(\hat{\mathbf{x}}_0) = \mathbf{F}\,\tilde{\mathbf{x}}_0 - \tilde{\mathbf{x}}_1 + \mathbf{w}_0 = \begin{bmatrix} \mathbf{F} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} + \mathbf{w}_0. \tag{473}$$

Measurement model. From $\mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n}_1$, linearizing $\mathbf{h}$ at $\hat{\mathbf{x}}_1$ with Jacobian $\mathbf{H}$,

$$\mathbf{z}_1 = \mathbf{h}(\hat{\mathbf{x}}_1) + \mathbf{H}\,(\mathbf{x}_1 - \hat{\mathbf{x}}_1) + \mathbf{n}_1 \quad \Rightarrow \quad \tilde{\mathbf{z}}_1 \triangleq \mathbf{z}_1 - \mathbf{h}(\hat{\mathbf{x}}_1) = \mathbf{H}\,\tilde{\mathbf{x}}_1 + \mathbf{n}_1 = \begin{bmatrix} \mathbf{0} & \mathbf{H} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} + \mathbf{n}_1. \tag{474}$$

The linearized cost. Then the cost function (470) becomes

$$\boxed{C = \|\tilde{\mathbf{z}}_1 - \mathbf{H}\tilde{\mathbf{x}}_1\|^2_{\mathbf{R}} + \|\mathbf{r}_x - \begin{bmatrix} \mathbf{F} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}\|^2_{\mathbf{Q}_0} + \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0}.} \tag{475}$$

6.3 Solution 1: solve it incrementally — the (E)KF

Solve (475) in two steps:

$$\begin{cases} \text{step 1:} \quad \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0} + \|\mathbf{r}_x - \begin{bmatrix} \mathbf{F} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}\|^2_{\mathbf{Q}_0} & \rightarrow \quad \mathbf{x}_{1|0} \sim \mathcal{N}(\hat{\mathbf{x}}_{1|0}, \mathbf{P}_{1|0}), \\ \text{step 2:} \quad \|\tilde{\mathbf{x}}_{1|0}\|^2_{\mathbf{P}_{1|0}} + \|\tilde{\mathbf{z}}_1 - \mathbf{H}\tilde{\mathbf{x}}_1\|^2_{\mathbf{R}} & \rightarrow \quad \mathbf{x}_{1|1} \sim \mathcal{N}(\hat{\mathbf{x}}_{1|1}, \mathbf{P}_{1|1}). \end{cases} \tag{476}$$

Step 1: prior + motion. We solve

$$\min_{\tilde{\mathbf{x}}} \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0} + \|\mathbf{r}_x - \begin{bmatrix} \mathbf{F} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}\|^2_{\mathbf{Q}_0}. \tag{477}$$

Expanding, the cost is

$$C_1 = \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}^\top \begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} - 2 \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}^\top \begin{bmatrix} \mathbf{F}^\top \\ -\mathbf{I} \end{bmatrix} \mathbf{Q}_0^{-1}\,\mathbf{r}_x + \mathbf{r}_x^\top\mathbf{Q}_0^{-1}\mathbf{r}_x. \tag{478}$$

Setting $\frac{\partial C_1}{\partial \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix}} = \mathbf{0}$ gives the *information system*:

$$\begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^\top\mathbf{Q}_0^{-1}\,\mathbf{r}_x \\ -\mathbf{Q}_0^{-1}\,\mathbf{r}_x \end{bmatrix}. \tag{479}$$

Then, marginalizing $\tilde{\mathbf{x}}_0$ (Schur complement onto the $\tilde{\mathbf{x}}_1$ block, Lecture 4),

$$\left[\mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\right]\tilde{\mathbf{x}}_1 = -\mathbf{Q}_0^{-1}\,\mathbf{r}_x + \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\,\mathbf{r}_x, \tag{480}$$

and applying the matrix inversion lemma to both sides,

$$\underbrace{\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}}_{\mathbf{P}_{1|0}^{-1}}\tilde{\mathbf{x}}_1 = -\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}\,\mathbf{r}_x. \tag{481}$$

Hence

$$\boxed{\tilde{\mathbf{x}}_1 = -\mathbf{r}_x, \qquad \mathbf{P}_{1|0} = \mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top.} \tag{482}$$

Interpreting $\tilde{\mathbf{x}}_1 = -\mathbf{r}_x$:

$$\begin{cases} \text{if } \mathbf{r}_x = \mathbf{0}, \quad \hat{\mathbf{x}}_{1|0} = \hat{\mathbf{x}}_1 = \mathbf{f}(\hat{\mathbf{x}}_0), \\ \text{if } \mathbf{r}_x \neq \mathbf{0}, \quad \hat{\mathbf{x}}_{1|0} = \hat{\mathbf{x}}_1 - \mathbf{r}_x = \hat{\mathbf{x}}_1 - \big(\hat{\mathbf{x}}_1 - \mathbf{f}(\hat{\mathbf{x}}_0)\big) = \mathbf{f}(\hat{\mathbf{x}}_0). \end{cases} \tag{483}$$

Either way the propagated mean is $\mathbf{f}(\hat{\mathbf{x}}_0)$.

Step 2: + update. Since step 1 left $\tilde{\mathbf{x}}_1$ with mean $-\mathbf{r}_x$ and covariance $\mathbf{P}_{1|0}$, the remaining problem is

$$\min_{\tilde{\mathbf{x}}_1} \|\tilde{\mathbf{x}}_1 + \mathbf{r}_x\|^2_{\mathbf{P}_{1|0}} + \|\tilde{\mathbf{z}}_1 - \mathbf{H}\tilde{\mathbf{x}}_1\|^2_{\mathbf{R}}. \tag{484}$$

Expanding (one step per row, dropping the tilde-1 subscript on $\tilde{\mathbf{z}}$ for brevity):

$$\begin{aligned}
C_2 &= (\tilde{\mathbf{x}}_1 + \mathbf{r}_x)^\top \mathbf{P}_{1|0}^{-1}(\tilde{\mathbf{x}}_1 + \mathbf{r}_x) + (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}_1)^\top \mathbf{R}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}_1) && (485)\\
&= \tilde{\mathbf{x}}_1^\top \mathbf{P}_{1|0}^{-1}\tilde{\mathbf{x}}_1 + 2\,\tilde{\mathbf{x}}_1^\top \mathbf{P}_{1|0}^{-1}\mathbf{r}_x + \mathbf{r}_x^\top \mathbf{P}_{1|0}^{-1}\mathbf{r}_x + \tilde{\mathbf{z}}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} - 2\,\tilde{\mathbf{z}}^\top \mathbf{R}^{-1}\mathbf{H}\tilde{\mathbf{x}}_1 + \tilde{\mathbf{x}}_1^\top \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\tilde{\mathbf{x}}_1 && (486)\\
&= \tilde{\mathbf{x}}_1^\top \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)\tilde{\mathbf{x}}_1 + 2\,\tilde{\mathbf{x}}_1^\top \big(\mathbf{P}_{1|0}^{-1}\mathbf{r}_x - \mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}}\big) + \tilde{\mathbf{z}}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} + \mathbf{r}_x^\top \mathbf{P}_{1|0}^{-1}\mathbf{r}_x. && (487)
\end{aligned}$$

Hence, $\frac{\partial C_2}{\partial \tilde{\mathbf{x}}_1} = \mathbf{0}$ gives

$$\underbrace{\big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)}_{\boldsymbol{\Sigma} = \mathbf{P}_{1|1}^{-1}}\,\tilde{\mathbf{x}}_1 = \mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{P}_{1|0}^{-1}\,\mathbf{r}_x, \tag{488}$$

so that

$$\begin{aligned}
\tilde{\mathbf{x}}_1 &= \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)^{-1}\big(\mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{P}_{1|0}^{-1}\mathbf{r}_x\big) && (489)\\
&= \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)^{-1}\mathbf{H}^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} - \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)^{-1}\mathbf{P}_{1|0}^{-1}\mathbf{r}_x. && (490)
\end{aligned}$$

The gain equivalence. For the first term of (490), apply the matrix inversion lemma, $\big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)^{-1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\mathbf{H}\mathbf{P}_{1|0}$, and simplify step by step:

$$\begin{aligned}
&\Big(\mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\mathbf{H}\mathbf{P}_{1|0}\Big)\mathbf{H}^\top \mathbf{R}^{-1}\,\tilde{\mathbf{z}}\\
&= \Big(\mathbf{P}_{1|0}\mathbf{H}^\top \mathbf{R}^{-1} - \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top \mathbf{R}^{-1}\Big)\,\tilde{\mathbf{z}} && (491)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top \Big(\mathbf{R}^{-1} - \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top \mathbf{R}^{-1}\Big)\,\tilde{\mathbf{z}} && (492)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\Big(\big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)\mathbf{R}^{-1} - \mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top \mathbf{R}^{-1}\Big)\,\tilde{\mathbf{z}} && (493)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\,\tilde{\mathbf{z}}. && (494)
\end{aligned}$$

Hence we got

$$\boxed{\tilde{\mathbf{x}}_1 = \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\,\tilde{\mathbf{z}} - \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)^{-1}\mathbf{P}_{1|0}^{-1}\,\mathbf{r}_x.} \tag{495}$$

If $\mathbf{r}_x \to \mathbf{0}$, then

$$\tilde{\mathbf{x}}_1 = \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\,\tilde{\mathbf{z}}, \tag{496}$$

and the posterior covariance follows from (488) (matrix inversion lemma again):

$$\mathbf{P}_{1|1} = \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H}\big)^{-1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top \big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}\mathbf{H}\mathbf{P}_{1|0}. \tag{497}$$

Therefore, we have the posterior $\mathbf{x}_{1|1} \sim \mathcal{N}(\hat{\mathbf{x}}_{1|1}, \mathbf{P}_{1|1})$.

Summary of the process. Collecting the two steps ($\mathbf{r}_x \to \mathbf{0}$):

$$
\boxed{\begin{array}{ll}
\text{prior:} & \mathbf{x}_0 \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0), \\
\text{propagation:} & \mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0, \\
& \hat{\mathbf{x}}_{1|0} = \mathbf{f}(\hat{\mathbf{x}}_0), \qquad \mathbf{P}_{1|0} = \mathbf{F}\mathbf{P}_0\mathbf{F}^\top + \mathbf{Q}_0, \\
\text{update:} & \mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n}_1, \\
& \tilde{\mathbf{z}}_1 = \mathbf{z}_1 - \mathbf{h}(\hat{\mathbf{x}}_{1|0}), \qquad \tilde{\mathbf{z}}_1 = \mathbf{H}\tilde{\mathbf{x}}_1 + \mathbf{n}_1, \\
& \mathbf{S}_1 = \mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R} \quad \to \text{ innovation covariance}, \\
& \mathbf{K} = \mathbf{P}_{1|0}\mathbf{H}^\top\mathbf{S}_1^{-1} \quad \to \text{ Kalman gain}, \\
& \hat{\mathbf{x}}_{1|1} = \hat{\mathbf{x}}_{1|0} + \mathbf{K}\,\tilde{\mathbf{z}}_1, \\
& \mathbf{P}_{1|1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top\mathbf{S}_1^{-1}\mathbf{H}\mathbf{P}_{1|0}.
\end{array}} \tag{498}
$$

This is the EKF from the MLE/MAP formulation.

6.4 Solution 2: solve it by batch

Alternatively, solve the full linearized cost (475) in one shot. Setting the gradient with respect to $[\tilde{\mathbf{x}}_0; \tilde{\mathbf{x}}_1]$ to zero now includes the measurement term in the information system:

$$
\begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^\top\mathbf{Q}_0^{-1}\,\mathbf{r}_x \\ -\mathbf{Q}_0^{-1}\,\mathbf{r}_x + \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} \end{bmatrix}. \tag{499}
$$

Then, marginalizing $\tilde{\mathbf{x}}_0$ (Schur complement),

$$
\begin{aligned}
&\Big[\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} + \underbrace{\mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}}_{\mathbf{P}_{1|0}^{-1}}\Big]\tilde{\mathbf{x}}_1 \\
&\quad = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{Q}_0^{-1}\mathbf{r}_x + \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{r}_x \qquad (500) \\
&\quad = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} - \underbrace{\Big(\mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\Big)}_{\mathbf{P}_{1|0}^{-1}}\,\mathbf{r}_x. \qquad (501)
\end{aligned}
$$

If we define $\mathbf{P}_{1|0}^{-1} \triangleq \mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}$ — which by the matrix inversion lemma is exactly $\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}$, the same $\mathbf{P}_{1|0}$ as in (482) — then

$$
\big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} + \mathbf{P}_{1|0}^{-1}\big)\,\tilde{\mathbf{x}}_1 = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{P}_{1|0}^{-1}\,\mathbf{r}_x, \tag{502}
$$

and hence

$$
\boxed{\mathbf{P}_{1|1} = \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}, \qquad \tilde{\mathbf{x}}_1 = \big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} + \mathbf{P}_{1|0}^{-1}\big)^{-1}\big(\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{P}_{1|0}^{-1}\,\mathbf{r}_x\big).} \tag{503}
$$

It's the same as the EKF as $\mathbf{r}_x \to \mathbf{0}$.

6.5 Discussion: the EKF, MLE/MAP, and the VINS system

(1) We need to *linearize* the nonlinear motion and measurement models — the KF is actually a *linear* estimator. We derived it based on MLE/MAP (nonlinear least squares); the Gaussian assumption is the foundation.

(2) The EKF is the way to solve the MLE/MAP *incrementally*; (nonlinear) least squares solve it in *batch*.

(3) The VINS system has exactly the structure of (465), repeated over time:

$$\left.\begin{cases} \mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k) & \rightarrow \text{ prior of the IMU state,} \\ \mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k) + \mathbf{w}_k & \rightarrow \text{ IMU integration model,} \\ \mathbf{z}_{k+1} = \mathbf{h}(\mathbf{x}_{k+1}) + \mathbf{n}_{k+1} & \rightarrow \text{ vision measurements} \end{cases}\right\} \text{ VINS,} \tag{504}$$

the visual-inertial navigation system (when used for odometry: VIO, visual-inertial odometry).

(4) The two models are the ones we have built in the previous lectures:

- IMU model (Lecture 5): $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \;\Rightarrow\; \mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_{Id})$;
- vision model (Lectures 3–4): $\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c$.

(5) The Kalman gain can be written in equivalent forms — by (494),

$$\mathbf{K} = \mathbf{P}_{1|0}\mathbf{H}^\top\mathbf{S}_1^{-1} = \mathbf{P}_{1|0}\mathbf{H}^\top\big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1} = \big(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}, \tag{505}$$

where $\mathbf{P}_{1|0}$ is the *propagated covariance* of $\mathbf{x}_1$; $\mathbf{S}_1$ is the *residual covariance* (of $\tilde{\mathbf{z}}_1$); $\mathbf{R}$ is the measurement noise; and $\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} = \mathbf{\Sigma}_{1|1} = \mathbf{P}_{1|1}^{-1}$ is the posterior information matrix.

(6) The covariance update can likewise be written in equivalent forms:

$$\mathbf{P}_{1|1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top\underbrace{\big(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\big)^{-1}}_{\mathbf{S}_1^{-1}}\mathbf{H}\mathbf{P}_{1|0} = \mathbf{P}_{1|0} - \mathbf{K}\mathbf{S}_1\mathbf{K}^\top. \tag{506}$$

Moreover, substituting the linearized measurement into the state update,

$$\hat{\mathbf{x}}_{1|1} = \hat{\mathbf{x}}_{1|0} + \tilde{\mathbf{x}}_1 = \hat{\mathbf{x}}_{1|0} + \mathbf{K}\,\tilde{\mathbf{z}}_1 = \hat{\mathbf{x}}_{1|0} + \mathbf{K}\big(\mathbf{H}\tilde{\mathbf{x}}_{1|0} + \mathbf{n}_1\big), \tag{507}$$

the posterior covariance follows step by step:

$$\begin{aligned}
\mathbf{P}_{1|1} &= \mathbb{E}\Big(\big(\mathbf{x} - \hat{\mathbf{x}}_{1|1}\big)\big(\mathbf{x} - \hat{\mathbf{x}}_{1|1}\big)^\top\Big) && (508)\\
&= \mathbb{E}\Big(\big[\underbrace{\mathbf{x} - \hat{\mathbf{x}}_{1|0}}_{\tilde{\mathbf{x}}_{1|0}} - \mathbf{K}\big(\mathbf{H}\tilde{\mathbf{x}}_{1|0} + \mathbf{n}_1\big)\big]\big[\mathbf{x} - \hat{\mathbf{x}}_{1|0} - \mathbf{K}\big(\mathbf{H}\tilde{\mathbf{x}}_{1|0} + \mathbf{n}_1\big)\big]^\top\Big) && (509)\\
&= \mathbb{E}\Big(\big[(\mathbf{I} - \mathbf{K}\mathbf{H})\,\tilde{\mathbf{x}}_{1|0} - \mathbf{K}\mathbf{n}_1\big]\big[(\mathbf{I} - \mathbf{K}\mathbf{H})\,\tilde{\mathbf{x}}_{1|0} - \mathbf{K}\mathbf{n}_1\big]^\top\Big), && (510)
\end{aligned}$$

hence

$$\boxed{\mathbf{P}_{1|1} = (\mathbf{I} - \mathbf{K}\mathbf{H})\,\mathbf{P}_{1|0}\,(\mathbf{I} - \mathbf{K}\mathbf{H})^\top + \mathbf{K}\mathbf{R}\mathbf{K}^\top \qquad \text{(Joseph form).}} \tag{511}$$

(7) **If the system is linear, the KF and (weighted) least squares are exactly the same.** With a linear motion model $\mathbf{x}_1 = \mathbf{F}\mathbf{x}_0 + \mathbf{w}_0$ and a linear measurement model $\mathbf{z}_1 = \mathbf{H}\mathbf{x}_1 + \mathbf{n}_1$, the cost (470) is exactly quadratic: there is no linearization step, and $\mathbf{r}_x = \hat{\mathbf{x}}_1 - \mathbf{F}\hat{\mathbf{x}}_0 = \mathbf{0}$ holds by construction, so the "as $\mathbf{r}_x \to \mathbf{0}$" caveat disappears. The incremental solution (§6.3) and the batch solution (§6.4) are then two orderings of the *same* Gaussian elimination on the *same* information system, and they agree *exactly*: the KF posterior $\mathcal{N}(\hat{\mathbf{x}}_{1|1}, \mathbf{P}_{1|1})$ equals the batch MAP estimate of $\mathbf{x}_1$ with its marginal covariance. Moreover, the posterior is Gaussian, so its mean equals its mode: the MAP, MMSE, and KF estimates all coincide. Two qualifications:

- the equivalence is with the *weighted* least squares (470) — including the prior and motion terms, with the inverse covariances $\mathbf{P}_0^{-1}$, $\mathbf{Q}_0^{-1}$, $\mathbf{R}^{-1}$ as weights; ordinary unweighted least squares (or dropping the prior) solves a different problem;
- batch over a whole trajectory $\mathbf{x}_0, \ldots, \mathbf{x}_N$ conditions every state on *all* measurements — it is a *smoother*. It matches the KF at the *final* state $\mathbf{x}_N$; at the intermediate states it returns the smoothed estimates, which differ from (and improve on) the filtered ones.

Once $\mathbf{f}$, $\mathbf{h}$ are nonlinear, the equivalence breaks: the EKF takes a *single* Gauss–Newton step, linearized at the predicted estimate, and immediately marginalizes — the linearization points are frozen and can never be revisited — while batch nonlinear least squares re-linearizes all states and iterates. Both the accuracy gap and the consistency gap of the EKF trace back to these frozen linearization points.

(8) **A practical problem for the KF: initialization.** To run the filter we need the prior

$$\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k) \quad \Rightarrow \quad \text{how to get } \hat{\mathbf{x}}_k,\, \mathbf{P}_k\,? \tag{512}$$

For the IMU state (Lecture 5)

$$\mathbf{x}_I = \left\{{}^G_I\mathbf{R},\ {}^G\mathbf{p}_I,\ {}^G\mathbf{v}_I,\ \mathbf{b}_g,\ \mathbf{b}_a\right\}, \tag{513}$$

this is the subject of the next subsection.

6.6 Visual-inertial initialization

Consider three consecutive images and the IMU between them (Fig. 17).

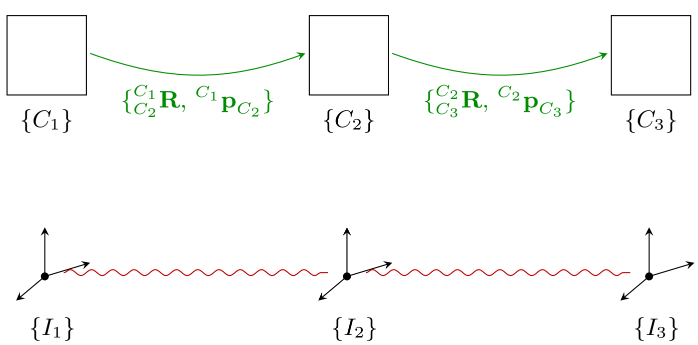


Figure 17: Visual-inertial initialization: the relative camera poses from the visual SLAM pipeline (top, green) are aligned with the IMU integration over the same intervals (bottom, red wavy links) through the camera–IMU extrinsics.

From these, we want to **initialize**:

$$\boxed{\begin{cases} (1) & {}^{I_1}\mathbf{g} \\ (2) & {}^{I_1}\mathbf{v}_{I_1},\ {}^{I_1}\mathbf{v}_{I_2},\ {}^{I_1}\mathbf{v}_{I_3},\ \dots \\ (3) & s \quad \text{(mono camera)} \\ (4) & \mathbf{b}_g,\ \mathbf{b}_a \end{cases}} \tag{514}$$

From vision. Using the visual SLAM pipeline (Lectures 3–4), we obtain the relative camera poses

$$\{{}^{C_1}_{C_2}\mathbf{R},\ {}^{C_1}\mathbf{p}_{C_2}\}, \qquad \{{}^{C_2}_{C_3}\mathbf{R},\ {}^{C_2}\mathbf{p}_{C_3}\}, \tag{515}$$

where for a monocular camera the translations are recovered only *up to a scale s*.

From IMU integration. Recall the exact integration results of Lecture 5 (writing $\Delta\mathbf{p}_k, \Delta\mathbf{v}_k$ for the preintegrated measurements over $[t_k, t_{k+1}]$, $\delta t_k = t_{k+1} - t_k$):

$$ {}^{I_k}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R}^\top\, {}^{G}_{I_{k+1}}\mathbf{R}, \tag{516}$$

$$ {}^{G}\mathbf{p}_{I_{k+1}} = {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\,\delta t_k + {}^{G}_{I_k}\mathbf{R}\,\Delta\mathbf{p}_k + \frac{1}{2}\,{}^{G}\mathbf{g}\,\delta t_k^2, \tag{517}$$

$$ {}^{G}\mathbf{v}_{I_{k+1}} = {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\,\Delta\mathbf{v}_k + {}^{G}\mathbf{g}\,\delta t_k. \tag{518}$$

Rotate the position update into the local frame $\{I_k\}$, defining ${}^{I_k}\mathbf{v}_{I_k} \triangleq {}^{G}_{I_k}\mathbf{R}^\top\,{}^{G}\mathbf{v}_{I_k}$ and ${}^{I_k}\mathbf{g} \triangleq {}^{G}_{I_k}\mathbf{R}^\top\,{}^{G}\mathbf{g}$:

$$ {}^{I_k}\mathbf{p}_{I_{k+1}} = {}^{G}_{I_k}\mathbf{R}^\top\left({}^{G}\mathbf{p}_{I_{k+1}} - {}^{G}\mathbf{p}_{I_k}\right) = {}^{I_k}\mathbf{v}_{I_k}\,\delta t_k + \Delta\mathbf{p}_k + \frac{1}{2}\,{}^{I_k}\mathbf{g}\,\delta t_k^2, \tag{519}$$

and, one interval later, expressing everything with $\{I_k\}$-frame quantities via the (gyro-integrated, hence known) rotation ${}^{I_{k+1}}_{I_k}\mathbf{R} = \left({}^{I_k}_{I_{k+1}}\mathbf{R}\right)^\top$:

$$\begin{aligned} {}^{I_{k+1}}\mathbf{p}_{I_{k+2}} &= {}^{G}_{I_{k+1}}\mathbf{R}^\top\left({}^{G}\mathbf{p}_{I_{k+2}} - {}^{G}\mathbf{p}_{I_{k+1}}\right) = {}^{I_{k+1}}\mathbf{v}_{I_{k+1}}\,\delta t_{k+1} + \Delta\mathbf{p}_{k+1} + \frac{1}{2}\,{}^{I_{k+1}}\mathbf{g}\,\delta t_{k+1}^2 && (520)\\ &= \Delta\mathbf{p}_{k+1} + {}^{I_{k+1}}_{I_k}\mathbf{R}\left({}^{I_k}\mathbf{v}_{I_{k+1}}\,\delta t_{k+1} + \frac{1}{2}\,{}^{I_k}\mathbf{g}\,\delta t_{k+1}^2\right). && (521)\end{aligned}$$

Connecting the two. With the camera–IMU extrinsics $\{{}^{I}_{C}\mathbf{R}, {}^{I}\mathbf{p}_C\}$ (inverse $\{{}^{C}_{I}\mathbf{R}, {}^{C}\mathbf{p}_I\}$) and the monocular scale s on the vision translation, chain the transformations:

$$\begin{aligned} \begin{bmatrix} {}^{I_k}_{I_{k+1}}\mathbf{R} & {}^{I_k}\mathbf{p}_{I_{k+1}} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} &= \begin{bmatrix} {}^{I}_{C}\mathbf{R} & {}^{I}\mathbf{p}_C \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} {}^{C_k}_{C_{k+1}}\mathbf{R} & s\,{}^{C_k}\mathbf{p}_{C_{k+1}} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} {}^{C}_{I}\mathbf{R} & {}^{C}\mathbf{p}_I \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} && (522)\\ &= \begin{bmatrix} {}^{I}_{C}\mathbf{R}\,{}^{C_k}_{C_{k+1}}\mathbf{R} & s\,{}^{I}_{C}\mathbf{R}\,{}^{C_k}\mathbf{p}_{C_{k+1}} + {}^{I}\mathbf{p}_C \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \begin{bmatrix} {}^{C}_{I}\mathbf{R} & {}^{C}\mathbf{p}_I \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} && (523)\\ &= \begin{bmatrix} {}^{I}_{C}\mathbf{R}\,{}^{C_k}_{C_{k+1}}\mathbf{R}\,{}^{C}_{I}\mathbf{R} & {}^{I}_{C}\mathbf{R}\,{}^{C_k}_{C_{k+1}}\mathbf{R}\,{}^{C}\mathbf{p}_I + s\,{}^{I}_{C}\mathbf{R}\,{}^{C_k}\mathbf{p}_{C_{k+1}} + {}^{I}\mathbf{p}_C \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix}. && (524)\end{aligned}$$

Hence, equating the translation of (524) with (519) and (521), we get the two *position constraints*

$$ {}^{I}_{C}\mathbf{R}\,{}^{C_k}_{C_{k+1}}\mathbf{R}\,{}^{C}\mathbf{p}_I + s\,{}^{I}_{C}\mathbf{R}\,{}^{C_k}\mathbf{p}_{C_{k+1}} + {}^{I}\mathbf{p}_C = \Delta\mathbf{p}_k + {}^{I_k}\mathbf{v}_{I_k}\,\delta t_k + \frac{1}{2}\,{}^{I_k}\mathbf{g}\,\delta t_k^2, \tag{525} $$

$$ {}^{I}_{C}\mathbf{R}\,{}^{C_{k+1}}_{C_{k+2}}\mathbf{R}\,{}^{C}\mathbf{p}_I + s\,{}^{I}_{C}\mathbf{R}\,{}^{C_{k+1}}\mathbf{p}_{C_{k+2}} + {}^{I}\mathbf{p}_C = \Delta\mathbf{p}_{k+1} + {}^{I_{k+1}}_{I_k}\mathbf{R}\left({}^{I_k}\mathbf{v}_{I_{k+1}}\,\delta t_{k+1} + \frac{1}{2}\,{}^{I_k}\mathbf{g}\,\delta t_{k+1}^2\right). \tag{526} $$

At the same time, rotating the velocity updates into $\{I_k\}$ gives the two *velocity constraints*

$$ {}^{I_k}\mathbf{v}_{I_{k+1}} - {}^{I_k}\mathbf{v}_{I_k} - {}^{I_k}\mathbf{g}\,\delta t_k = \Delta\mathbf{v}_k, \tag{527} $$

$$ {}^{I_{k+1}}_{I_k}\mathbf{R}\left({}^{I_k}\mathbf{v}_{I_{k+2}} - {}^{I_k}\mathbf{v}_{I_{k+1}} - {}^{I_k}\mathbf{g}\,\delta t_{k+1}\right) = \Delta\mathbf{v}_{k+1}. \tag{528} $$

The linear system. Stacking the four constraints into $\mathbf{A}\mathbf{u} = \mathbf{b}$ with the unknowns $\mathbf{u} = \begin{bmatrix} s & {}^{I_k}\mathbf{g}^\top & {}^{I_k}\mathbf{v}_{I_k}^\top & {}^{I_k}\mathbf{v}_{I_{k+1}}^\top & {}^{I_k}\mathbf{v}_{I_{k+2}}^\top \end{bmatrix}^\top$ (13×1), and abbreviating $\bar{\mathbf{R}} \triangleq {}^{I_{k+1}}_{I_k}\mathbf{R}$:

$$ \boxed{\begin{bmatrix} -{}^{I}_{C}\mathbf{R}\,{}^{C_k}\mathbf{p}_{C_{k+1}} & \frac{1}{2}\delta t_k^2\,\mathbf{I}_3 & \delta t_k\,\mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ -{}^{I}_{C}\mathbf{R}\,{}^{C_{k+1}}\mathbf{p}_{C_{k+2}} & \frac{1}{2}\delta t_{k+1}^2\,\bar{\mathbf{R}} & \mathbf{0}_3 & \delta t_{k+1}\,\bar{\mathbf{R}} & \mathbf{0}_3 \\ \mathbf{0}_{3\times 1} & -\delta t_k\,\mathbf{I}_3 & -\mathbf{I}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_{3\times 1} & -\delta t_{k+1}\,\bar{\mathbf{R}} & \mathbf{0}_3 & -\bar{\mathbf{R}} & \bar{\mathbf{R}} \end{bmatrix} \begin{bmatrix} s \\ {}^{I_k}\mathbf{g} \\ {}^{I_k}\mathbf{v}_{I_k} \\ {}^{I_k}\mathbf{v}_{I_{k+1}} \\ {}^{I_k}\mathbf{v}_{I_{k+2}} \end{bmatrix} = \begin{bmatrix} {}^{I}_{C}\mathbf{R}\,{}^{C_k}_{C_{k+1}}\mathbf{R}\,{}^{C}\mathbf{p}_I + {}^{I}\mathbf{p}_C - \Delta\mathbf{p}_k \\ {}^{I}_{C}\mathbf{R}\,{}^{C_{k+1}}_{C_{k+2}}\mathbf{R}\,{}^{C}\mathbf{p}_I + {}^{I}\mathbf{p}_C - \Delta\mathbf{p}_{k+1} \\ \Delta\mathbf{v}_k \\ \Delta\mathbf{v}_{k+1} \end{bmatrix}.} \tag{529} $$

Solving this linear system recovers targets (1)–(3) of (514): the scale, the gravity direction in $\{I_k\}$ (hence the initial orientation up to yaw), and the velocities.

How many frames do we need? The system (529) has 12 constraints and 13 unknowns. In general, every pair of consecutive frames gives 6 constraints (3 position + 3 velocity), so n frames give $6(n-1)$, while we totally need to solve $1+3+3n$ unknowns (s, gravity, and n velocities):

- **if s is known** (stereo camera): 3 frames are good to solve the initialization;
- **if s is unknown** (mono camera):

$$ 6(n-1) - (4+3n) > 0 \;\;\Rightarrow\;\; n > \tfrac{10}{3} \;\;\Rightarrow\;\; n = 4 \text{ frames minimum (3 intervals).} \tag{530} $$

6.7 The EKF summary for the VINS

We now put the pieces together: the EKF for the VINS system, fusing **IMU + vision**.

The state vector. For simplicity, take the 15-DoF IMU state plus one 3D feature:

$$ \mathbf{x} = \begin{bmatrix} \mathbf{x}_I \\ \mathbf{x}_f \end{bmatrix} = \begin{bmatrix} {}^{G}_{I}\boldsymbol{\theta} \\ {}^{G}\mathbf{p}_I \\ {}^{G}\mathbf{v}_I \\ \mathbf{b}_g \\ \mathbf{b}_a \\ {}^{G}\mathbf{p}_f \end{bmatrix}, \tag{531} $$

each block 3×1: the first five blocks are $\mathbf{x}_I$, the 15-DoF IMU state on $SO(3)+\mathbb{R}^3+\mathbb{R}^3+\mathbb{R}^3+\mathbb{R}^3$, and the last is the feature (3 per 3D feature).

The error state. With $\mathbf{x} = \hat{\mathbf{x}} \boxplus \tilde{\mathbf{x}}$,

$$\tilde{\mathbf{x}} = \begin{bmatrix} \delta\boldsymbol{\theta} \\ {}^G\tilde{\mathbf{p}}_I \\ {}^G\tilde{\mathbf{v}}_I \\ \tilde{\mathbf{b}}_g \\ \tilde{\mathbf{b}}_a \\ {}^G\tilde{\mathbf{p}}_f \end{bmatrix}, \qquad \hat{\mathbf{x}} = \begin{bmatrix} {}^G_I\hat{\boldsymbol{\theta}} \\ {}^G\hat{\mathbf{p}}_I \\ {}^G\hat{\mathbf{v}}_I \\ \hat{\mathbf{b}}_g \\ \hat{\mathbf{b}}_a \\ {}^G\hat{\mathbf{p}}_f \end{bmatrix}, \tag{532}$$

where the rotation error is *multiplicative*,

$$\boxed{{}^G_I\mathbf{R} = {}^G_I\hat{\mathbf{R}} \cdot \delta\mathbf{R} \triangleq \operatorname{Exp}\left({}^G_I\hat{\boldsymbol{\theta}}\right) \cdot \operatorname{Exp}(\delta\boldsymbol{\theta}),} \tag{533}$$

and the rest are additive: ${}^G\mathbf{p}_I = {}^G\hat{\mathbf{p}}_I + {}^G\tilde{\mathbf{p}}_I$, ${}^G\mathbf{v}_I = {}^G\hat{\mathbf{v}}_I + {}^G\tilde{\mathbf{v}}_I$, $\mathbf{b}_g = \hat{\mathbf{b}}_g + \tilde{\mathbf{b}}_g$, $\mathbf{b}_a = \hat{\mathbf{b}}_a + \tilde{\mathbf{b}}_a$, ${}^G\mathbf{p}_f = {}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f$. Two remarks:

1. From $\operatorname{Exp}\left({}^G_I\boldsymbol{\theta}\right) = \operatorname{Exp}\left({}^G_I\hat{\boldsymbol{\theta}}\right) \cdot \operatorname{Exp}(\delta\boldsymbol{\theta})$ and the right Jacobian (Lecture 2), ${}^G_I\boldsymbol{\theta} = {}^G_I\hat{\boldsymbol{\theta}} + \mathbf{J}_r\left({}^G_I\hat{\boldsymbol{\theta}}\right)\delta\boldsymbol{\theta}$, so the boxplus reads, in column form,

$$\mathbf{x} = \hat{\mathbf{x}} \boxplus \tilde{\mathbf{x}} \;\; \Rightarrow \;\; \begin{bmatrix} {}^G_I\boldsymbol{\theta} \\ {}^G\mathbf{p}_I \\ {}^G\mathbf{v}_I \\ \mathbf{b}_g \\ \mathbf{b}_a \\ {}^G\mathbf{p}_f \end{bmatrix} = \begin{bmatrix} {}^G_I\hat{\boldsymbol{\theta}} \\ {}^G\hat{\mathbf{p}}_I \\ {}^G\hat{\mathbf{v}}_I \\ \hat{\mathbf{b}}_g \\ \hat{\mathbf{b}}_a \\ {}^G\hat{\mathbf{p}}_f \end{bmatrix} \boxplus \begin{bmatrix} \delta\boldsymbol{\theta} \\ {}^G\tilde{\mathbf{p}}_I \\ {}^G\tilde{\mathbf{v}}_I \\ \tilde{\mathbf{b}}_g \\ \tilde{\mathbf{b}}_a \\ {}^G\tilde{\mathbf{p}}_f \end{bmatrix} = \begin{bmatrix} {}^G_I\hat{\boldsymbol{\theta}} + \mathbf{J}_r\left({}^G_I\hat{\boldsymbol{\theta}}\right)\delta\boldsymbol{\theta} \\ {}^G\hat{\mathbf{p}}_I + {}^G\tilde{\mathbf{p}}_I \\ {}^G\hat{\mathbf{v}}_I + {}^G\tilde{\mathbf{v}}_I \\ \hat{\mathbf{b}}_g + \tilde{\mathbf{b}}_g \\ \hat{\mathbf{b}}_a + \tilde{\mathbf{b}}_a \\ {}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f \end{bmatrix}. \tag{534}$$

2. In quaternion form (Lecture 2), the same multiplicative error is ${}^G_I\bar{q} = {}^G_I\hat{\bar{q}} \otimes \delta\bar{q}$ with $\delta\bar{q} = \begin{bmatrix} 1 \\ \frac{1}{2}\delta\boldsymbol{\theta} \end{bmatrix}$ (to first order).

The IMU (motion) model. From Lecture 5, $\boldsymbol{\omega}_m = \boldsymbol{\omega} + \mathbf{b}_g + \mathbf{n}_g$ and $\mathbf{a}_m = \mathbf{a} + \mathbf{b}_a + \mathbf{n}_a$, so the IMU dynamics (appending the static feature) are

$$\left.\begin{aligned} {}^G_I\dot{\mathbf{R}} &= {}^G_I\mathbf{R}\,\lfloor \boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g \rfloor_\times \\ {}^G\dot{\mathbf{p}}_I &= {}^G\mathbf{v}_I \\ {}^G\dot{\mathbf{v}}_I &= {}^G_I\mathbf{R}\,(\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a) + {}^G\mathbf{g}, \qquad {}^G\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -9.81 \end{bmatrix} \\ \dot{\mathbf{b}}_g &= \mathbf{n}_{wg} \\ \dot{\mathbf{b}}_a &= \mathbf{n}_{wa} \\ {}^G\dot{\mathbf{p}}_f &= \mathbf{0}_{3\times 1} \end{aligned}\right\} \Rightarrow \quad \begin{gathered} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \\ \Downarrow \\ \mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_{I_k}), \end{gathered} \tag{535}$$

where $\mathbf{f}_{\text{int}}(\cdot)$ is the integration of $\mathbf{f}(\cdot)$ over $[t_k, t_{k+1}]$ (Lecture 5).

The vision (measurement) model. From Lecture 3, $\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c$ with

$$
\mathbf{z}_c = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix}, \qquad {}^C\mathbf{p}_f = \begin{bmatrix} c_x \\ c_y \\ c_z \end{bmatrix} = {}^G_C\mathbf{R}^\top \left({}^G\mathbf{p}_f - {}^G\mathbf{p}_C\right)
$$

$$
\begin{aligned}
&= \left({}^G_I\mathbf{R}\,{}^I_C\mathbf{R}\right)^\top \Big({}^G\mathbf{p}_f - \left({}^G_I\mathbf{R}\,{}^I\mathbf{p}_C + {}^G\mathbf{p}_I\right)\Big) && (536)\\
&= {}^I_C\mathbf{R}^\top\,{}^G_I\mathbf{R}^\top \left({}^G\mathbf{p}_f - {}^G\mathbf{p}_I - {}^G_I\mathbf{R}\,{}^I\mathbf{p}_C\right) && (537)\\
&= {}^I_C\mathbf{R}^\top\,{}^G_I\mathbf{R}^\top \left({}^G\mathbf{p}_f - {}^G\mathbf{p}_I\right) - {}^I_C\mathbf{R}^\top\,{}^I\mathbf{p}_C && (538)\\
&= {}^I_C\mathbf{R}^\top\,{}^G_I\mathbf{R}^\top \left({}^G\mathbf{p}_f - {}^G\mathbf{p}_I\right) + {}^C\mathbf{p}_I. && (539)
\end{aligned}
$$

EKF–VINS: propagation. Given the prior information $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_{k|k})$ and the motion model $\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_{I_k})$,

$$
\hat{\mathbf{x}}_{k+1|k} = \mathbf{f}_{\text{int}}(\hat{\mathbf{x}}_k, \mathbf{0}), \qquad \tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\,\tilde{\mathbf{x}}_k + \mathbf{G}_k\,\mathbf{n}_{Id_k}, \tag{540}
$$

$$
\boxed{\mathbf{P}_{k+1|k} = \boldsymbol{\Phi}_{(k+1,k)}\,\mathbf{P}_{k|k}\,\boldsymbol{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k\,\mathbf{R}_{Id}\,\mathbf{G}_k^\top,} \tag{541}
$$

where $\boldsymbol{\Phi}_{(k+1,k)}$, $\mathbf{G}_k$, and the integrals $\boldsymbol{\Xi}_{1,2,3,4}$ are exactly those of Lecture 5, (437)–(438). Written with the propagated estimates (equivalent to (437) via $\lfloor \mathbf{R}\mathbf{a}\rfloor_\times \mathbf{R} = \mathbf{R}\lfloor \mathbf{a}\rfloor_\times$), and appending $\mathbf{I}_3$ for the static feature block,

$$
\boldsymbol{\Phi}_{(k+1,k)} = \begin{bmatrix}
{}^G_{I_{k+1|k}}\hat{\mathbf{R}}^\top\,{}^G_{I_k}\hat{\mathbf{R}} & \mathbf{0}_3 & \mathbf{0}_3 & \underbrace{-\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t_k\right)\delta t_k}_{\boldsymbol{\Phi}_{14}} & \mathbf{0}_3 \\
-\left\lfloor {}^G\hat{\mathbf{p}}_{I_{k+1|k}} - {}^G\hat{\mathbf{p}}_{I_k} - {}^G\hat{\mathbf{v}}_{I_k}\delta t_k - \frac{1}{2}{}^G\mathbf{g}\,\delta t_k^2 \right\rfloor_\times {}^G_{I_k}\hat{\mathbf{R}} & \mathbf{I}_3 & \mathbf{I}_3\,\delta t_k & \underbrace{{}^G_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_4}_{\boldsymbol{\Phi}_{24}} & \underbrace{-{}^G_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_2}_{\boldsymbol{\Phi}_{25}} \\
-\left\lfloor {}^G\hat{\mathbf{v}}_{I_{k+1|k}} - {}^G\hat{\mathbf{v}}_{I_k} - {}^G\mathbf{g}\,\delta t_k \right\rfloor_\times {}^G_{I_k}\hat{\mathbf{R}} & \mathbf{0}_3 & \mathbf{I}_3 & {}^G_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_3 & -{}^G_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_1 \\
\mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\
\mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3
\end{bmatrix}, \tag{542}
$$

where the marked blocks $\boldsymbol{\Phi}_{14}, \boldsymbol{\Phi}_{24}, \boldsymbol{\Phi}_{25}$ will be used in the observability analysis below.

EKF–VINS: update. With the vision measurement $\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c$, $\mathbf{n}_c \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$:

$$
\hat{\mathbf{z}}_c = \mathbf{h}_c\left(\hat{\mathbf{x}}_{k+1|k}\right), \qquad \tilde{\mathbf{z}}_c = \mathbf{z}_c - \hat{\mathbf{z}}_c = \mathbf{H}\,\tilde{\mathbf{x}}_{k+1} + \mathbf{n}_c, \qquad \mathbf{S}_{k+1} = \mathbf{H}\,\mathbf{P}_{k+1|k}\,\mathbf{H}^\top + \mathbf{R}_{k+1}, \tag{543}
$$

where, by the chain rule through ${}^C\tilde{\mathbf{p}}_f$,

$$
\mathbf{H} = \frac{\partial \tilde{\mathbf{z}}_c}{\partial {}^C\tilde{\mathbf{p}}_f} \cdot \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \tilde{\mathbf{x}}}, \qquad \frac{\partial \tilde{\mathbf{z}}_c}{\partial {}^C\tilde{\mathbf{p}}_f} = \begin{bmatrix} \frac{1}{\hat{c}_z} & 0 & -\frac{\hat{c}_x}{\hat{c}_z^2} \\ 0 & \frac{1}{\hat{c}_z} & -\frac{\hat{c}_y}{\hat{c}_z^2} \end{bmatrix}, \tag{544}
$$

$$
\frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \tilde{\mathbf{x}}} = \left[{}^I_C\hat{\mathbf{R}}^\top \left\lfloor {}^G_I\hat{\mathbf{R}}^\top \left({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_I\right) \right\rfloor_\times \;\; -{}^I_C\hat{\mathbf{R}}^\top\,{}^G_I\hat{\mathbf{R}}^\top \;\; \mathbf{0}_3 \;\; \mathbf{0}_3 \;\; \mathbf{0}_3 \;\; {}^I_C\hat{\mathbf{R}}^\top\,{}^G_I\hat{\mathbf{R}}^\top \right]. \tag{545}
$$

Then the Kalman gain and the update:

$$
\begin{aligned}
\mathbf{K} &= \mathbf{P}_{k+1|k}\,\mathbf{H}^\top\,\mathbf{S}_{k+1}^{-1} = \mathbf{P}_{k+1|k}\,\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_{k+1|k}\mathbf{H}^\top + \mathbf{R}_{k+1}\right)^{-1}, && (546)\\
\hat{\mathbf{x}}_{k+1|k+1} &= \hat{\mathbf{x}}_{k+1|k} \boxplus \tilde{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{k+1|k} \boxplus \mathbf{K}\,\tilde{\mathbf{z}}_c, && (547)\\
\mathbf{P}_{k+1|k+1} &= \mathbf{P}_{k+1|k} - \mathbf{P}_{k+1|k}\,\mathbf{H}^\top\,\mathbf{S}_{k+1}^{-1}\,\mathbf{H}\,\mathbf{P}_{k+1|k}. && (548)
\end{aligned}
$$

Beyond the basic EKF. The above is the basic idea of the EKF; a family of VINS estimators refines it:

1. sliding-window filter;
2. iterative EKF;
3. square-root inverse filter;
4. square-root filter;
5. Schmidt filter;
6. invariant EKF;
7. equivariant filter.

6.8 Observability analysis

Observability describes whether the system can recover its initial state based on all the history measurements. Assuming no noise, the error-state system evolves as

$$\begin{aligned}
t_1 :&\quad \tilde{\mathbf{z}}_1 = \mathbf{H}_1\,\tilde{\mathbf{x}}_1,\\
t_2 :&\quad \tilde{\mathbf{x}}_2 = \mathbf{\Phi}_{(2,1)}\,\tilde{\mathbf{x}}_1,\quad \tilde{\mathbf{z}}_2 = \mathbf{H}_2\,\tilde{\mathbf{x}}_2 \;\Rightarrow\; \tilde{\mathbf{z}}_2 = \mathbf{H}_2\,\mathbf{\Phi}_{(2,1)}\,\tilde{\mathbf{x}}_1,\\
t_3 :&\quad \tilde{\mathbf{x}}_3 = \mathbf{\Phi}_{(3,2)}\,\tilde{\mathbf{x}}_2,\quad \tilde{\mathbf{z}}_3 = \mathbf{H}_3\,\tilde{\mathbf{x}}_3 \;\Rightarrow\; \tilde{\mathbf{z}}_3 = \mathbf{H}_3\,\mathbf{\Phi}_{(3,2)}\,\mathbf{\Phi}_{(2,1)}\,\tilde{\mathbf{x}}_1 \triangleq \mathbf{H}_3\,\mathbf{\Phi}_{(3,1)}\,\tilde{\mathbf{x}}_1,\\
&\quad\vdots\\
t_k :&\quad \tilde{\mathbf{x}}_k = \mathbf{\Phi}_{(k,k-1)}\,\tilde{\mathbf{x}}_{k-1},\quad \tilde{\mathbf{z}}_k = \mathbf{H}_k\,\tilde{\mathbf{x}}_k \;\Rightarrow\; \tilde{\mathbf{z}}_k = \mathbf{H}_k\,\mathbf{\Phi}_{(k,k-1)}\cdots\mathbf{\Phi}_{(2,1)}\,\tilde{\mathbf{x}}_1 = \mathbf{H}_k\,\mathbf{\Phi}_{(k,1)}\,\tilde{\mathbf{x}}_1.
\end{aligned} \tag{549}$$

Hence, stacking all the measurements,

$$\begin{bmatrix}\tilde{\mathbf{z}}_1\\ \tilde{\mathbf{z}}_2\\ \tilde{\mathbf{z}}_3\\ \vdots\\ \tilde{\mathbf{z}}_k\end{bmatrix} = \begin{bmatrix}\mathbf{H}_1\\ \mathbf{H}_2\,\mathbf{\Phi}_{(2,1)}\\ \mathbf{H}_3\,\mathbf{\Phi}_{(3,1)}\\ \vdots\\ \mathbf{H}_k\,\mathbf{\Phi}_{(k,1)}\end{bmatrix}\tilde{\mathbf{x}}_1, \qquad \mathbf{\Theta} \triangleq \begin{bmatrix}\mathbf{H}_1\\ \mathbf{H}_2\,\mathbf{\Phi}_{(2,1)}\\ \mathbf{H}_3\,\mathbf{\Phi}_{(3,1)}\\ \vdots\\ \mathbf{H}_k\,\mathbf{\Phi}_{(k,1)}\end{bmatrix} \triangleq \begin{bmatrix}\mathbf{M}_1\\ \mathbf{M}_2\\ \mathbf{M}_3\\ \vdots\\ \mathbf{M}_k\end{bmatrix} \quad \leftarrow \text{the } \textit{observability matrix}. \tag{550}$$

The block rows. Let us study $\mathbf{M}_k \triangleq \mathbf{H}_k\,\mathbf{\Phi}_{(k,1)}$. Using $\mathbf{H}_k$ from (545) and the accumulated state transition $\mathbf{\Phi}_{(k,1)}$ from (542),

$$\mathbf{M}_k = \frac{\partial \tilde{\mathbf{z}}_c}{\partial {}^C\tilde{\mathbf{p}}_f}\,{}^I_C\hat{\mathbf{R}}^\top\,{}^G_{I_k}\hat{\mathbf{R}}^\top \begin{bmatrix}\mathbf{\Gamma}_1 & \mathbf{\Gamma}_2 & \mathbf{\Gamma}_3 & \mathbf{\Gamma}_4 & \mathbf{\Gamma}_5 & \mathbf{\Gamma}_6\end{bmatrix}, \tag{551}$$

with (writing $\delta t_k \triangleq t_k - t_1$ for the accumulated interval)

$$\mathbf{\Gamma}_1 = \left\lfloor {}^G\mathbf{p}_f - {}^G\hat{\mathbf{p}}_{I_1} - {}^G\hat{\mathbf{v}}_{I_1}\,\delta t_k - \tfrac{1}{2}{}^G\mathbf{g}\,\delta t_k^2 \right\rfloor_\times {}^G_{I_1}\hat{\mathbf{R}}, \tag{552}$$

$$\mathbf{\Gamma}_2 = -\mathbf{I}_3, \qquad \mathbf{\Gamma}_3 = -\mathbf{I}_3\,\delta t_k, \tag{553}$$

$$\mathbf{\Gamma}_4 = \left\lfloor {}^G\mathbf{p}_f - {}^G\mathbf{p}_{I_k} \right\rfloor_\times {}^G_{I_k}\hat{\mathbf{R}}\,\mathbf{\Phi}_{14} - \mathbf{\Phi}_{24}, \qquad \mathbf{\Gamma}_5 = -\mathbf{\Phi}_{25}, \qquad \mathbf{\Gamma}_6 = \mathbf{I}_3, \tag{554}$$

where $\mathbf{\Phi}_{14}, \mathbf{\Phi}_{24}, \mathbf{\Phi}_{25}$ are the marked blocks of $\mathbf{\Phi}_{(k,1)}$ (see (542)).

The unobservable directions. We can easily find $\mathbf{N}$ such that $\mathbf{M}_k \mathbf{N} = \mathbf{0}$ for all k:

$$\mathbf{N} = \begin{bmatrix} {}^{G}_{I_1}\hat{\mathbf{R}}^{\top}\, {}^{G}\mathbf{g} & \mathbf{0}_3 \\ -\left\lfloor {}^{G}\mathbf{p}_{I_1} \right\rfloor_{\times} {}^{G}\mathbf{g} & \mathbf{I}_3 \\ -\left\lfloor {}^{G}\mathbf{v}_{I_1} \right\rfloor_{\times} {}^{G}\mathbf{g} & \mathbf{0}_3 \\ \mathbf{0}_{3\times 1} & \mathbf{0}_3 \\ \mathbf{0}_{3\times 1} & \mathbf{0}_3 \\ -\left\lfloor {}^{G}\mathbf{p}_{f} \right\rfloor_{\times} {}^{G}\mathbf{g} & \mathbf{I}_3 \end{bmatrix}, \tag{555}$$

an 18×4 matrix (the first column is 18×1, the second block 18×3). Hence the VINS has **4-DoF unobservable directions**:

$$\boxed{\begin{array}{l} \text{1 DoF relates to the global yaw (rotation about gravity),} \\ \text{3 DoF relate to the global translation.} \end{array}}$$

7 Lecture 7: IMU Preintegration, Optimization-Based VINS, and Multi-Visual-Inertial Systems

Topics. (1) preintegration; (2) optimization-based VINS (graph VINS) with the IMU factor; (3) multi-visual-inertial systems (asynchronous multi-camera and multi-IMU).

7.1 Preintegration

Recall the IMU kinematics $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ (Lecture 5). Integrating over one IMU interval gives

$$\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_I) \qquad \leftarrow \text{ 1 measurement: } k \to k+1, \tag{556}$$

while integrating across *multiple* IMU measurements up to t_j gives

$$\mathbf{x}_j = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_I) \qquad \leftarrow \text{ multiple measurements: } k \to j. \tag{557}$$

By the exact integration results of Lecture 5, (557) can be described as (with $\delta t_k \triangleq t_j - t_k$)

$$\begin{cases} {}^G_{I_j}\mathbf{R} = {}^G_{I_k}\mathbf{R} \cdot \Delta\mathbf{R}, \\ {}^G\mathbf{p}_{I_j} = {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k}\,\delta t_k + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{p} + \frac{1}{2}{}^G\mathbf{g}\,\delta t_k^2, \\ {}^G\mathbf{v}_{I_j} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{v} + {}^G\mathbf{g}\,\delta t_k, \\ \mathbf{b}_{g_j} = \mathbf{b}_{g_k} + \Delta\mathbf{b}_g, \\ \mathbf{b}_{a_j} = \mathbf{b}_{a_k} + \Delta\mathbf{b}_a. \end{cases} \tag{558}$$

It is clear that $\Delta\mathbf{R}, \Delta\mathbf{p}$ and $\Delta\mathbf{v}$ are *integration terms*, driven by the bias-corrected inertial readings

$${}^I\mathbf{a} = {}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a, \qquad {}^I\boldsymbol{\omega} = {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g, \tag{559}$$

while $\Delta\mathbf{b}_g, \Delta\mathbf{b}_a$ are the integrated random walks. We can then reorganize the measurements:

$$\Delta\mathbf{R} \triangleq \Delta\mathbf{R}(\mathbf{b}_g, \mathbf{n}_I) \triangleq {}^{I_k}_{I_j}\mathbf{R} = \underbrace{{}^G_{I_k}\mathbf{R}^\top\, {}^G_{I_j}\mathbf{R}}_{\mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j)}, \tag{560}$$

$$\Delta\mathbf{p} \triangleq \Delta\mathbf{p}(\mathbf{b}_a, \mathbf{b}_g, \mathbf{n}_I) \triangleq \int_{t_k}^{t_j}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\, {}^{I_s}\mathbf{a}\, ds\, d\tau = \underbrace{{}^G_{I_k}\mathbf{R}^\top\left({}^G\mathbf{p}_{I_j} - {}^G\mathbf{p}_{I_k} - {}^G\mathbf{v}_{I_k}\,\delta t_k - \frac{1}{2}{}^G\mathbf{g}\,\delta t_k^2\right)}_{\mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j)}, \tag{561}$$

$$\Delta\mathbf{v} \triangleq \Delta\mathbf{v}(\mathbf{b}_a, \mathbf{b}_g, \mathbf{n}_I) \triangleq \int_{t_k}^{t_j} {}^{I_k}_{I_\tau}\mathbf{R}\, {}^{I_\tau}\mathbf{a}\, d\tau = \underbrace{{}^G_{I_k}\mathbf{R}^\top\left({}^G\mathbf{v}_{I_j} - {}^G\mathbf{v}_{I_k} - {}^G\mathbf{g}\,\delta t_k\right)}_{\mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j)}, \tag{562}$$

$$\Delta\mathbf{b}_g \triangleq \Delta\mathbf{b}_g(\mathbf{n}_I) \triangleq \int_{t_k}^{t_j} \mathbf{n}_{wg}\, d\tau = \mathbf{b}_{g_j} - \mathbf{b}_{g_k}, \tag{563}$$

$$\Delta\mathbf{b}_a \triangleq \Delta\mathbf{b}_a(\mathbf{n}_I) \triangleq \int_{t_k}^{t_j} \mathbf{n}_{wa}\, d\tau = \mathbf{b}_{a_j} - \mathbf{b}_{a_k}. \tag{564}$$

Two sides of the same equations:

- **Left-hand side:** the measurements from the IMU (relates to the IMU biases and noises);

- **Right-hand side:** the nonlinear functions containing the states $\mathbf{x}$:

$$\begin{cases} \Delta\mathbf{R}(\mathbf{b}_g, \mathbf{n}_I) = \mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j), \\ \Delta\mathbf{p}(\mathbf{b}_a, \mathbf{b}_g, \mathbf{n}_I) = \mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j), \\ \Delta\mathbf{v}(\mathbf{b}_a, \mathbf{b}_g, \mathbf{n}_I) = \mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j), \\ \Delta\mathbf{b}_g(\mathbf{n}_I) = \mathbf{b}_{g_j} - \mathbf{b}_{g_k}, \\ \Delta\mathbf{b}_a(\mathbf{n}_I) = \mathbf{b}_{a_j} - \mathbf{b}_{a_k} \end{cases} \quad \Rightarrow \text{ the IMU measurements, no assumption yet.} \tag{565}$$

Linearizing about the bias estimates. Give an estimate of $\hat{\mathbf{b}}_g$ and $\hat{\mathbf{b}}_a$. Since $\Delta\mathbf{p}$ and $\Delta\mathbf{v}$ depend on *both* biases, it is convenient to collect them into a single bias state and to stack the corresponding Jacobians:

$$\mathbf{x}_b \triangleq \begin{bmatrix} \mathbf{b}_g \\ \mathbf{b}_a \end{bmatrix}, \qquad \mathbf{H}_b^p \triangleq \begin{bmatrix} \mathbf{H}_g^p & \mathbf{H}_a^p \end{bmatrix}, \qquad \mathbf{H}_b^v \triangleq \begin{bmatrix} \mathbf{H}_g^v & \mathbf{H}_a^v \end{bmatrix}, \tag{566}$$

so that e.g. $\mathbf{H}_b^p\,\tilde{\mathbf{x}}_b = \mathbf{H}_g^p\,\tilde{\mathbf{b}}_g + \mathbf{H}_a^p\,\tilde{\mathbf{b}}_a$. ($\Delta\mathbf{R}$ depends on $\mathbf{b}_g$ only, so it keeps the single Jacobian $\mathbf{H}_g^R$.) To first order, then,

$$\begin{aligned} &\Delta\mathbf{R}(\mathbf{b}_g, \mathbf{n}_I) \approx \Delta\mathbf{R}(\hat{\mathbf{b}}_g, \mathbf{0}) \cdot \mathrm{Exp}\left(\mathbf{H}_g^R\,\tilde{\mathbf{b}}_g + \mathbf{H}_n^R\,\mathbf{n}_I\right) \\ \Rightarrow \quad &\Delta\mathbf{R}(\hat{\mathbf{b}}_g, \mathbf{0}) = \mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j) \cdot \mathrm{Exp}\left(-\mathbf{H}_g^R\,\tilde{\mathbf{b}}_g - \mathbf{H}_n^R\,\mathbf{n}_I\right); \end{aligned} \tag{567}$$

similarly,

$$\begin{aligned} &\Delta\mathbf{p}(\mathbf{b}_g, \mathbf{b}_a, \mathbf{n}_I) \approx \Delta\mathbf{p}(\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a, \mathbf{0}) + \mathbf{H}_b^p\,\tilde{\mathbf{x}}_b + \mathbf{H}_n^p\,\mathbf{n}_I \\ \Rightarrow \quad &\Delta\mathbf{p}(\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a, \mathbf{0}) = \mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j) - \mathbf{H}_b^p\,\tilde{\mathbf{x}}_b - \mathbf{H}_n^p\,\mathbf{n}_I, \end{aligned} \tag{568}$$

$$\begin{aligned} &\Delta\mathbf{v}(\mathbf{b}_g, \mathbf{b}_a, \mathbf{n}_I) \approx \Delta\mathbf{v}(\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a, \mathbf{0}) + \mathbf{H}_b^v\,\tilde{\mathbf{x}}_b + \mathbf{H}_n^v\,\mathbf{n}_I \\ \Rightarrow \quad &\Delta\mathbf{v}(\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a, \mathbf{0}) = \mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j) - \mathbf{H}_b^v\,\tilde{\mathbf{x}}_b - \mathbf{H}_n^v\,\mathbf{n}_I. \end{aligned} \tag{569}$$

Then, for the biases, we get

$$\Delta\mathbf{b}_g(\mathbf{n}_I) = \mathbf{0} + \int_{t_k}^{t_j} \mathbf{n}_{wg}\,dt = \mathbf{0} + \mathbf{n}_{wgd}\,\delta t_k, \qquad \Delta\mathbf{b}_a(\mathbf{n}_I) = \mathbf{0} + \int_{t_k}^{t_j} \mathbf{n}_{wa}\,dt = \mathbf{0} + \mathbf{n}_{wad}\,\delta t_k, \tag{570}$$

with $\mathbf{n}_{wgd}, \mathbf{n}_{wad}$ the discrete noises. Hence **the cleaner IMU measurement** is:

$$\boxed{\begin{aligned} \Delta\mathbf{R}(\hat{\mathbf{b}}_g, \mathbf{0}) &= \mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j) \cdot \mathrm{Exp}\left(-\mathbf{H}_g^R\,\tilde{\mathbf{b}}_g - \mathbf{H}_n^R\,\mathbf{n}_I\right), \\ \Delta\mathbf{p}(\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a, \mathbf{0}) &= \mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j) - \mathbf{H}_b^p\,\tilde{\mathbf{x}}_b - \mathbf{H}_n^p\,\mathbf{n}_I, \\ \Delta\mathbf{v}(\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a, \mathbf{0}) &= \mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j) - \mathbf{H}_b^v\,\tilde{\mathbf{x}}_b - \mathbf{H}_n^v\,\mathbf{n}_I, \\ \mathbf{0} &= \mathbf{b}_{g_j} - \mathbf{b}_{g_k} - \mathbf{n}_{wgd}\,\delta t_k, \\ \mathbf{0} &= \mathbf{b}_{a_j} - \mathbf{b}_{a_k} - \mathbf{n}_{wad}\,\delta t_k. \end{aligned}}$$

Stacking the left-hand sides as the *preintegrated measurement* $\mathbf{z}_I$, with $\mathbf{n}_I \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{ij})$:

$$\begin{aligned} \mathbf{z}_I = \mathbf{h}_I(\mathbf{x}_k, \mathbf{x}_j, \mathbf{n}_I) \quad \Rightarrow \quad &\mathbf{z}_I = \underbrace{\mathbf{h}_I(\hat{\mathbf{x}}_k, \hat{\mathbf{x}}_j, \mathbf{0})}_{\hat{\mathbf{z}}_I} + \mathbf{H}_k\,\tilde{\mathbf{x}}_k + \mathbf{H}_j\,\tilde{\mathbf{x}}_j + \mathbf{G}_I\,\mathbf{n}_I \\ \Rightarrow \quad &\tilde{\mathbf{z}}_I = \mathbf{H}_k\,\tilde{\mathbf{x}}_k + \mathbf{H}_j\,\tilde{\mathbf{x}}_j + \mathbf{G}_I\,\mathbf{n}_I. \end{aligned} \tag{571}$$

Why would we like this change? Because this is exactly the format a least-squares solver wants: a residual that is linear in the state errors. Writing the covariance of the preintegrated measurement as

$$\mathbf{Q}_I \triangleq \mathbf{G}_I\,\mathbf{R}_{ij}\,\mathbf{G}_I^\top, \tag{572}$$

we can form a *cost term*

$$\|\mathbf{z}_I - \mathbf{h}_I(\mathbf{x}_k, \mathbf{x}_j, \mathbf{0})\|^2_{\mathbf{Q}_I} \quad\Rightarrow\quad \|\tilde{\mathbf{z}}_I - \mathbf{H}_k\,\tilde{\mathbf{x}}_k - \mathbf{H}_j\,\tilde{\mathbf{x}}_j\|^2_{\mathbf{Q}_I}, \tag{573}$$

which can be used in BA / least squares together with the vision factors easily.

But there is a **caveat** for the above formulation:

1. for the least squares / MLE, we have multiple iterations;
2. for multi-iterations, $\hat{\mathbf{b}}_g$ is changing; then $\Delta\mathbf{p}, \Delta\mathbf{R}, \Delta\mathbf{v}$ and $\mathbf{G}_I, \mathbf{R}_{ij}$ will change;
3. if so, for every iteration we need to recompute $\mathbf{z}_I$ and $\mathbf{G}_I, \mathbf{R}_{ij}$;
4. if we have a high-frequency IMU and a lot of such IMU costs in the MLE, it will cause substantial computations.

How to solve this? The question is: *what is the way to only integrate once and avoid repeated integration?* Consider a generic measurement $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, and let $\hat{\mathbf{x}}_B^0$ be a *fixed* linearization point for $\mathbf{x}_B$ — the *initial* estimate — while $\hat{\mathbf{x}}_B$ is the current estimate. Writing the error with respect to the fixed point,

$$\tilde{\mathbf{x}}_B^0 \triangleq \mathbf{x}_B - \hat{\mathbf{x}}_B^0 = \big(\mathbf{x}_B - \hat{\mathbf{x}}_B\big) + \big(\hat{\mathbf{x}}_B - \hat{\mathbf{x}}_B^0\big) = \tilde{\mathbf{x}}_B + \Delta\hat{\mathbf{x}}_B, \qquad \Delta\hat{\mathbf{x}}_B \triangleq \hat{\mathbf{x}}_B - \hat{\mathbf{x}}_B^0. \tag{574}$$

Linearizing about $\hat{\mathbf{x}}_B^0$ — so that $\mathbf{H}_B^0$ is evaluated at the initial estimate and never re-evaluated —

$$\begin{aligned} \tilde{\mathbf{z}} &= \mathbf{H}_A\,\tilde{\mathbf{x}}_A + \mathbf{H}_B^0\,\tilde{\mathbf{x}}_B^0 + \mathbf{n} = \mathbf{H}_A\,\tilde{\mathbf{x}}_A + \mathbf{H}_B^0\,\tilde{\mathbf{x}}_B + \mathbf{H}_B^0\,\Delta\hat{\mathbf{x}}_B + \mathbf{n} \\ \Rightarrow\quad \underbrace{\tilde{\mathbf{z}}}_{\text{unchanged}} &- \boxed{\mathbf{H}_B^0\,\Delta\hat{\mathbf{x}}_B} = \mathbf{H}_A\,\tilde{\mathbf{x}}_A + \underbrace{\mathbf{H}_B^0\,\tilde{\mathbf{x}}_B}_{\text{unchanged}} + \underbrace{\mathbf{n}}_{\text{unchanged}}, \end{aligned} \tag{575}$$

where across iterations *only the boxed term* $\mathbf{H}_B^0\,\Delta\hat{\mathbf{x}}_B$ *changes* — a small computation: the Jacobians (and hence the preintegrated terms and noise mapping) stay at the fixed linearization point, and only the correction $\Delta\hat{\mathbf{x}}_B$ is updated.

7.2 Preintegration with a Fixed Bias Linearization Point

Everything is now in place: the only change from §7.1 is *where* the bias Jacobians are evaluated. Instead of linearizing about the *current* estimates $\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a$ — which move at every Gauss–Newton / LM iteration and therefore force a re-integration — we linearize about the *fixed initial* estimates $\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0$ and pay for it with one extra correction term per row, exactly as in (575).

Applying the fixed linearization point. Using the bias state $\mathbf{x}_b$ of (566), define the corrections

$$\Delta\hat{\mathbf{x}}_b \triangleq \begin{bmatrix} \hat{\mathbf{b}}_g - \hat{\mathbf{b}}_g^0 \\ \hat{\mathbf{b}}_a - \hat{\mathbf{b}}_a^0 \end{bmatrix}, \qquad \Delta\hat{\mathbf{x}}_{b_g} \triangleq \hat{\mathbf{b}}_g - \hat{\mathbf{b}}_g^0. \tag{576}$$

For the rotation, splitting $\tilde{\mathbf{b}}_g^0 = \Delta\hat{\mathbf{x}}_{b_g} + \tilde{\mathbf{b}}_g$ exactly as in (574),

$$\begin{aligned}
\Delta\mathbf{R}(\mathbf{b}_g, \mathbf{n}_I) &= \Delta\mathbf{R}\big(\hat{\mathbf{b}}_g^0 + \tilde{\mathbf{b}}_g^0, \mathbf{n}_I\big) \;\approx\; \Delta\mathbf{R}(\hat{\mathbf{b}}_g^0, \mathbf{0}) \cdot \mathrm{Exp}\left(\mathbf{H}_g^R\, \tilde{\mathbf{b}}_g^0 + \mathbf{H}_n^R\, \mathbf{n}_I\right) \\
&= \Delta\mathbf{R}(\hat{\mathbf{b}}_g^0, \mathbf{0}) \cdot \mathrm{Exp}\left(\mathbf{H}_g^R\, \Delta\hat{\mathbf{x}}_{b_g} + \mathbf{H}_g^R\, \tilde{\mathbf{b}}_g + \mathbf{H}_n^R\, \mathbf{n}_I\right) \;=\; \mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j) \\
\Rightarrow \quad \Delta\mathbf{R}(\hat{\mathbf{b}}_g^0, \mathbf{0}) &= \mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j) \cdot \mathrm{Exp}\left(-\mathbf{H}_g^R\, \Delta\hat{\mathbf{x}}_{b_g} - \mathbf{H}_g^R\, \tilde{\mathbf{b}}_g - \mathbf{H}_n^R\, \mathbf{n}_I\right).
\end{aligned} \tag{577}$$

Likewise for position and velocity, with the same stacked Jacobians $\mathbf{H}_b^p, \mathbf{H}_b^v$ of (566) now evaluated at the fixed point:

$$\begin{aligned}
\Delta\mathbf{p}(\mathbf{b}_g, \mathbf{b}_a, \mathbf{n}_I) &= \Delta\mathbf{p}(\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0, \mathbf{0}) + \mathbf{H}_b^p\, \Delta\hat{\mathbf{x}}_b + \mathbf{H}_b^p\, \tilde{\mathbf{x}}_b + \mathbf{H}_n^p\, \mathbf{n}_I \;=\; \mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j) \\
\Rightarrow \quad \Delta\mathbf{p}(\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0, \mathbf{0}) &= \mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j) - \mathbf{H}_b^p\, \Delta\hat{\mathbf{x}}_b - \mathbf{H}_b^p\, \tilde{\mathbf{x}}_b - \mathbf{H}_n^p\, \mathbf{n}_I,
\end{aligned} \tag{578}$$

$$\begin{aligned}
\Delta\mathbf{v}(\mathbf{b}_g, \mathbf{b}_a, \mathbf{n}_I) &= \Delta\mathbf{v}(\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0, \mathbf{0}) + \mathbf{H}_b^v\, \Delta\hat{\mathbf{x}}_b + \mathbf{H}_b^v\, \tilde{\mathbf{x}}_b + \mathbf{H}_n^v\, \mathbf{n}_I \;=\; \mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j) \\
\Rightarrow \quad \Delta\mathbf{v}(\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0, \mathbf{0}) &= \mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j) - \mathbf{H}_b^v\, \Delta\hat{\mathbf{x}}_b - \mathbf{H}_b^v\, \tilde{\mathbf{x}}_b - \mathbf{H}_n^v\, \mathbf{n}_I.
\end{aligned} \tag{579}$$

The final IMU measurement. Stacking the five rows gives the preintegrated measurement. This is row-for-row the "cleaner IMU measurement" box of §7.1, with $\hat{\mathbf{b}}_g, \hat{\mathbf{b}}_a \to \hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0$ and one extra $\mathbf{H}\,\Delta\hat{\mathbf{x}}$ term per row — and those boxed terms are the *only* ones that change from iteration to iteration:

$$\begin{bmatrix} \Delta\mathbf{R}(\hat{\mathbf{b}}_g^0, \mathbf{0}) \\ \Delta\mathbf{p}(\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0, \mathbf{0}) \\ \Delta\mathbf{v}(\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0, \mathbf{0}) \\ \Delta\mathbf{b}_g \\ \Delta\mathbf{b}_a \end{bmatrix} = \begin{bmatrix} \mathbf{h}_R(\mathbf{x}_k, \mathbf{x}_j) \cdot \mathrm{Exp}\left(-\boxed{\mathbf{H}_g^R\, \Delta\hat{\mathbf{x}}_{b_g}} - \mathbf{H}_g^R\, \tilde{\mathbf{b}}_g - \mathbf{H}_n^R\, \mathbf{n}_I\right) \\ \mathbf{h}_p(\mathbf{x}_k, \mathbf{x}_j) - \boxed{\mathbf{H}_b^p\, \Delta\hat{\mathbf{x}}_b} - \mathbf{H}_b^p\, \tilde{\mathbf{x}}_b - \mathbf{H}_n^p\, \mathbf{n}_I \\ \mathbf{h}_v(\mathbf{x}_k, \mathbf{x}_j) - \boxed{\mathbf{H}_b^v\, \Delta\hat{\mathbf{x}}_b} - \mathbf{H}_b^v\, \tilde{\mathbf{x}}_b - \mathbf{H}_n^v\, \mathbf{n}_I \\ \mathbf{b}_{g_j} - \mathbf{b}_{g_k} \\ \mathbf{b}_{a_j} - \mathbf{b}_{a_k} \end{bmatrix} \triangleq \mathbf{z}_I = \mathbf{h}_I(\mathbf{x}_k, \mathbf{x}_j, \mathbf{n}_I). \tag{580}$$

The whole point is that

- $\mathbf{z}_I$: we only need to **integrate once**, given $\hat{\mathbf{b}}_g^0, \hat{\mathbf{b}}_a^0$;
- $\Delta\hat{\mathbf{x}}_b = \left[\hat{\mathbf{b}}_g - \hat{\mathbf{b}}_g^0;\; \hat{\mathbf{b}}_a - \hat{\mathbf{b}}_a^0\right]$ is the **bias linearization-change correction**, cheap to apply at each iteration.

The IMU factor. We can now construct the IMU factor (IMU cost term)

$$\boxed{C_I = \|\mathbf{z}_I - \mathbf{h}_I(\mathbf{x}_k, \mathbf{x}_j, \mathbf{0})\|^2_{\mathbf{Q}_I},}$$

where $\mathbf{z}_I$ is the IMU *preintegrated measurement* and $\mathbf{Q}_I$ of (572) the covariance of the IMU *preintegrated noise*.

7.3 Computing the Preintegration: Recursive Formulation

The remaining question is how to actually perform the IMU preintegration and obtain

$$\mathbf{z}_I, \quad \mathbf{Q}_I, \quad \mathbf{H}_b^p, \quad \mathbf{H}_b^v, \quad \mathbf{H}_g^R. \tag{581}$$

There are many IMU readings between k and j (Fig. 18); the way to solve this is to use a *recursive formulation* over the individual IMU intervals $i \in [k, j]$.

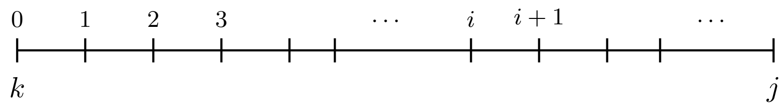


Figure 18: IMU readings between k and j: the preintegrated quantities are built up recursively over each IMU interval $i \to i+1$, $i \in [k, j]$.

Recursion. Propagating the preintegrated terms one IMU interval at a time,

$$
\begin{cases}
\Delta\mathbf{R}_{i+1} = \Delta\mathbf{R}_i \cdot \mathbf{R}_{i,i+1}, \\
\Delta\mathbf{p}_{i+1} = \Delta\mathbf{p}_i + \Delta\mathbf{v}_i\,\delta t_i + \Delta\mathbf{R}_i\,\mathbf{p}_{i,i+1}, \\
\Delta\mathbf{v}_{i+1} = \Delta\mathbf{v}_i + \Delta\mathbf{R}_i\,\mathbf{v}_{i,i+1}, \\
\Delta\mathbf{b}_{g,i+1} = \Delta\mathbf{b}_{g,i} + \int_{t_i}^{t_{i+1}} \mathbf{n}_{wg}\,dt, \\
\Delta\mathbf{b}_{a,i+1} = \Delta\mathbf{b}_{a,i} + \int_{t_i}^{t_{i+1}} \mathbf{n}_{wa}\,dt,
\end{cases}
\qquad
\begin{cases}
\mathbf{R}_{i,i+1} = \mathrm{Exp}\left(\hat{\boldsymbol{\omega}}_i\,\delta t_i\right), \\
\mathbf{p}_{i,i+1} = \displaystyle\int_{t_i}^{t_{i+1}}\int_{t_i}^{\tau} {}^{I_i}_{I_s}\mathbf{R}\;{}^{I_s}\mathbf{a}\,ds\,d\tau, \\
\mathbf{v}_{i,i+1} = \displaystyle\int_{t_i}^{t_{i+1}} {}^{I_i}_{I_\tau}\mathbf{R}\;{}^{I_\tau}\mathbf{a}\,d\tau,
\end{cases}
\tag{582}
$$

which advances the preintegrated measurement $\mathbf{z}_{I,i} \to \mathbf{z}_{I,i+1}$. The interval quantities $\mathbf{p}_{i,i+1}, \mathbf{v}_{i,i+1}$ are evaluated either in closed form under the ACI^2 / CPI models [14, 15] or a linear-in-time assumption on the readings, or numerically with RK4.

Linearized error and covariance propagation. If we linearize the recursion, the preintegrated error state and its covariance propagate as

$$
\tilde{\mathbf{z}}_{I,i+1} = \boldsymbol{\Phi}_{(i+1,i)}\,\tilde{\mathbf{z}}_{I,i} + \boxed{\boldsymbol{\Phi}_b}\,\tilde{\mathbf{x}}_{b_k} + \mathbf{G}_i\,\mathbf{n}_{Id}, \tag{583}
$$

$$
\mathbf{Q}_{I,i+1} = \boldsymbol{\Phi}_{(i+1,i)}\,\mathbf{Q}_{I,i}\,\boldsymbol{\Phi}_{(i+1,i)}^{\top} + \mathbf{G}_i\,\mathbf{Q}_d\,\mathbf{G}_i^{\top}, \tag{584}
$$

where $\mathbf{n}_{Id}$ is the discrete IMU noise

$$
\mathbf{n}_{Id} \sim \mathcal{N}\left(\mathbf{0},\ \begin{bmatrix} \sigma_g^2\mathbf{I}_3 & & & \\ & \sigma_a^2\mathbf{I}_3 & & \\ & & \sigma_{wg}^2\mathbf{I}_3 & \\ & & & \sigma_{wa}^2\mathbf{I}_3 \end{bmatrix} \frac{1}{\delta t_i}\right), \tag{585}
$$

and the boxed $\boldsymbol{\Phi}_b$ is exactly the block that **contains the Jacobians $\mathbf{H}_g^R, \mathbf{H}_b^p, \mathbf{H}_b^v$** needed in (580). With this, the IMU preintegration factor is complete.

7.4 Discussion

1. **The key idea** is to *control the linearization of the biases*; the assumption behind it is that the biases are *slowly varying*, so that $\Delta\hat{\mathbf{x}}_b$ stays small and the first-order correction remains valid.
2. This preintegration idea can be applied to **any other temporal signal**, e.g. wheel encoders or rotor RPM.
3. This preintegration keeps the linearization of the *biases*, but it can be extended to **more parameters** — e.g. the IMU intrinsics (scale / skew parameters).
4. **How to use the IMU factor?** One must initialize the velocity and the biases (cf. the visual-inertial initialization of Lecture 6).

7.5 Optimization-Based VINS (Graph VINS)

A typical optimization-based VINS combines a prior, the IMU factor, and the camera factors:

$$\begin{cases} \mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k), \\ \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \;\rightarrow\; \mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_{I_k}) \;\;\Rightarrow\;\; \mathbf{z}_I = \mathbf{h}(\mathbf{x}_k, \mathbf{x}_{k+1}, \mathbf{n}_I), \\ \mathbf{z}_{k+1} = \mathbf{h}(\mathbf{x}_{k+1}) + \mathbf{n}_{k+1}, \end{cases} \tag{586}$$

so that the cost is

$$\boxed{C_{\text{VINS}} = \underbrace{\|\mathbf{x}_k - \hat{\mathbf{x}}_k\|^2_{\mathbf{P}_k}}_{\text{prior}} + \underbrace{\|\mathbf{z}_I - \mathbf{h}(\mathbf{x}_k, \mathbf{x}_{k+1}, \mathbf{0})\|^2_{\mathbf{Q}_I}}_{C_{\text{IMU}}} + \underbrace{\|\mathbf{z}_{k+1} - \mathbf{h}(\mathbf{x}_{k+1})\|^2_{\mathbf{R}_{k+1}}}_{C_{\text{cam}}}.}$$

The corresponding factor graph is shown in Fig. 19.

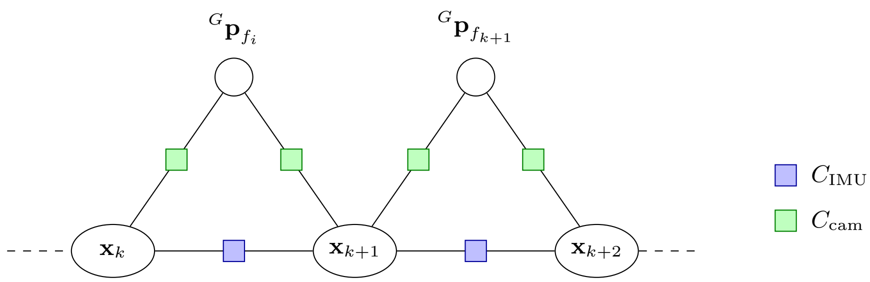


Figure 19: Factor graph of an optimization-based VINS: pose/navigation states $\mathbf{x}_k$ are linked by IMU (preintegration) factors, and to the features ${}^G\mathbf{p}_f$ by camera factors.

Two regimes follow directly from which factors are present:

- purely C_{cam} $\rightarrow$ **BA** (bundle adjustment);
- $C_{\text{IMU}} + C_{\text{cam}}$ $\rightarrow$ **visual-inertial BA**, where the *velocity* and the *biases* are added to the states.

7.6 Multi-Visual-Inertial Systems

7.6.1 Two asynchronous cameras

Suppose we have two cameras whose exposures are *not* synchronized, so that cam1 fires between the times at which the navigation states are defined (Fig. 20). Its pose is obtained by *pose interpolation*:

$${}^G_{C_1}\mathbf{T} = {}^G_{I_t}\mathbf{T} \cdot {}^I_{C_1}\mathbf{T}, \tag{587}$$

where ${}^G_{I_t}\mathbf{T}$ is the pose interpolation between ${}^G_{I_{k+1}}\mathbf{T}$ and ${}^G_{I_{k+2}}\mathbf{T}$.

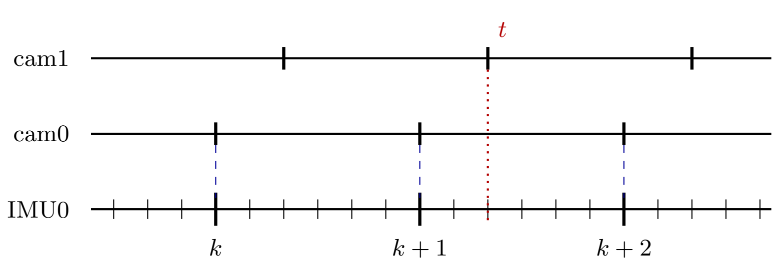


Figure 20: Asynchronous cameras. cam0 is triggered at the state times $k, k{+}1, k{+}2$; cam1 fires at an intermediate time t, so its pose must be interpolated between the two bracketing IMU poses.

In the factor graph this shows up as a third factor type — the *interpolated camera factor* — which, unlike an ordinary camera factor, depends on the *two* bracketing navigation states through ${}^{G}_{I_t}\mathbf{T}$ (Fig. 21).

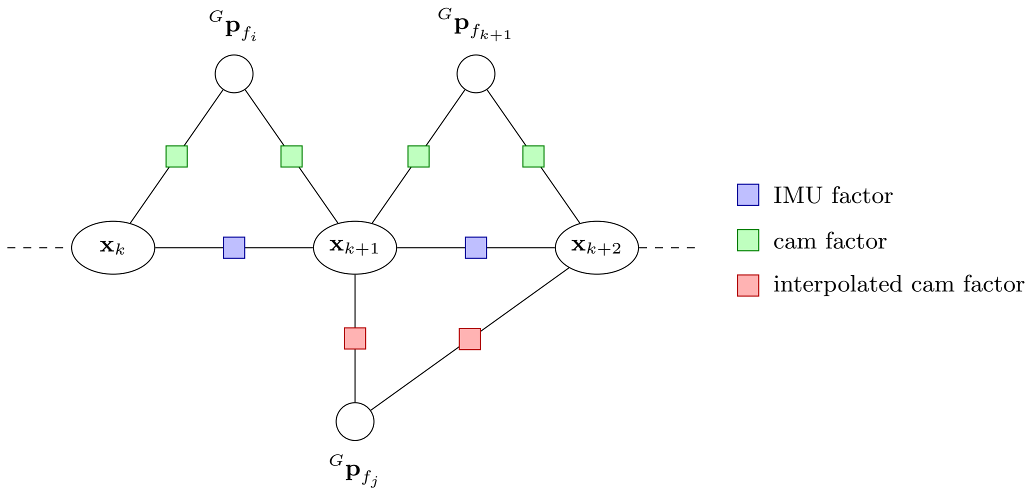


Figure 21: Factor graph with an asynchronous second camera: features seen by cam1 attach through *interpolated* camera factors, which couple the two navigation states bracketing the exposure time.

7.6.2 Multiple IMUs

Now suppose the platform carries more than one IMU, each running at its own rate (Fig. 22). There are two ways to handle the extra units.

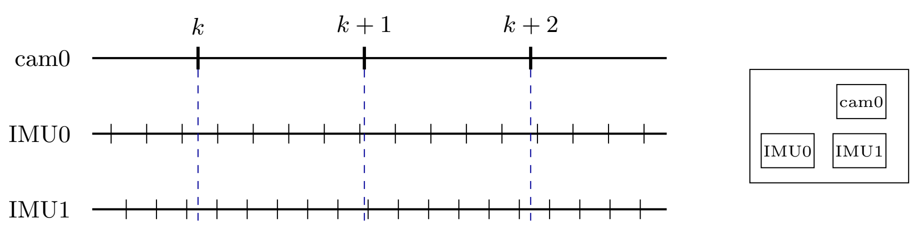


Figure 22: A multi-IMU rig: cam0 samples at the state times $k, k{+}1, k{+}2$ while IMU0 and IMU1 run asynchronously at their own rates.

Method 1: transfer the IMU readings from IMU1 onto IMU0. Using the rigid extrinsics ${}^{I_0}\mathbf{p}_{I_1}$ and differentiating the rigid-body kinematics,

$$
\begin{aligned}
{}^{G}\mathbf{p}_{I_1} &= {}^{G}\mathbf{p}_{I_0} + {}^{G}_{I_0}\mathbf{R}\,{}^{I_0}\mathbf{p}_{I_1}, && (588)\\
{}^{G}\dot{\mathbf{p}}_{I_1} &= {}^{G}\dot{\mathbf{p}}_{I_0} + {}^{G}_{I_0}\dot{\mathbf{R}}\,{}^{I_0}\mathbf{p}_{I_1}, && (589)\\
{}^{G}\mathbf{v}_{I_1} &= {}^{G}\mathbf{v}_{I_0} + {}^{G}_{I_0}\mathbf{R}\,\lfloor {}^{I_0}\boldsymbol{\omega} \rfloor_{\times}\,{}^{I_0}\mathbf{p}_{I_1}, && (590)\\
{}^{G}\dot{\mathbf{v}}_{I_1} &= {}^{G}\dot{\mathbf{v}}_{I_0} + {}^{G}_{I_0}\dot{\mathbf{R}}\,\lfloor {}^{I_0}\boldsymbol{\omega} \rfloor_{\times}\,{}^{I_0}\mathbf{p}_{I_1} + {}^{G}_{I_0}\mathbf{R}\,\lfloor {}^{I_0}\dot{\boldsymbol{\omega}} \rfloor_{\times}\,{}^{I_0}\mathbf{p}_{I_1}, && (591)\\
{}^{G}\mathbf{a}_{I_1} &= {}^{G}\mathbf{a}_{I_0} + {}^{G}_{I_0}\mathbf{R}\,\lfloor {}^{I_0}\boldsymbol{\omega} \rfloor_{\times}^{2}\,{}^{I_0}\mathbf{p}_{I_1} + {}^{G}_{I_0}\mathbf{R}\,\lfloor {}^{I_0}\boldsymbol{\alpha} \rfloor_{\times}\,{}^{I_0}\mathbf{p}_{I_1}, && (592)
\end{aligned}
$$

and hence, expressed in the second IMU's own frame,

$$^{I_1}\mathbf{a}_{I_1} = {}^{I_1}_{I_0}\mathbf{R}\Big({}^{I_0}\mathbf{a}_{I_0} + \lfloor{}^{I_0}\boldsymbol{\omega}\rfloor_\times^2\ {}^{I_0}\mathbf{p}_{I_1} + \underbrace{\lfloor{}^{I_0}\boldsymbol{\alpha}\rfloor_\times\ {}^{I_0}\mathbf{p}_{I_1}}_{\text{needs } \boldsymbol{\alpha}}\Big), \qquad \boldsymbol{\alpha} \triangleq \dot{\boldsymbol{\omega}}. \tag{593}$$

The caveat is explicit in the last term: the **angular acceleration $\boldsymbol{\alpha}$** is needed to do the transfer, and it is not directly measured by a gyroscope.

Method 2: keep IMU1 as its own state, tied by the rigid transform. Instead of transferring readings, we relate the two states directly:

$$\begin{cases} {}^G_{I_1}\mathbf{T} = {}^G_{I_0}\mathbf{T} \cdot {}^{I_0}_{I_1}\mathbf{T}, \\ {}^G\mathbf{v}_{I_1} = {}^G\mathbf{v}_{I_0} + {}^G_{I_0}\mathbf{R}\ \lfloor{}^{I_0}\boldsymbol{\omega}\rfloor_\times\ {}^{I_0}\mathbf{p}_{I_1}. \end{cases} \tag{594}$$

Based on the rigid transform we can therefore represent $\mathbf{x}_{\mathrm{IMU}_1}$ with $\mathbf{x}_{\mathrm{IMU}_0}$:

$$\mathbf{x}_{\mathrm{IMU}_0} = \begin{bmatrix} {}^G_{I_0}\mathbf{R} \\ {}^G\mathbf{p}_{I_0} \\ {}^G\mathbf{v}_{I_0} \\ {}^{I_0}\mathbf{b}_g \\ {}^{I_0}\mathbf{b}_a \end{bmatrix}, \qquad \mathbf{x}_{\mathrm{IMU}_1} = \begin{bmatrix} {}^G_{I_1}\mathbf{R} \\ {}^G\mathbf{p}_{I_1} \\ {}^G\mathbf{v}_{I_1} \\ {}^{I_1}\mathbf{b}_g \\ {}^{I_1}\mathbf{b}_a \end{bmatrix}, \tag{595}$$

The pose block of $\mathbf{x}_{\mathrm{IMU}_1}$ is fully determined by $\mathbf{x}_{\mathrm{IMU}_0}$ and the extrinsics, so the only extra *auxiliary* state that has to be kept per additional IMU is

$$\mathbf{x}_{I_1} = \begin{bmatrix} {}^G\mathbf{v}_{I_1} \\ {}^{I_1}\mathbf{b}_g \\ {}^{I_1}\mathbf{b}_a \end{bmatrix}. \tag{596}$$

The resulting graph (Fig. 23) has a *base* IMU chain (IMU0) carrying the navigation states plus one *auxiliary* IMU factor chain per additional unit [17].

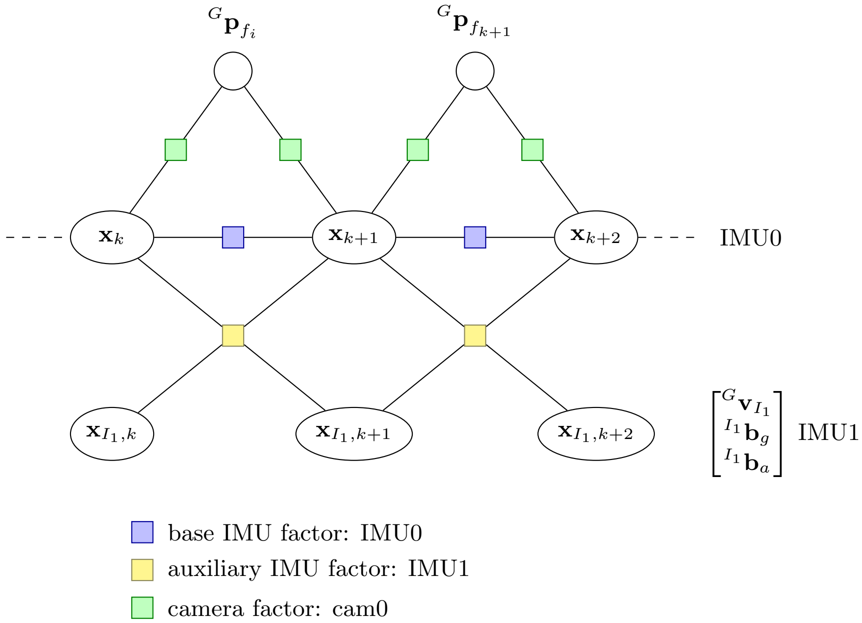


Figure 23: Multi-IMU factor graph. IMU0 is the *base* IMU whose preintegration factors link the navigation states $\mathbf{x}_k$; each additional IMU contributes an *auxiliary* factor chain over its own velocity and biases, tied to the base states through the rigid extrinsics.

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